

NOVA

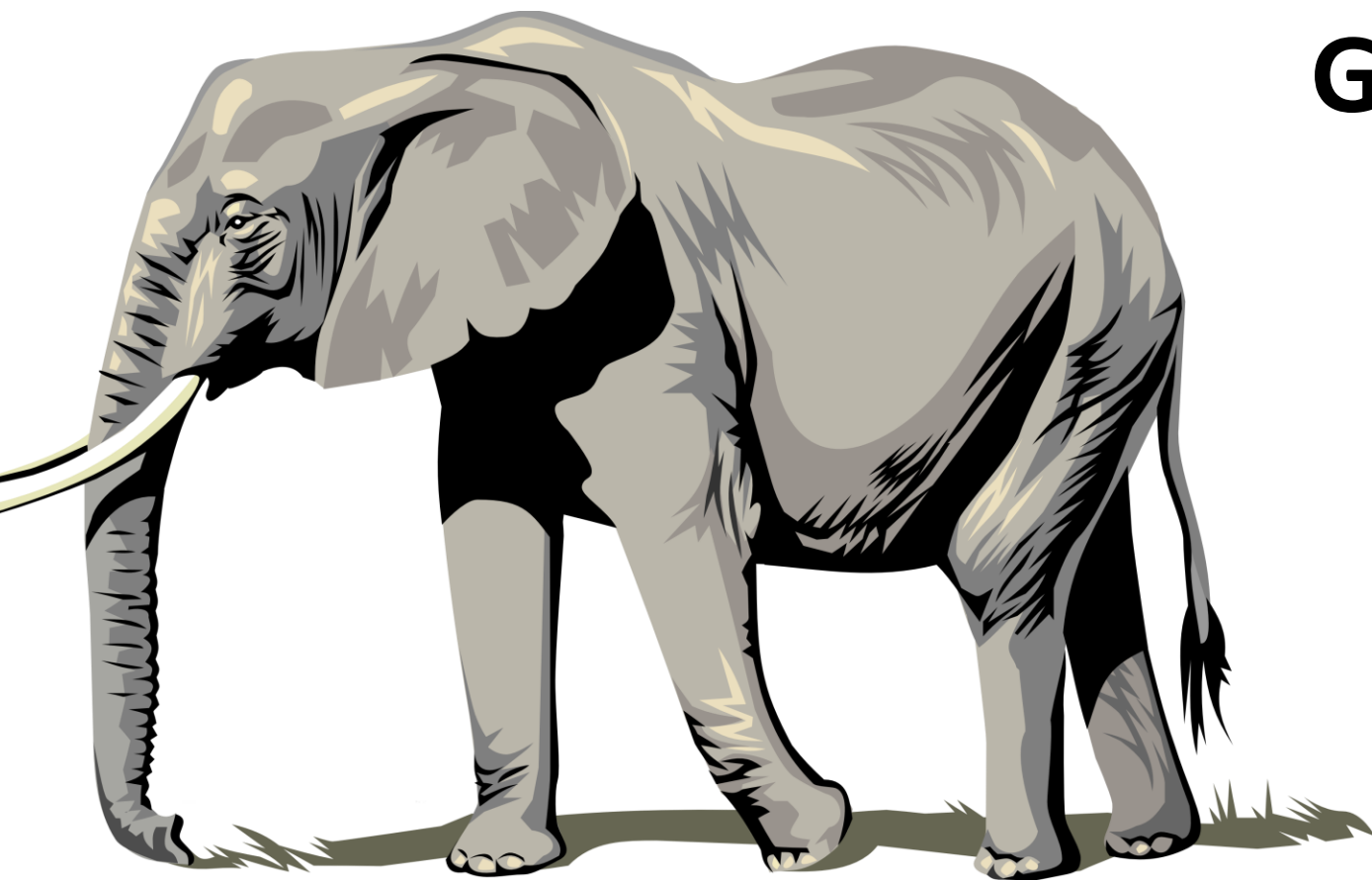
IMS

Information
Management
School

data with **purpose**

Big Data Analytics

Graph Data Analytics & Visualization



Lecturer:

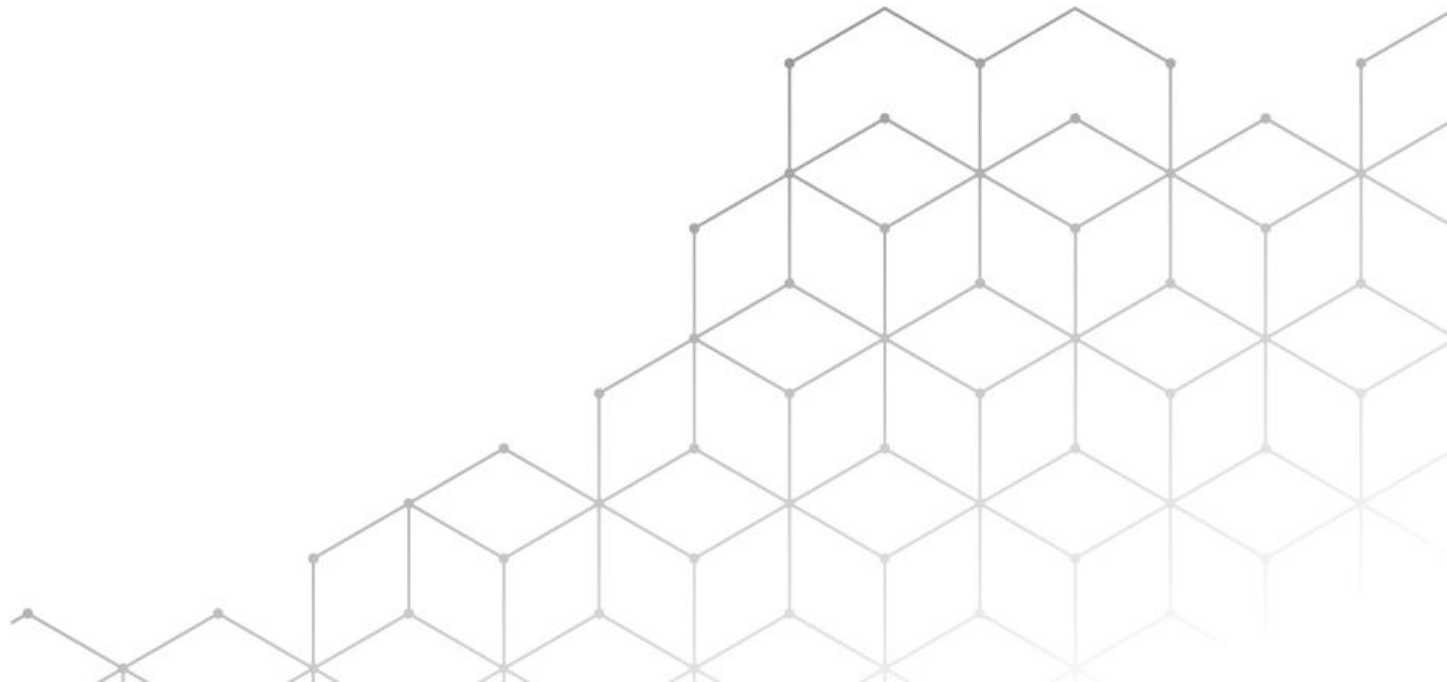
Márcia Lourenço Baptista

m.baptista@novaims.unl.pt

Agenda for Today



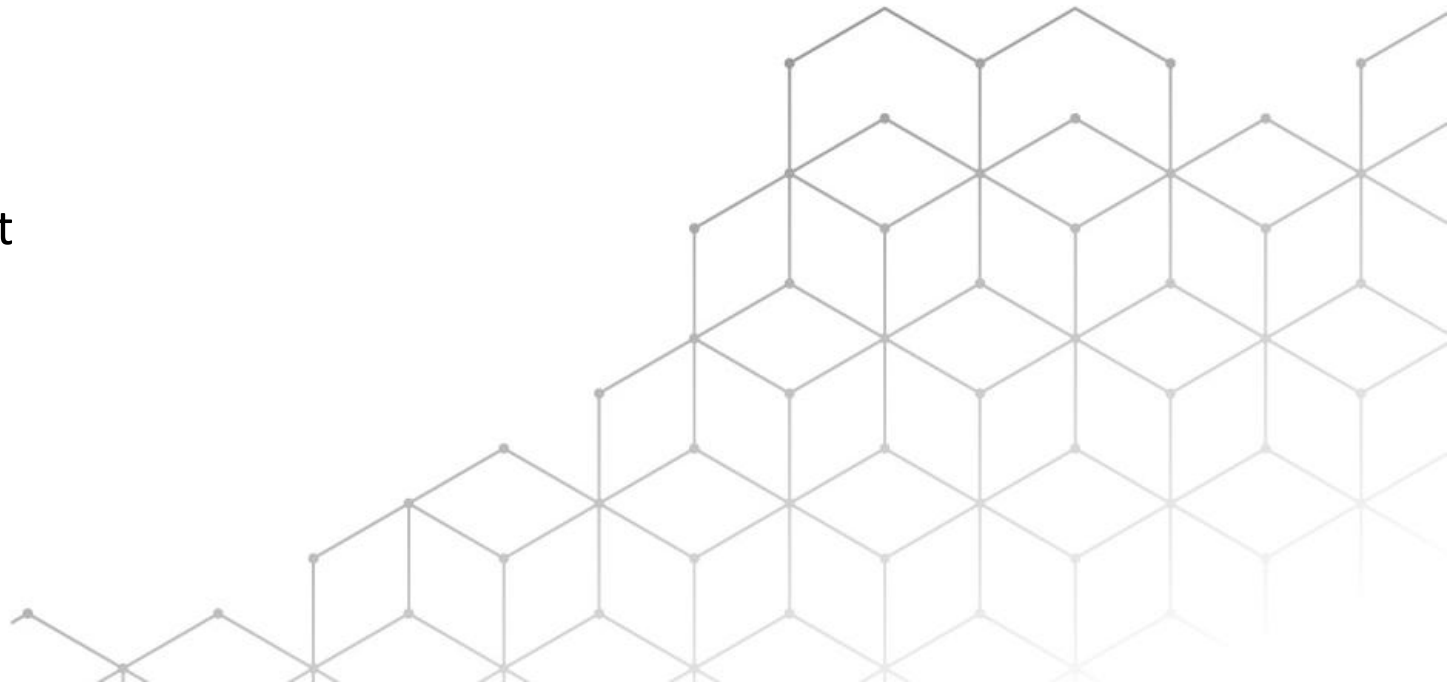
- Overview on Graphs
- Graph Analytics
- GraphX
- Big Data Visualization



Agenda for Today



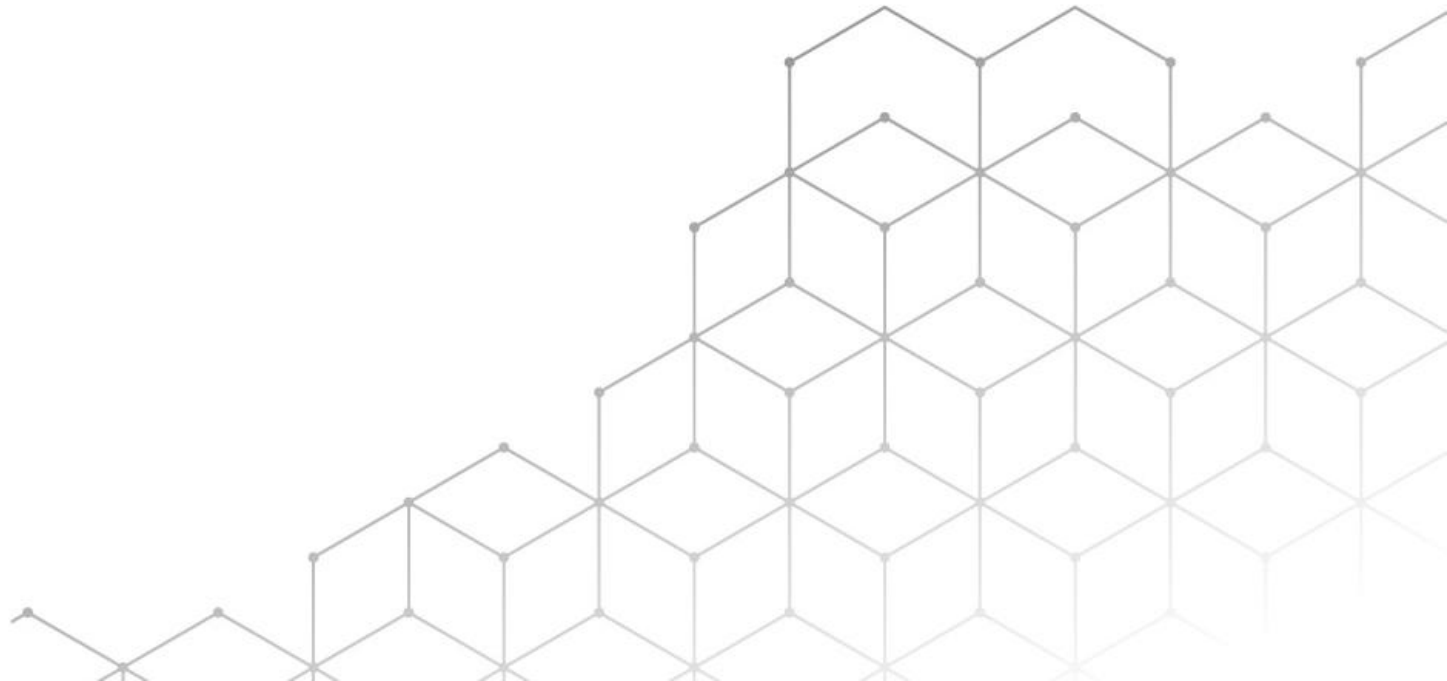
- Overview on Graphs
 - Motivation
 - History
 - Problems of Graph Analytics
 - Definition of Graph Analytics
 - Examples of Non-Graphs and Graphs
 - Nodes, Edges, Operations
 - Real-world Examples
 - Why Graphs?
 - Type of Graphs
 - Adjacency Matrix and Adjacency List
 - Information Rich Graphs
- Graph Analytics
- GraphX
- Big Data Visualization



Motivation

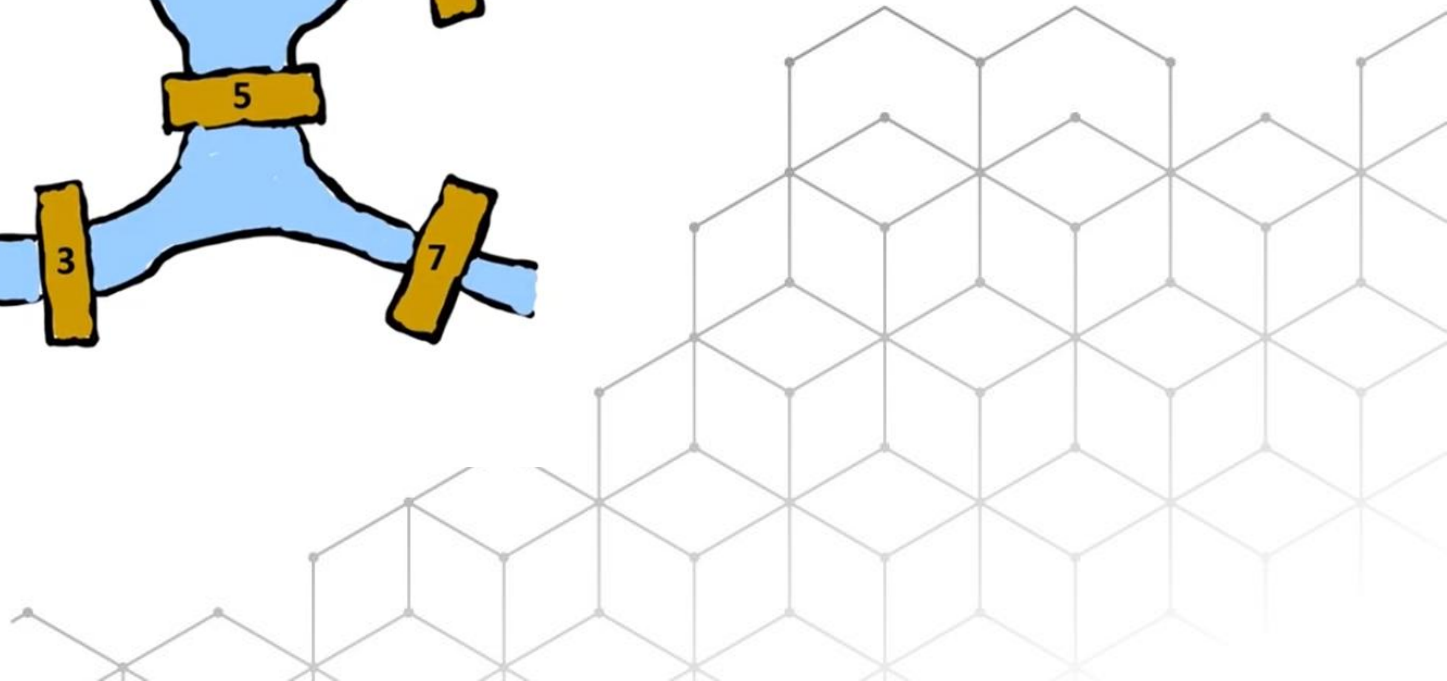
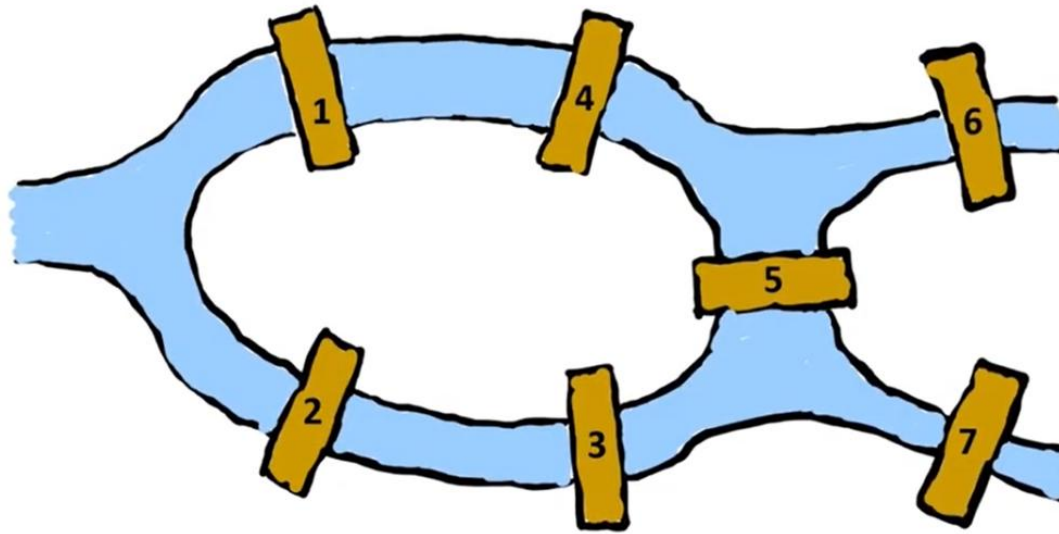


- What are Graphs?
- How can you use them?
- How do you deal with the daily challenges of their use?



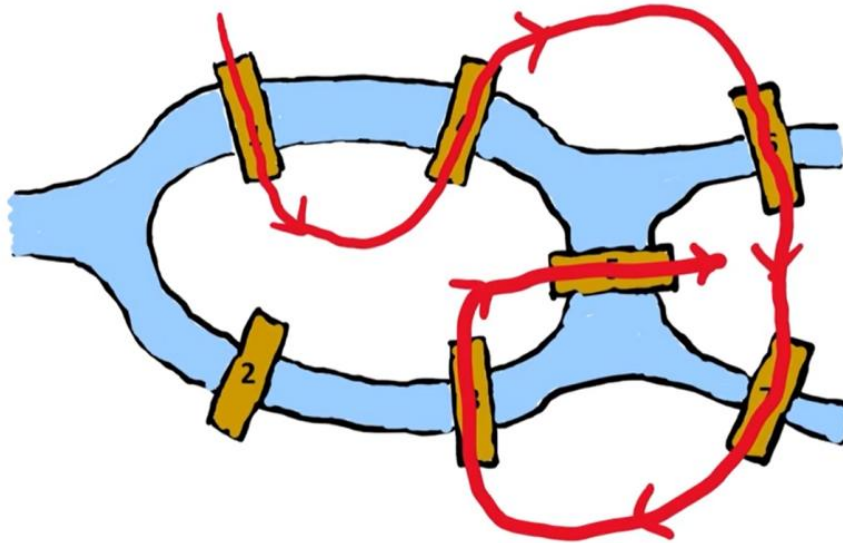
History

- ... the Seven Bridges of Königsberg

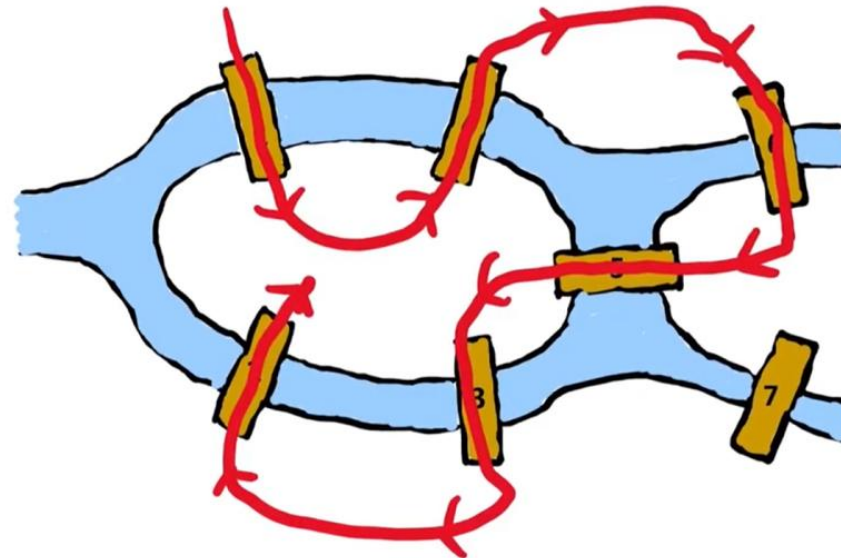


Possible Walks

- A **walk** is an arbitrary **sequence of edges** that start from some **node** and end at **another**.



Bridge 2 not used.



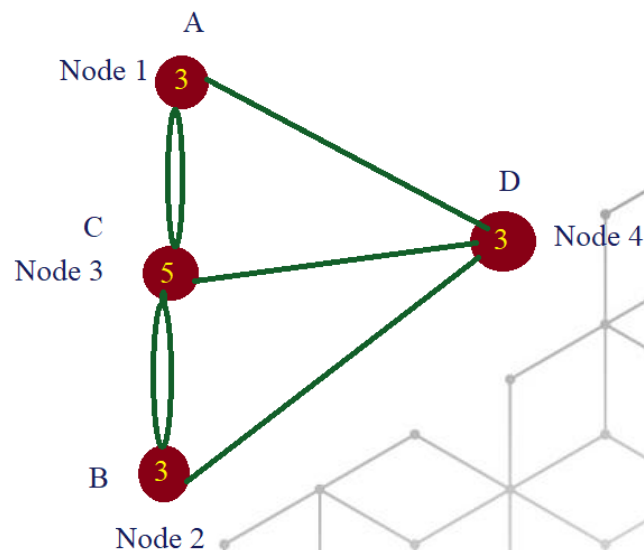
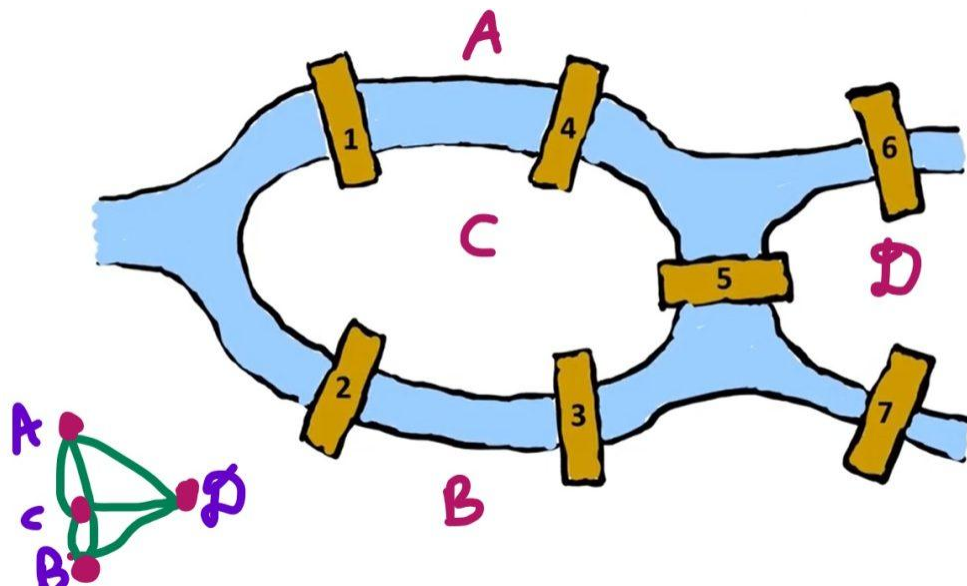
Bridge 7 not used.

Euler Path?

A **trail** is a **constrained walk** with no repeating edges.

Devise a **trail** through the city that would cross each of those **bridges** once and only once.

Euler proved that the problem **has no solution!**



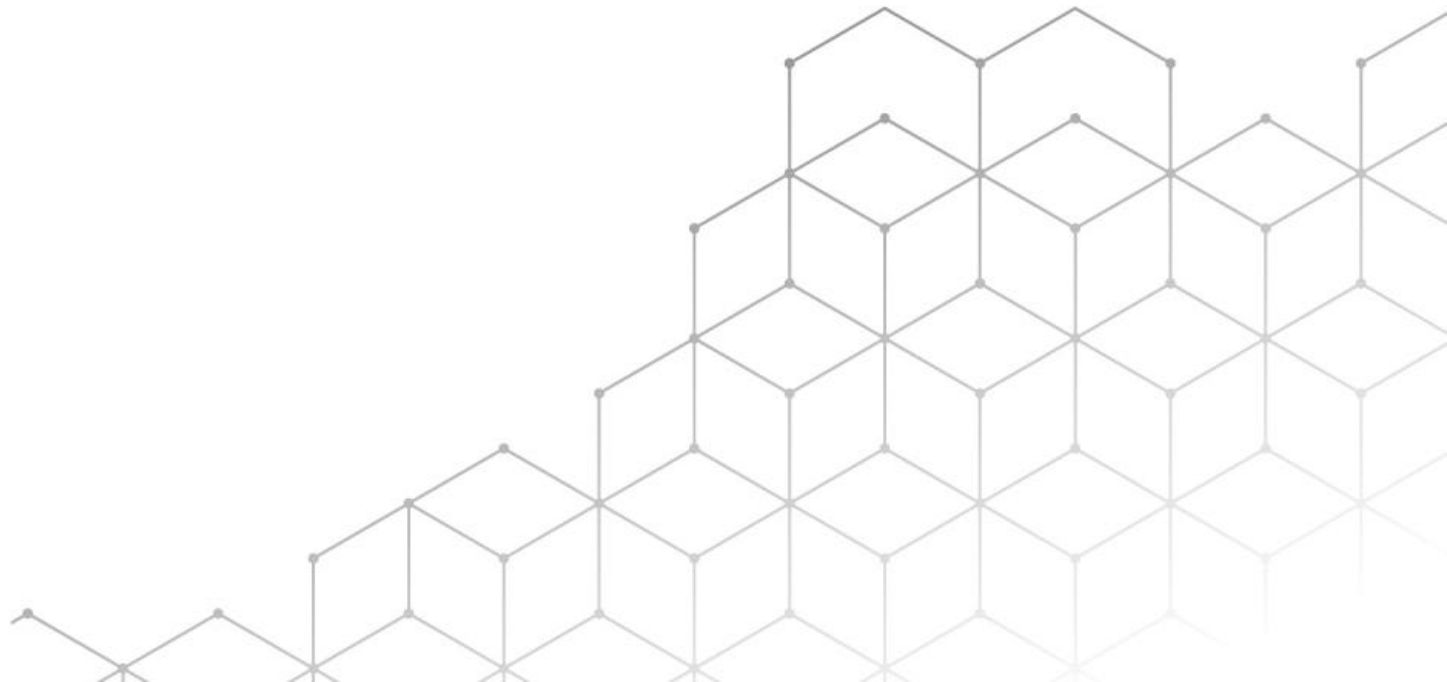
Curiosity

- After WWII two bridges were destroyed and Euler path created.
- However you can't find Königsberg now.
- It is renamed as Kaliningrad and is a part of Russia



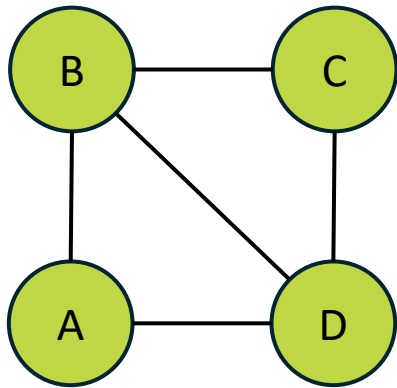
Other Problems

- The Hamiltonian Path
- The Travelling Salesman
- Graph Isomorphism
- Page Ranking

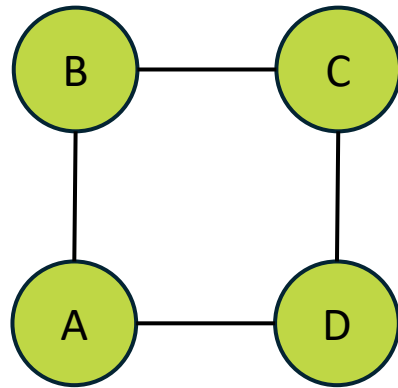


Hamiltonian Path

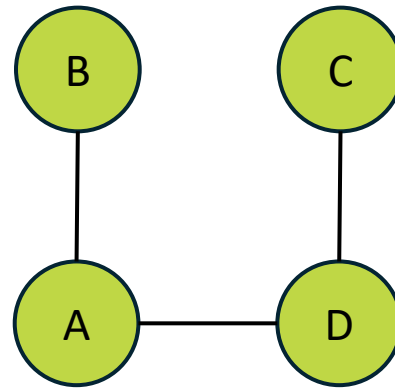
- If there exists a walk in the graph that visits every vertex exactly once without repeating the edges (What about the Hamiltonian cycle?)



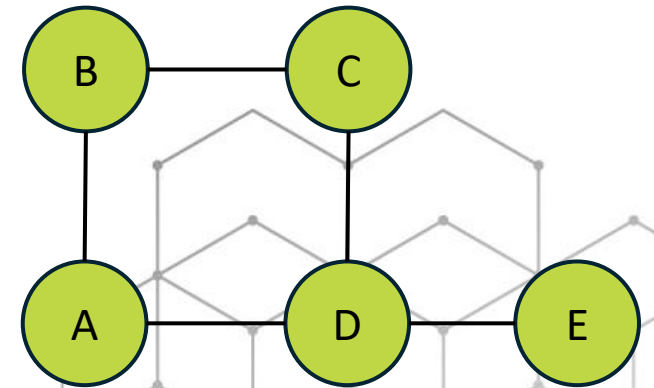
Graph 1



Graph 2



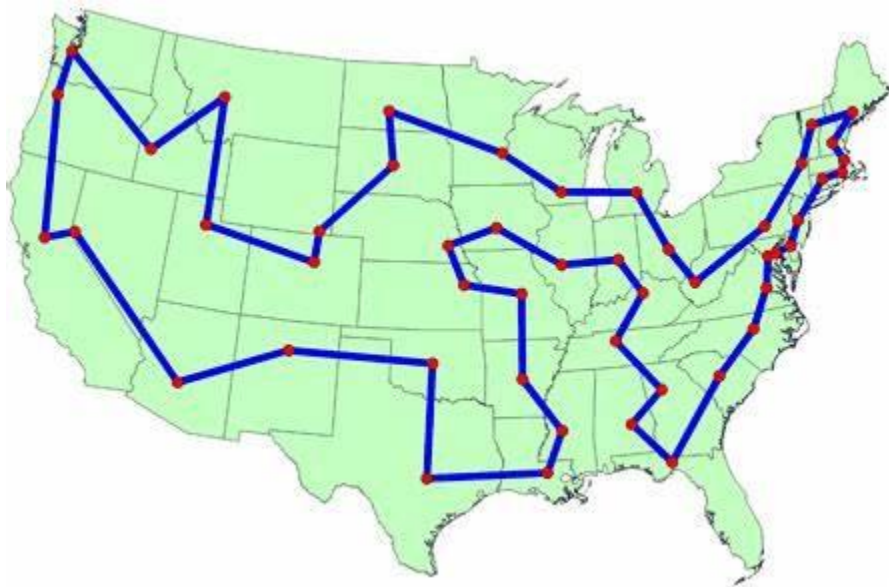
Graph 3



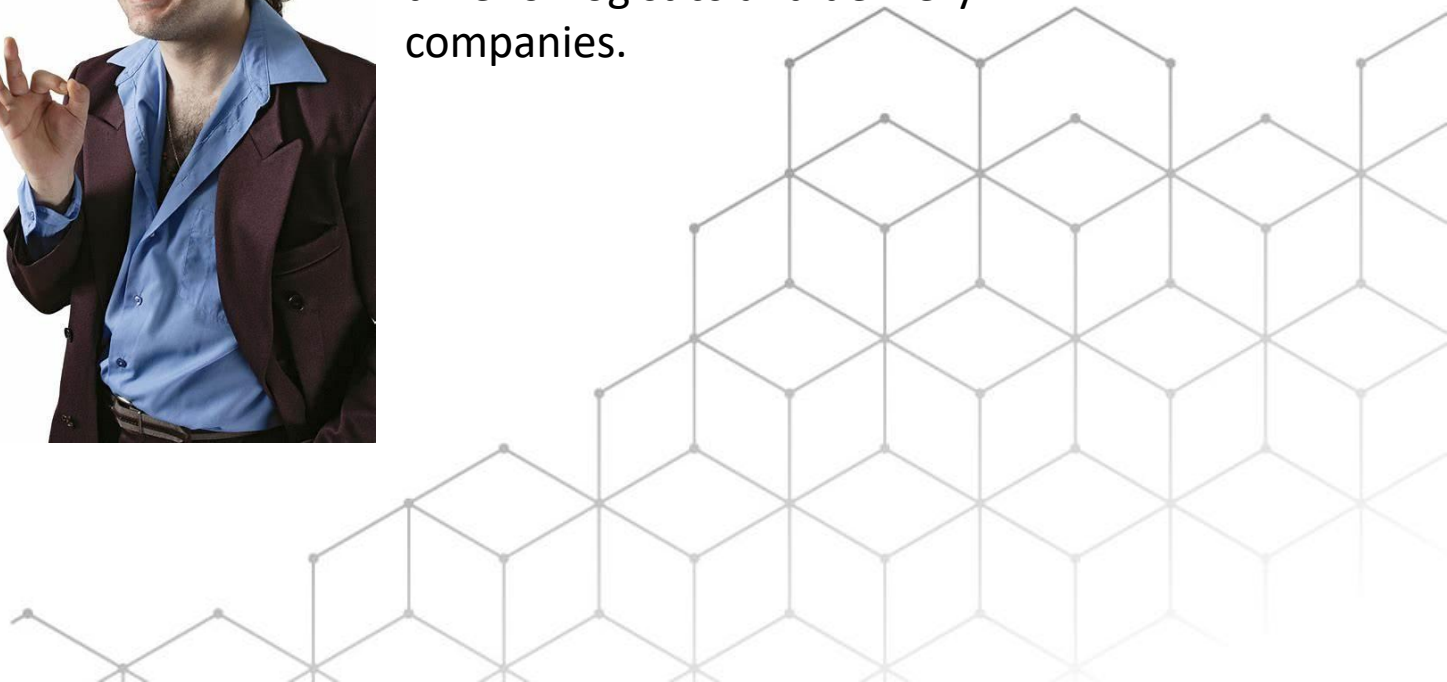
Graph 4

Traveling Salesman Problem

- A salesperson needs to visit a set of cities exactly once each, then return to the home city.

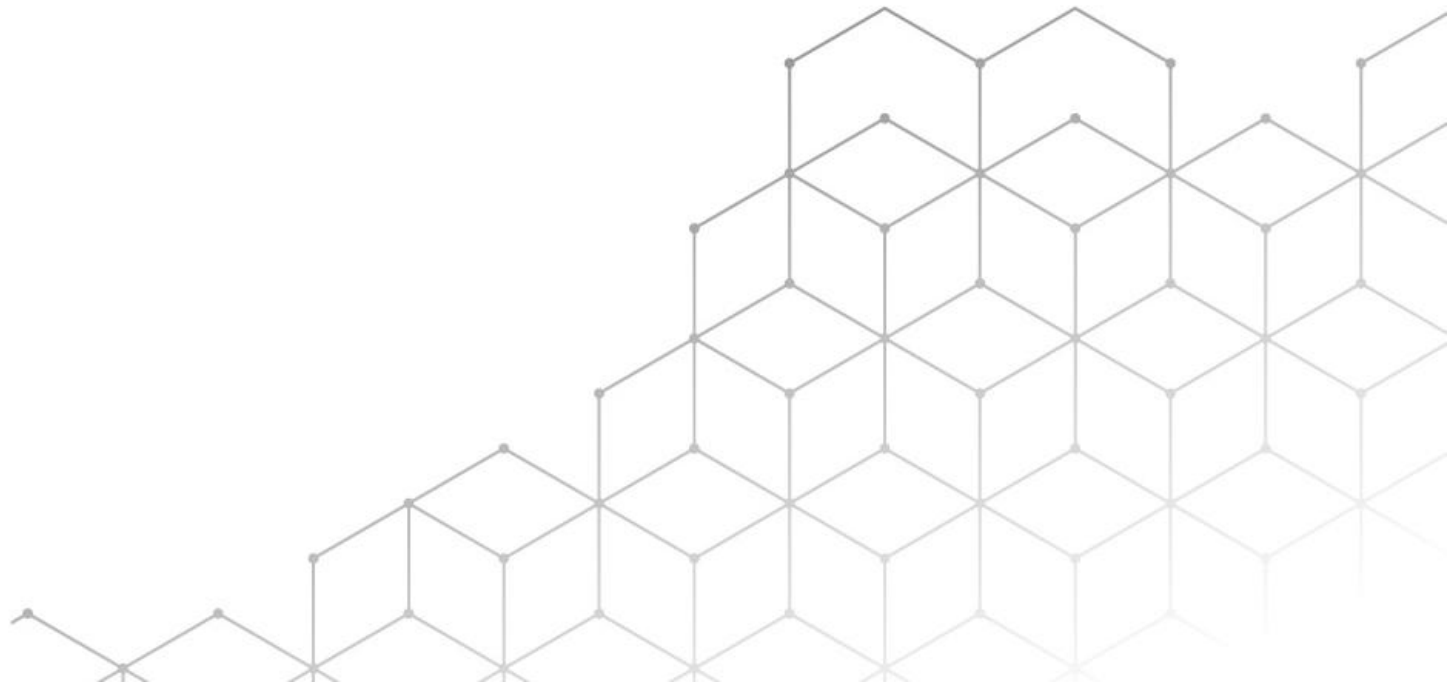


Route optimization can save transportation costs, fuel, and time for logistics and delivery companies.



Real-World Applications of Hamiltonian

- A need to traverse each “point” (city, station, sensor, etc.) exactly once in a single loop, making them a cornerstone of many scheduling and routing optimization problems.

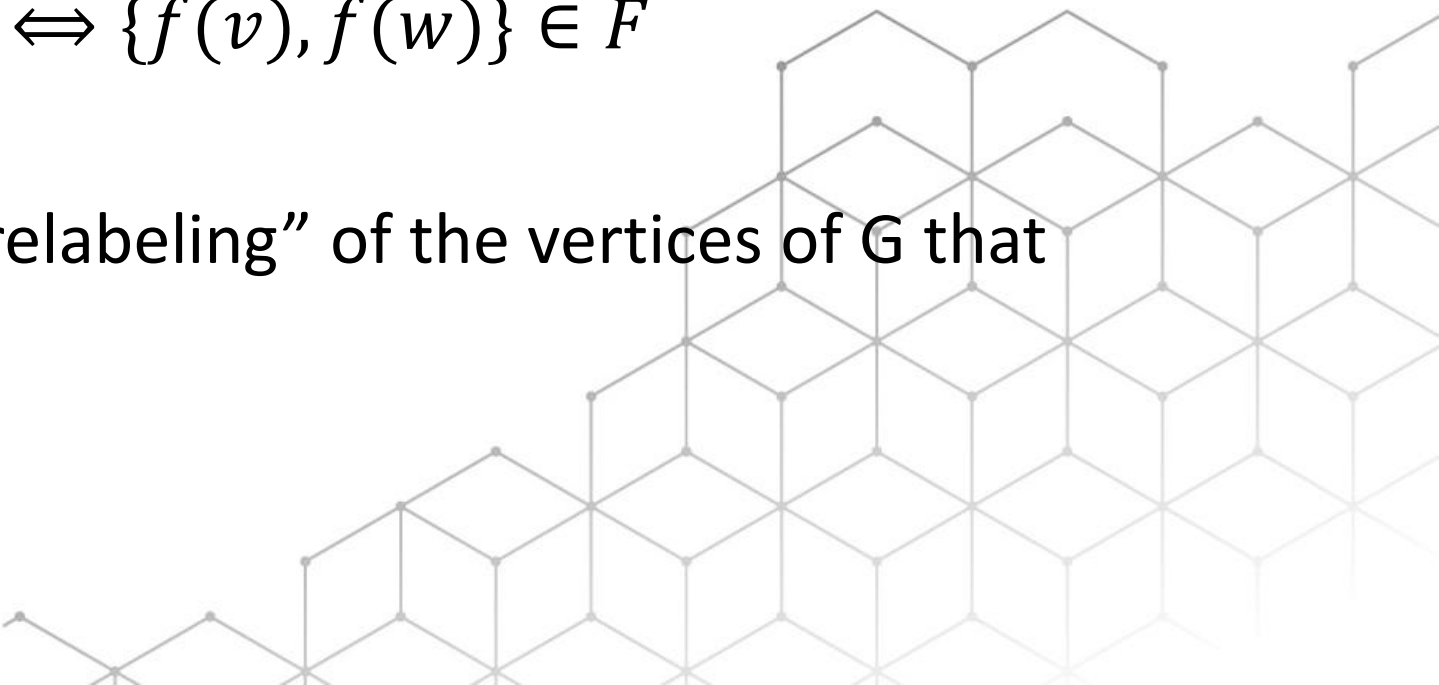


Graph Isomorphism

- Two graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic if there exists a bijective (one-to-one and onto) function $f: V \rightarrow W$ such that for every pair of vertices $v, w \in V$

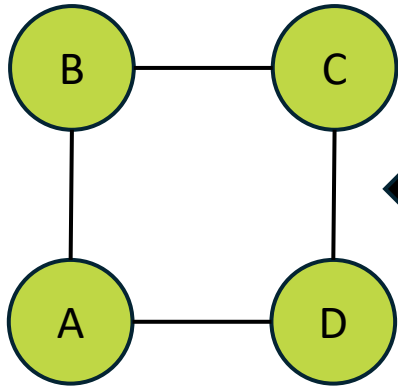
$$\{v, w\} \in E \iff \{f(v), f(w)\} \in F$$

- In simpler terms, there is a “relabeling” of the vertices of G that makes it look exactly like H .

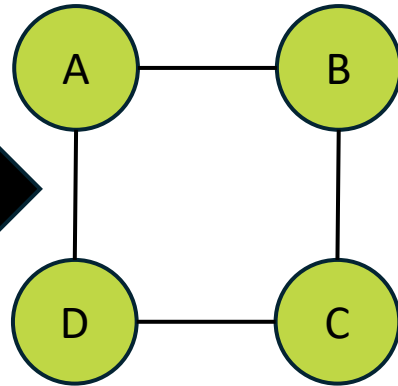


Isomorphic Graph Examples

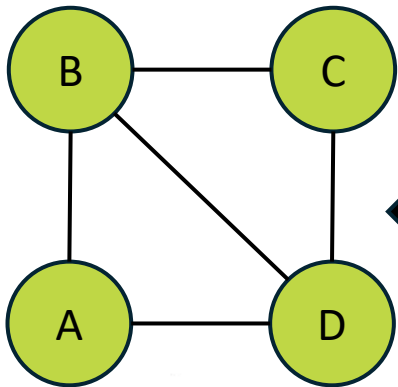
Graph 1



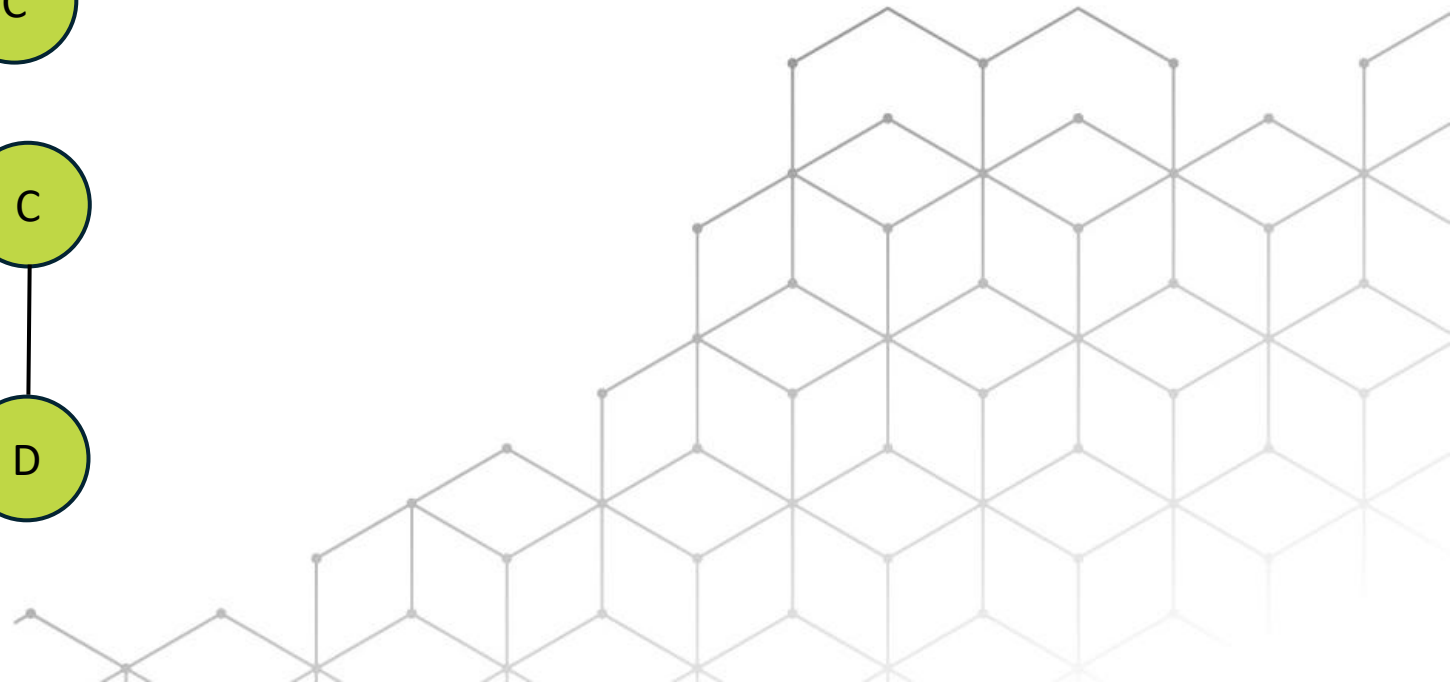
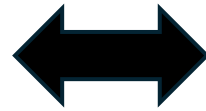
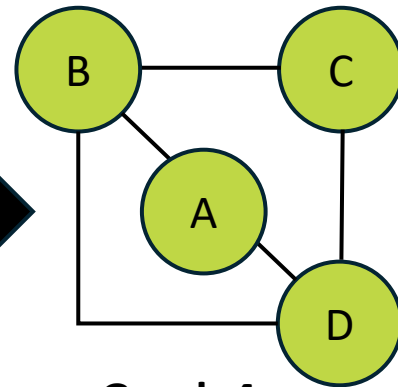
Graph 2



Graph 3

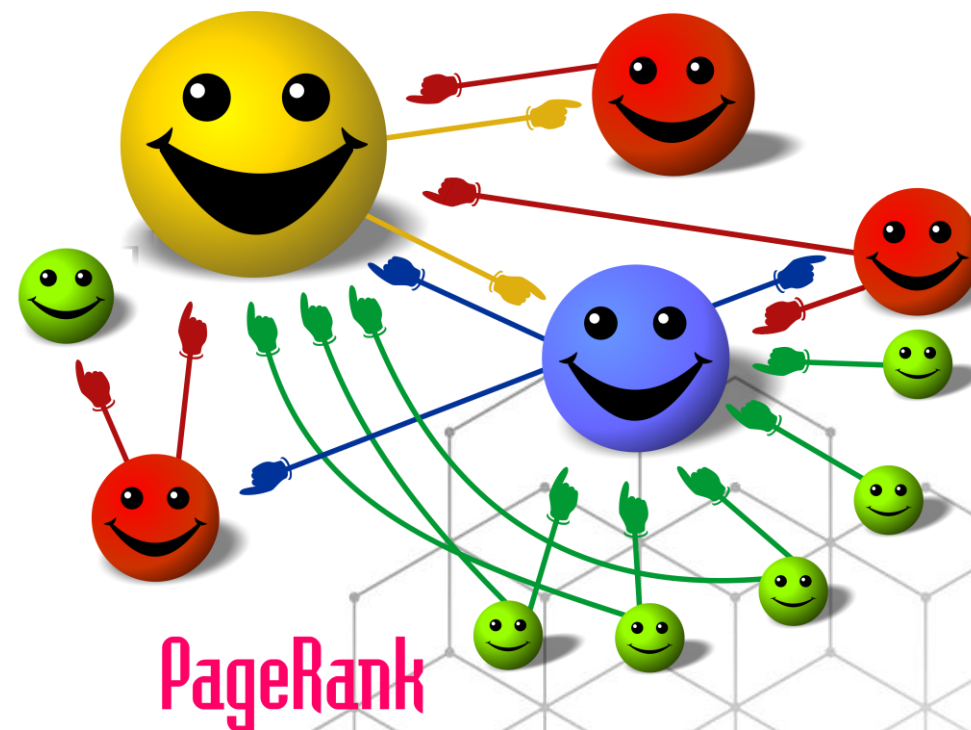


Graph 4



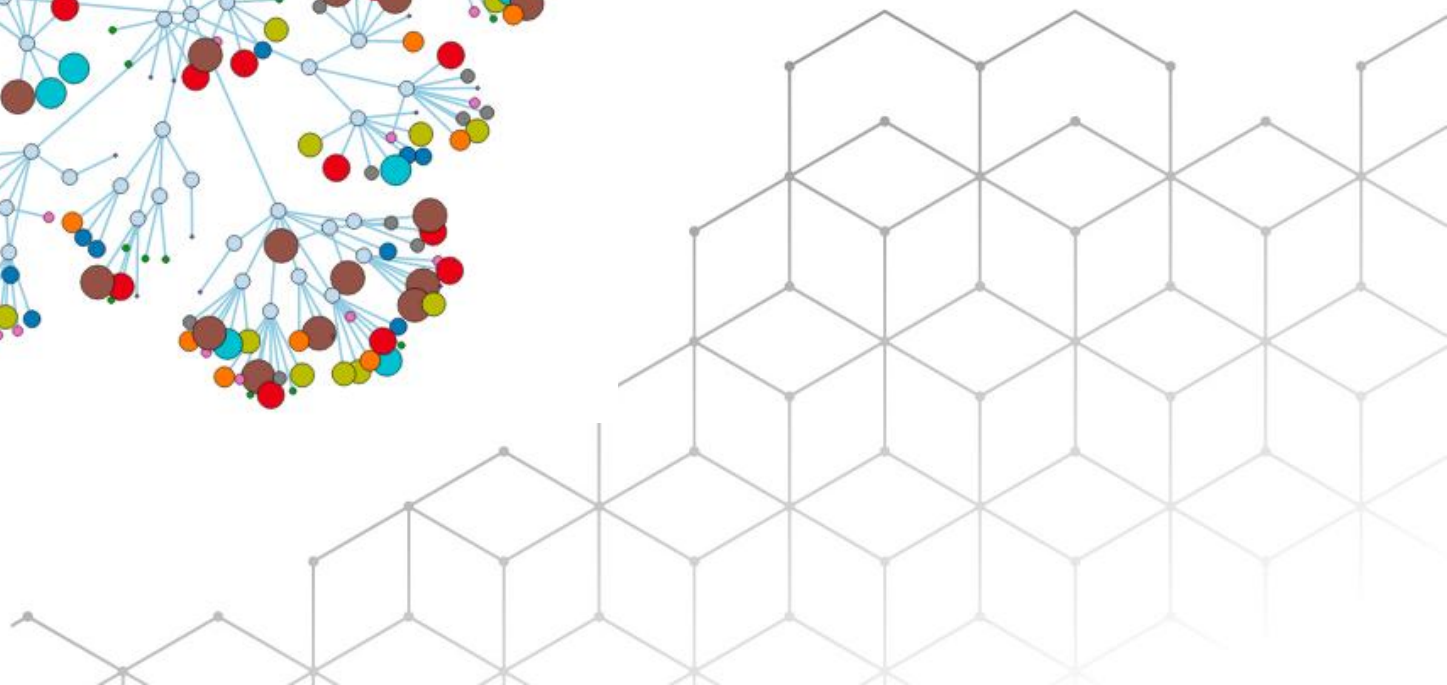
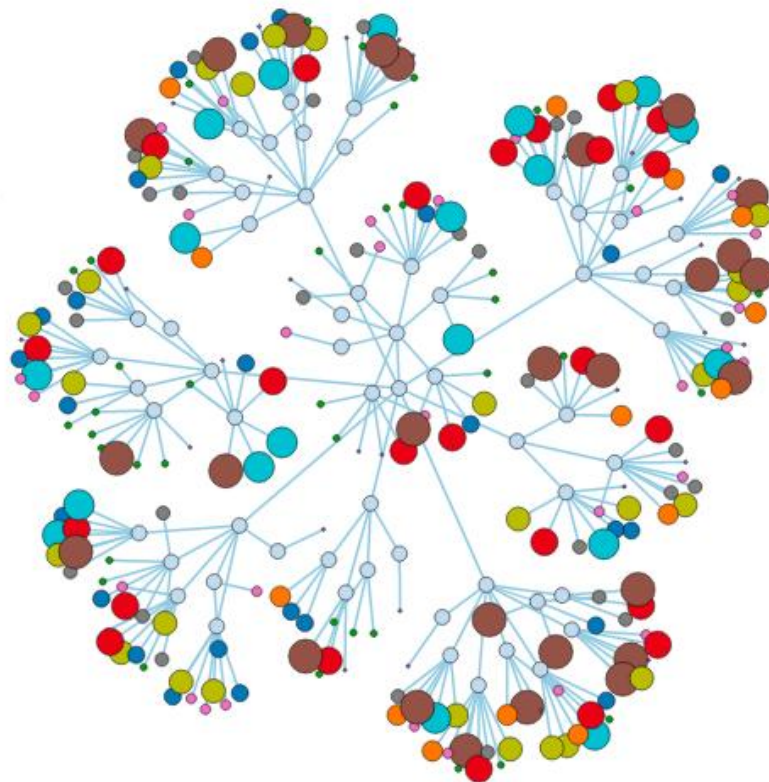
PageRank Algorithm

- PageRank works by counting the number and quality of links to a page to determine a rough estimate of how important the website is.
- The underlying assumption is that more important websites are likely to receive more links from other websites

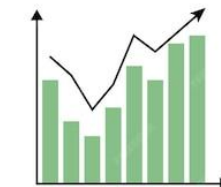
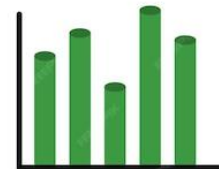
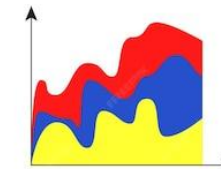
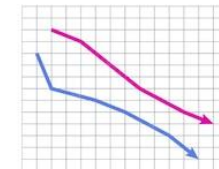
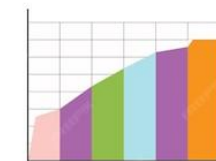
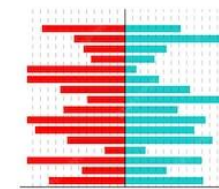
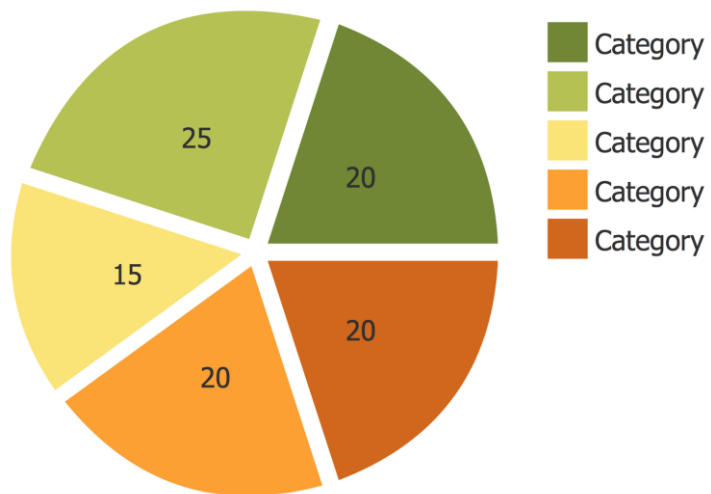
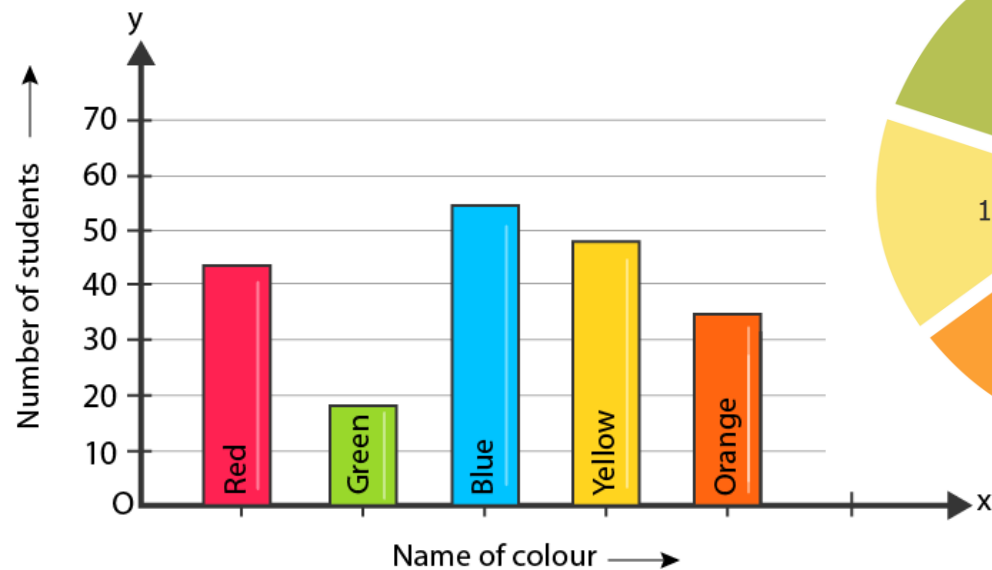


Graphs are Everywhere

- Government/Security
- Healthcare
- Energy/Resources
- Financial Services
- Life Sciences
- Telecom/Media



Examples of non-Graphs



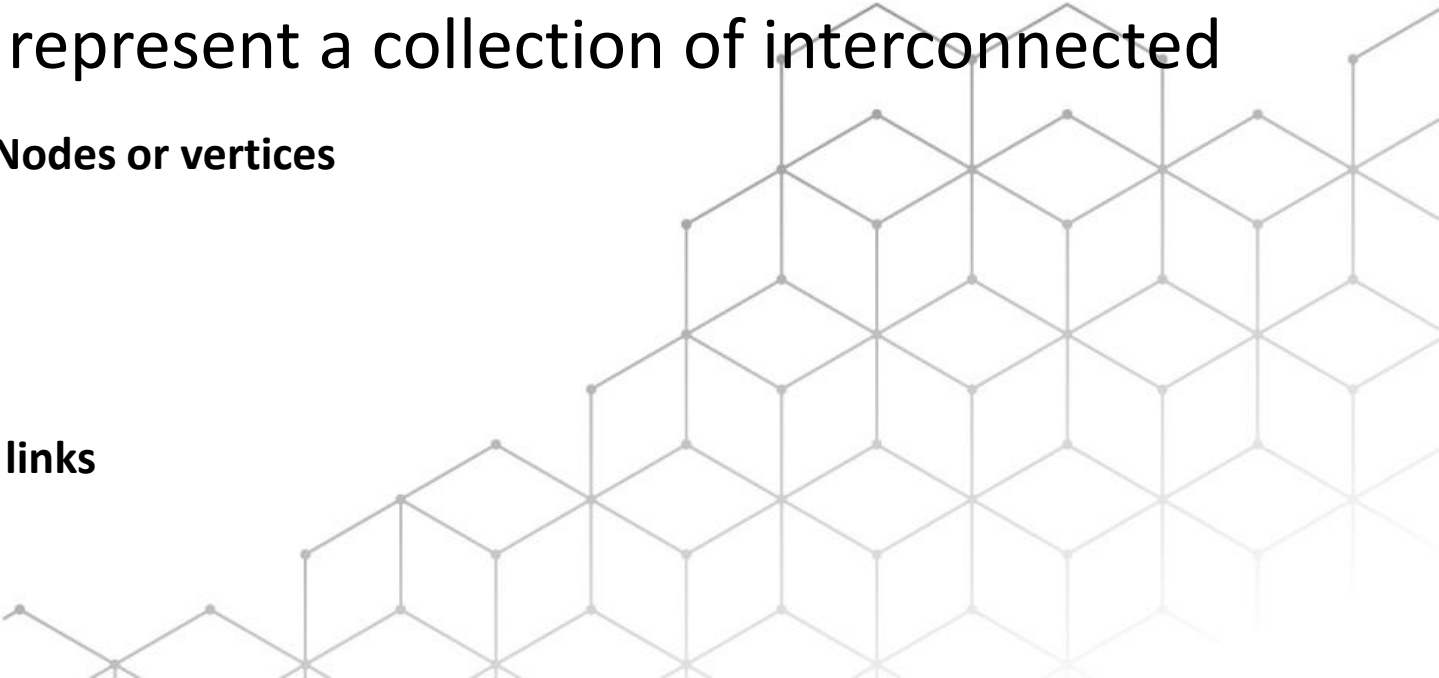
Definition of Graph Analytics

- Based on graph theory, graph analytics help to understand, codify, and visualize relationships that exist between objects in a given domain context.
- Graphs are a way to formally represent a collection of interconnected objects.

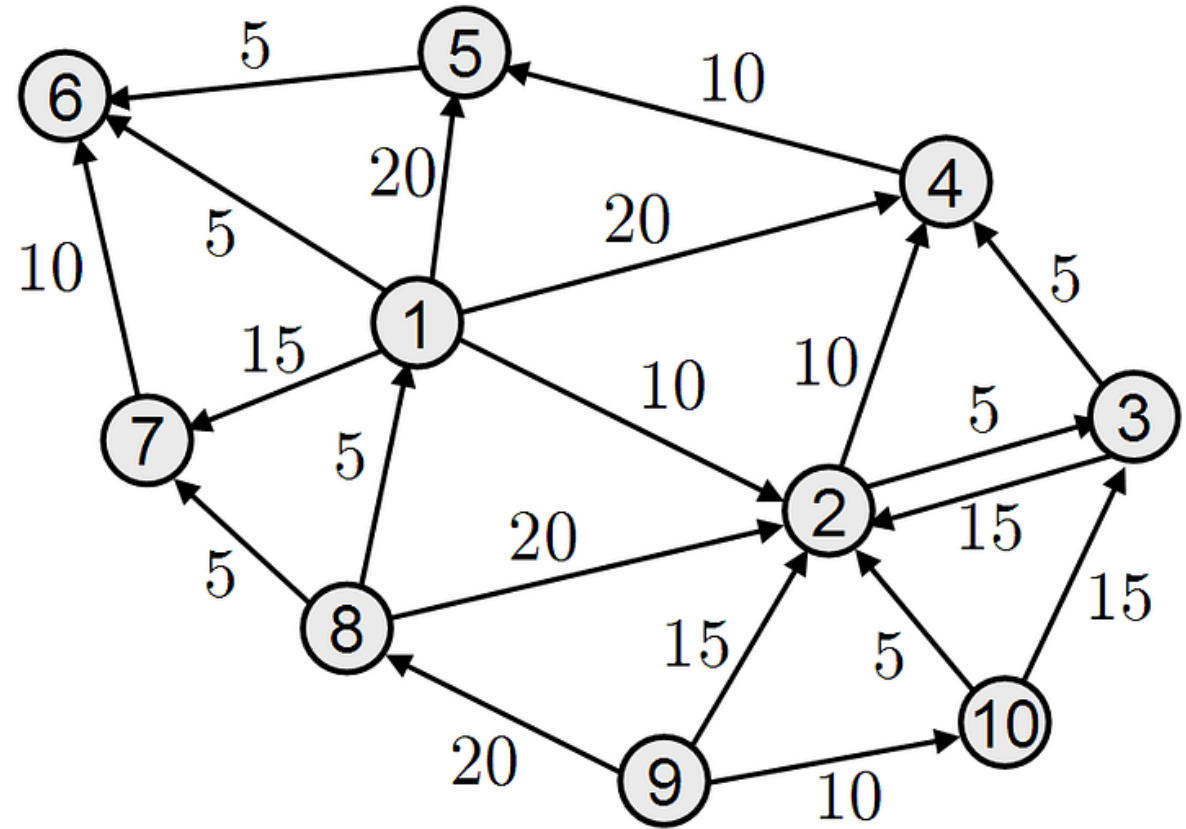
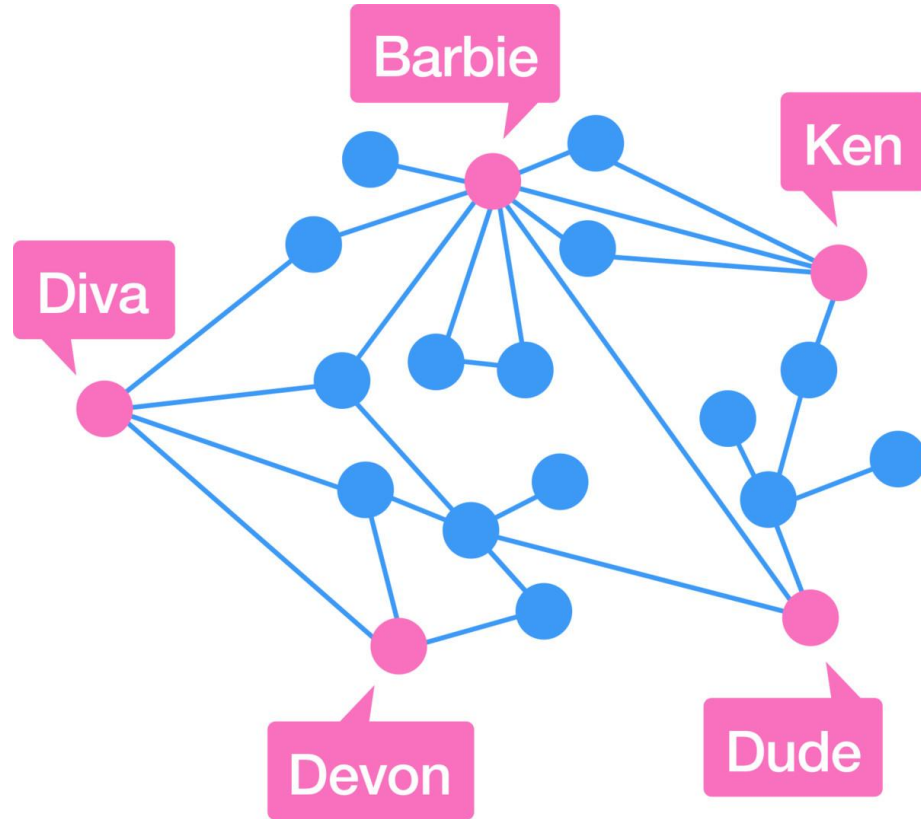
$$G = (V, E)$$

Nodes or vertices

Edges or links

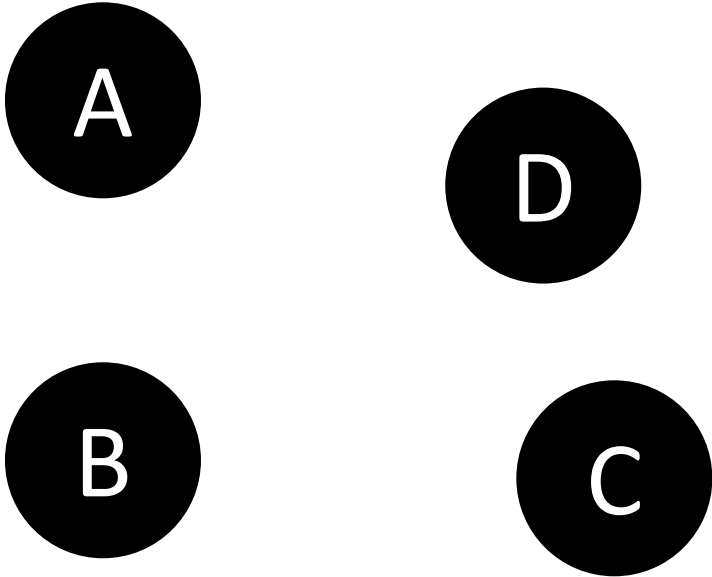


Examples of Graph



Nodes

- Nodes = {A, B, C, D}



A

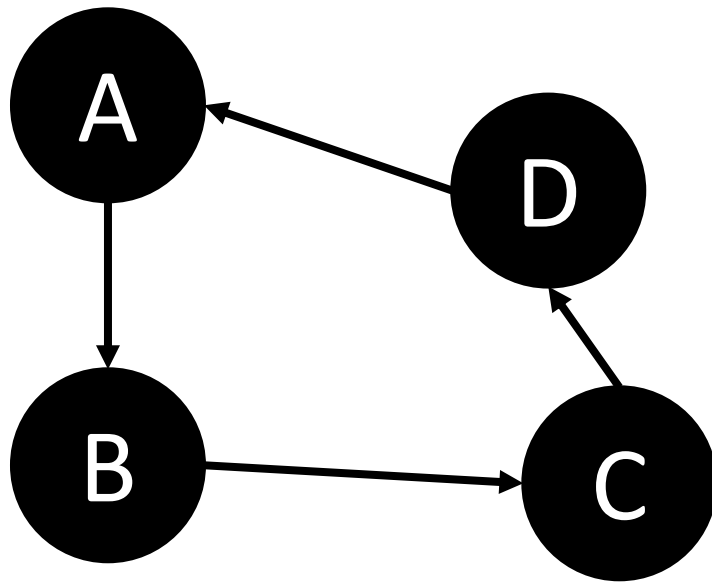
D

B

C

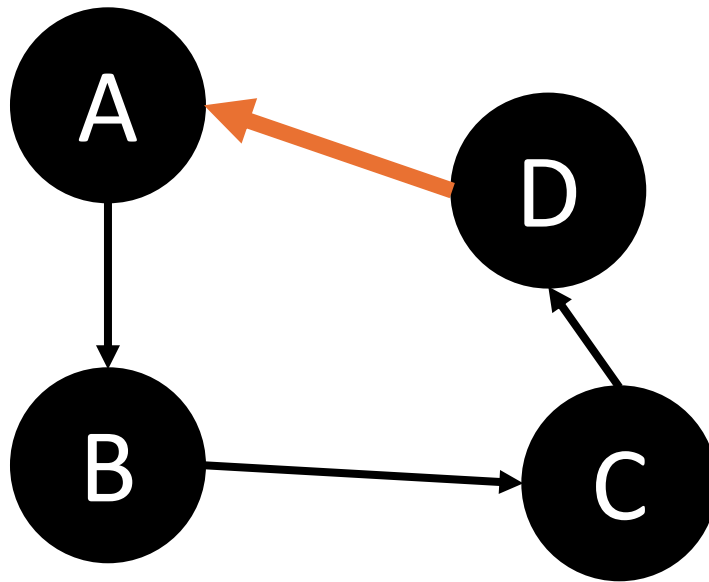
Edges

- Edges = {A-B, B-C, C-D, D-A}



Operations

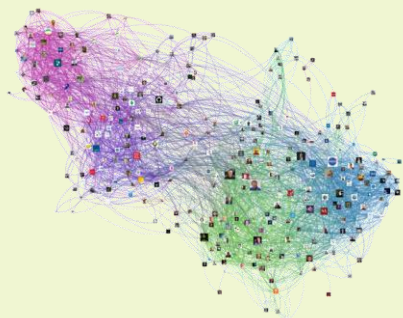
- ShortestPath(D, A)



Graph operation examples

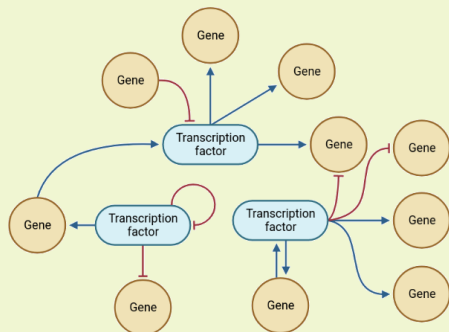
- `graph.GetNodes(<condition>)`
- `graph.GetEdges(<condition>)`
- `graph.AddNode(node)`
- `graph.AddEdge(node1,node2)`
- `graph.RemoveNode(node)`
- `graph.GetShortestPath(node1,node2)`
- `graph.Neighbours(node,level)`
- `graph.GetDistance(node1,node2)`
- `node.GetParents()`
- `node.GetChildren()`
- `node.GetAncestors(level)`
- `node.GetDescendants(level)`
- `node.IsAncestorTo(node2)`
- `node.IsDescendant(node2)`
- `node.AddParent(parentNode)`
- `node.AddChild(childeNode)`
- `node.IsReachable(node2)`

Real-World Examples



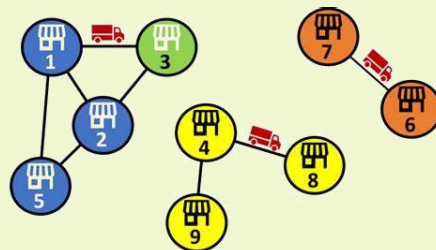
Social Media
facebook

- Identify groups
- Find influencers
- Extract topic interests



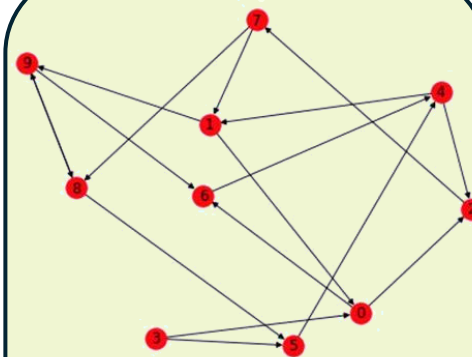
Biology
Gene Relationships

- Discover unknown relationships
- Gene to disease



Geo IS
Smart Cities

- Traffic flow, congestion prediction
- Coverage analysis



Industries
Predictive
Maintenance

- Anticipate failures
- Analyze component relationships

Why Graphs?

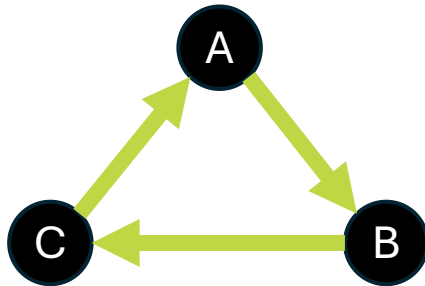
Intuitive / Natural
Representation

Efficient Data
Processing

Efficient **Query** /
Analytics

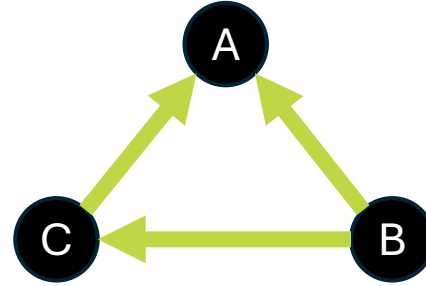
Types of Graphs

Directed Graph



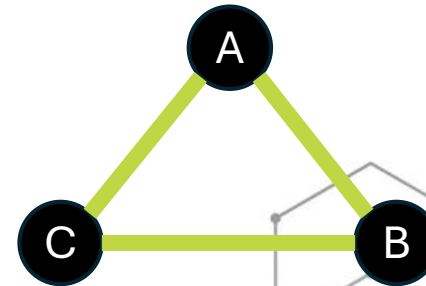
Finite State Machine

Directed Acyclic Graph



Can never go back home

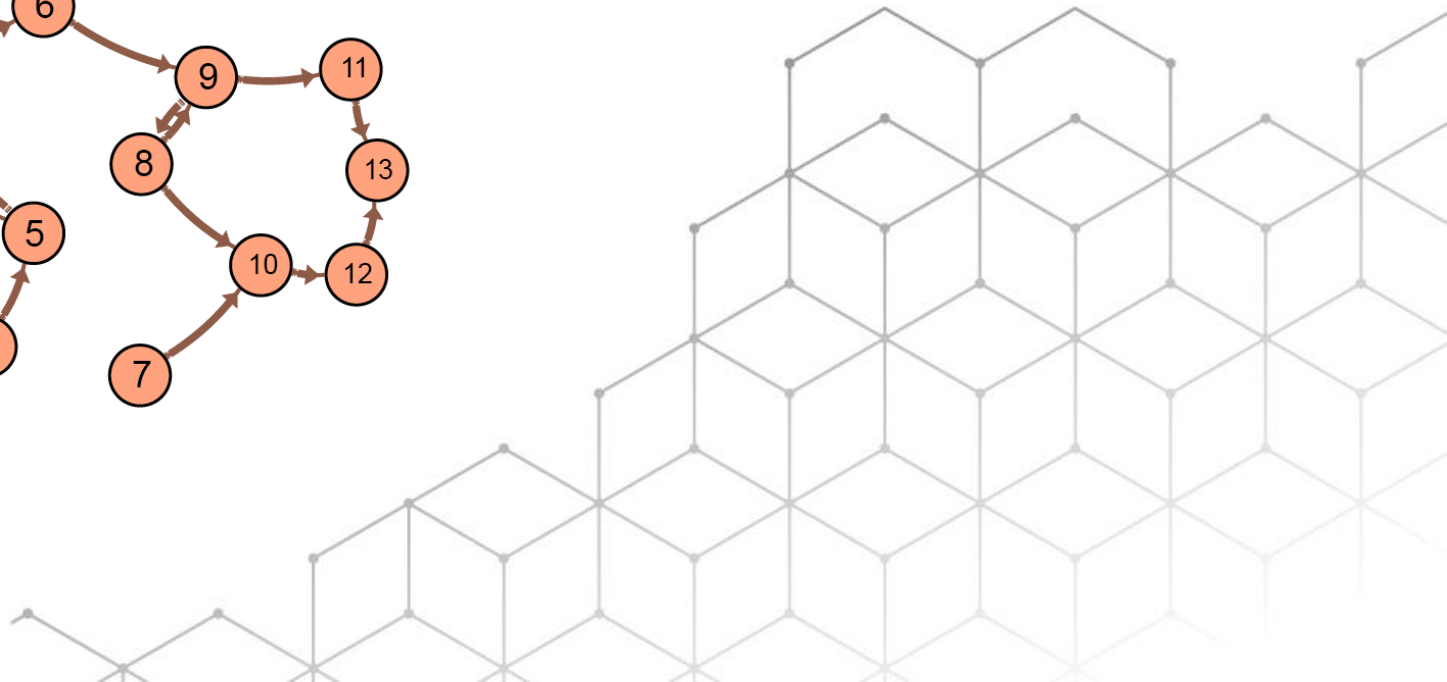
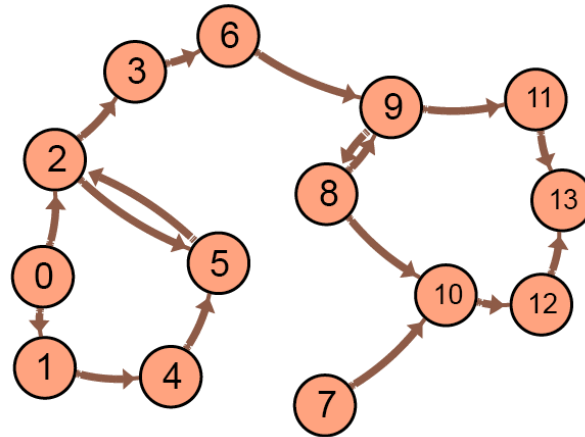
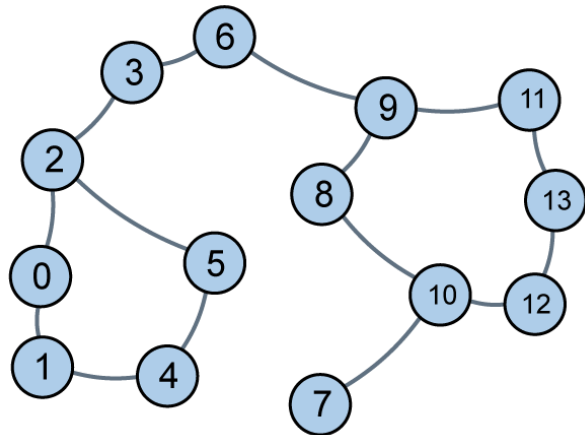
Undirected Graph



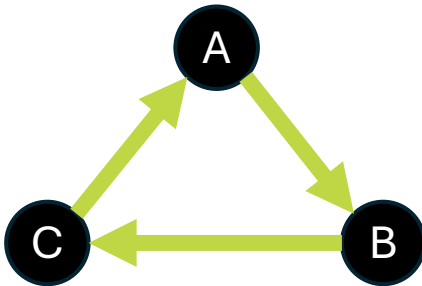
Facebook Connectivity

Undirected vs Directed Graph

- A **graph** may be **undirected**, meaning that there is **no distinction between the two vertices associated with each edge**, or its **edges** may be **directed** from one **vertex** to another.

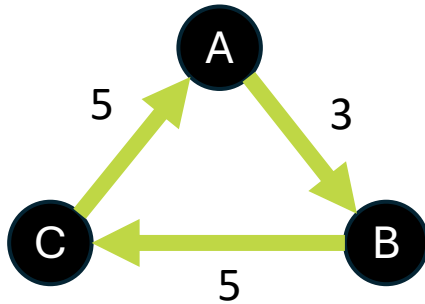


Adjacency Matrix



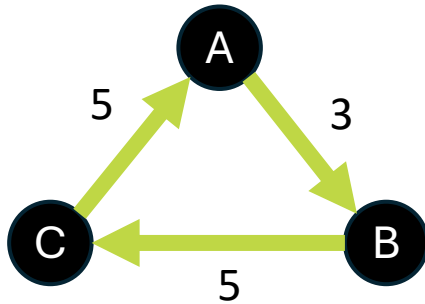
	A	B	C
A	0	1	0
B	0	0	1
C	1	0	0

Adjacency Matrix of Weighted Graph



	A	B	C
A	0	3	0
B	0	0	5
C	5	0	0

Edge Table



Useful for relational database

From:	To:	Edge Value
A	B	3
B	A	0
B	C	5
C	B	0
C	A	5
A	C	0

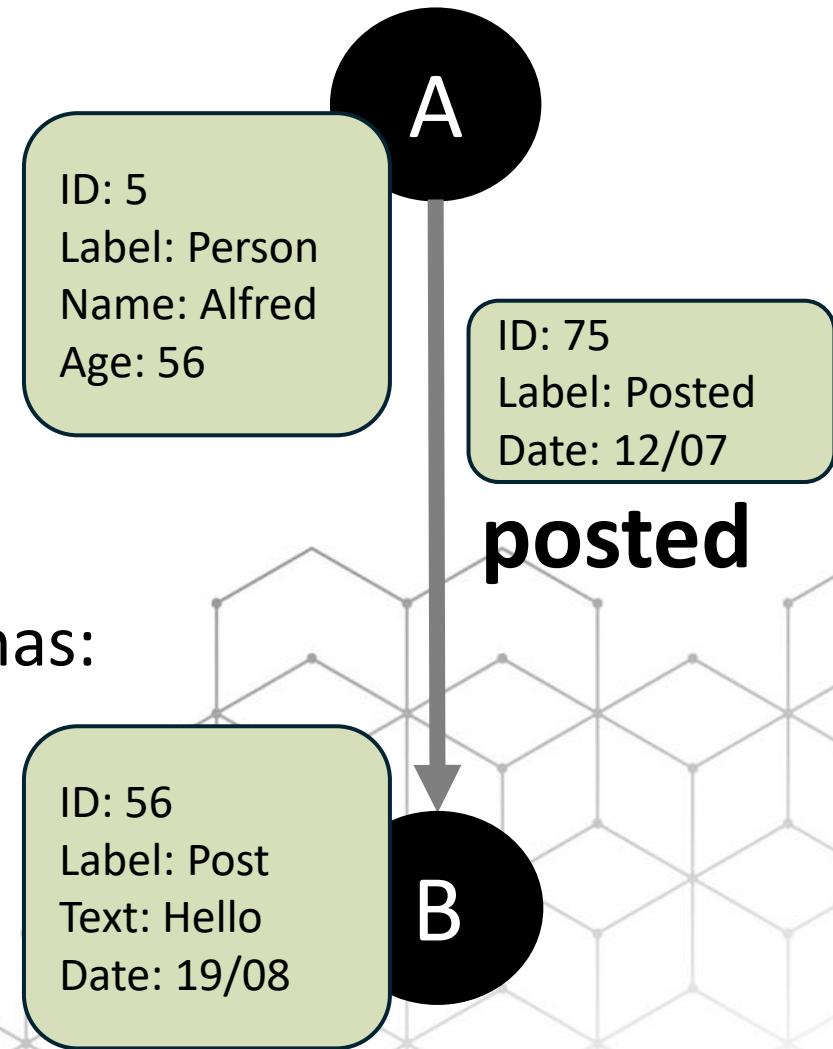
Information Rich Graphs

In a simple model, a graph consist of:

- A set of vertices (nodes)
- A set of edges (each connecting two nodes)

In a complex model each element (vertex/edge) has:

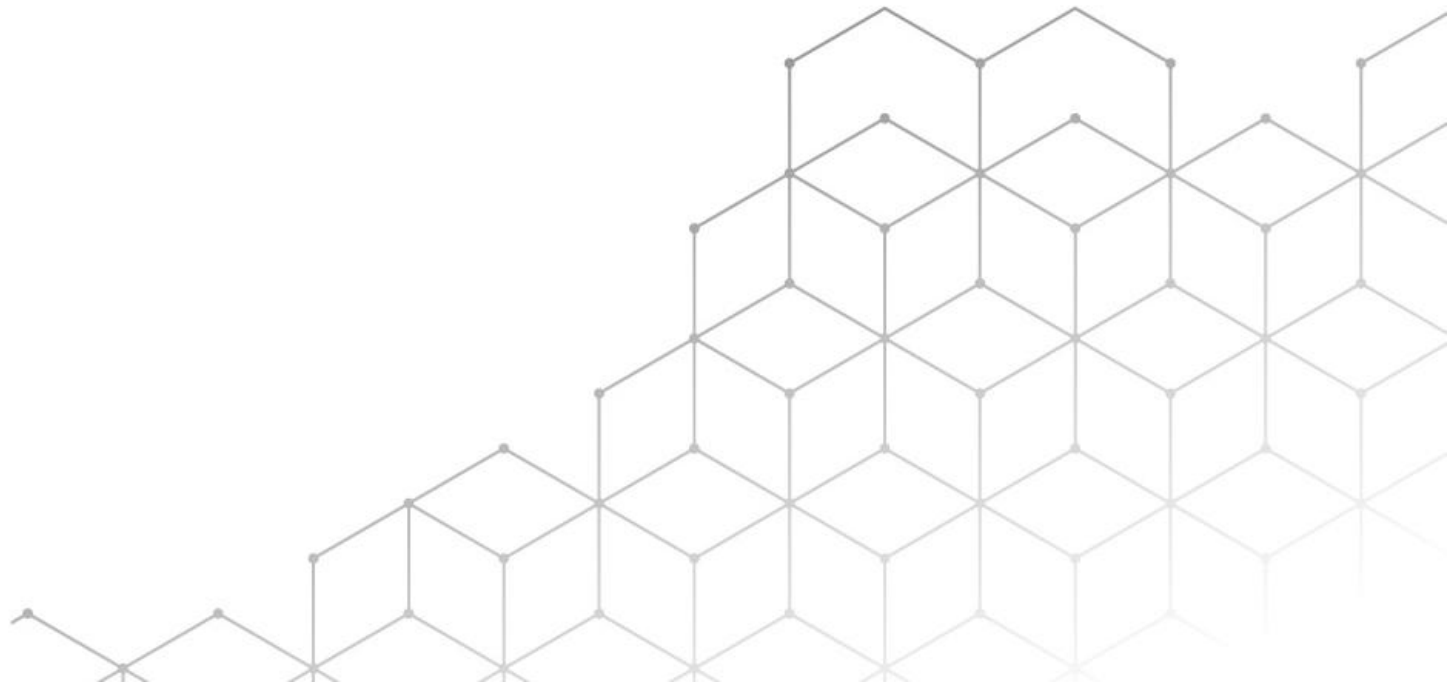
- Unique Identifier
- Class (label)
- A set of Key/Value pairs (properties)



Agenda for Today



- Overview on Graphs
- Graph Analytics
 - Connectivity Analytics
 - Path Analytics
 - Community Analytics
 - Centrality Analytics
 - Pattern Matching
- GraphX
- Big Data Visualization



Connectivity Analytics

How Big is the Graph?

Number of
Nodes

Number of
Edges

Degree
Distribution

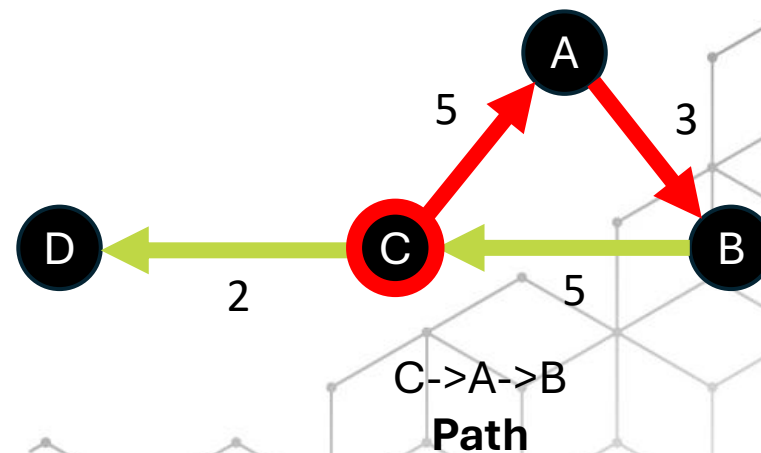
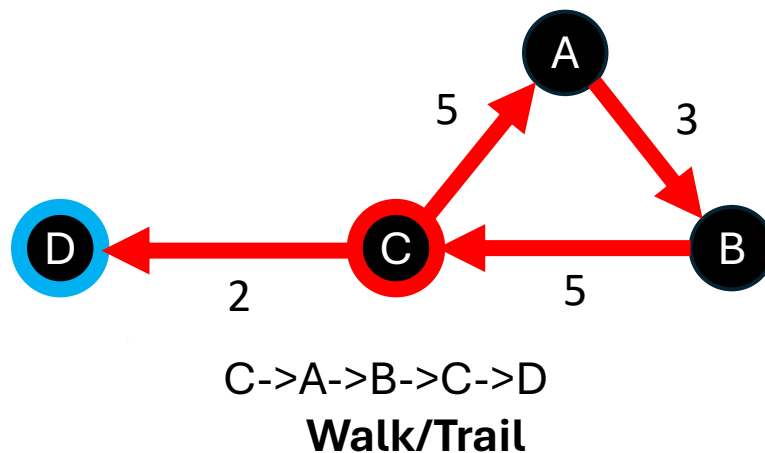
- Volume – Number of edges and nodes
- Velocity – How frequent a new node or edge is added to the graph

Degree

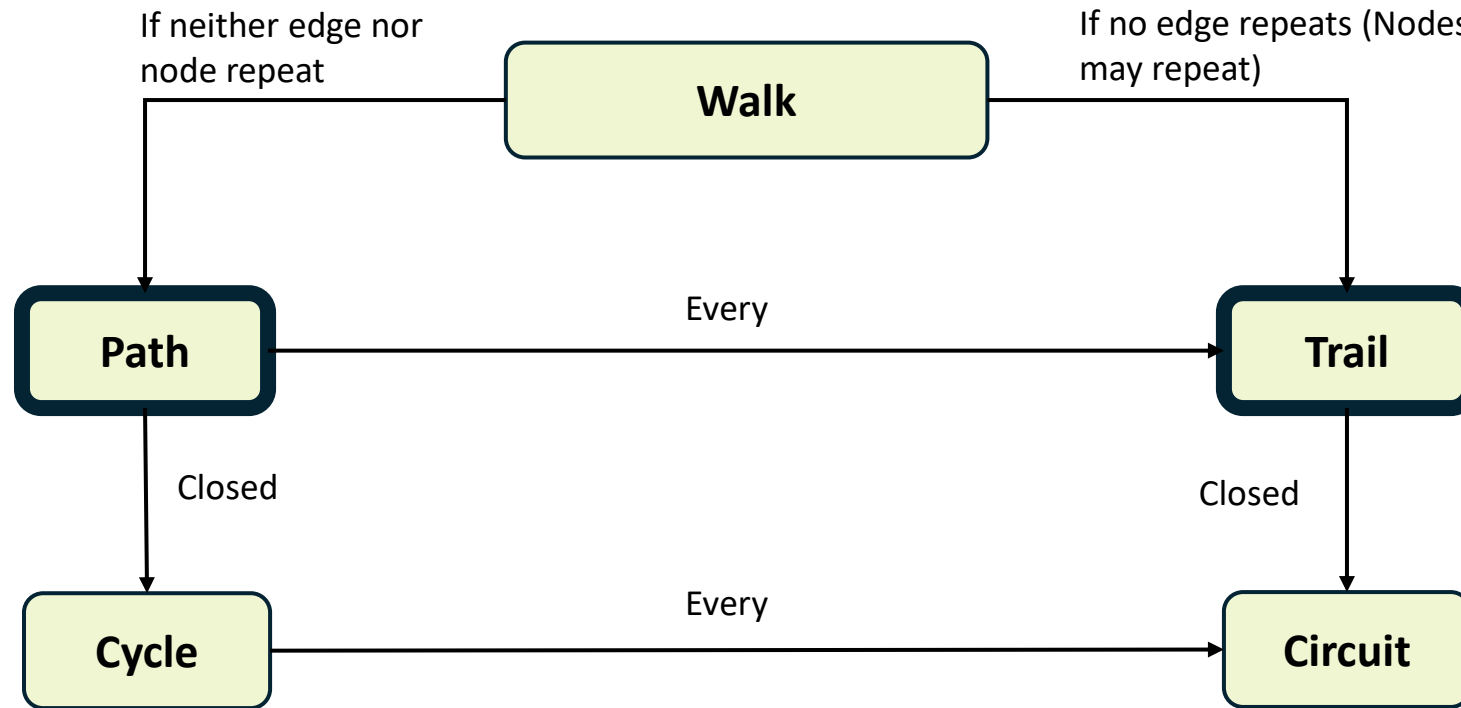
- In-degree of a node: number of edges pointing to the vertex (parents)
- Out-degree of a node: number of edges point out of the vertex (children)
- Degree of a node: number of neighbour of a vertex in an undirected graph

Path Analytics

- A **walk** is a sequence of edges
- A **trail** is a constrained walk with no repeating edges
- A **path** is a constrained walk with no node repetition (and edges)
 - Shortest path



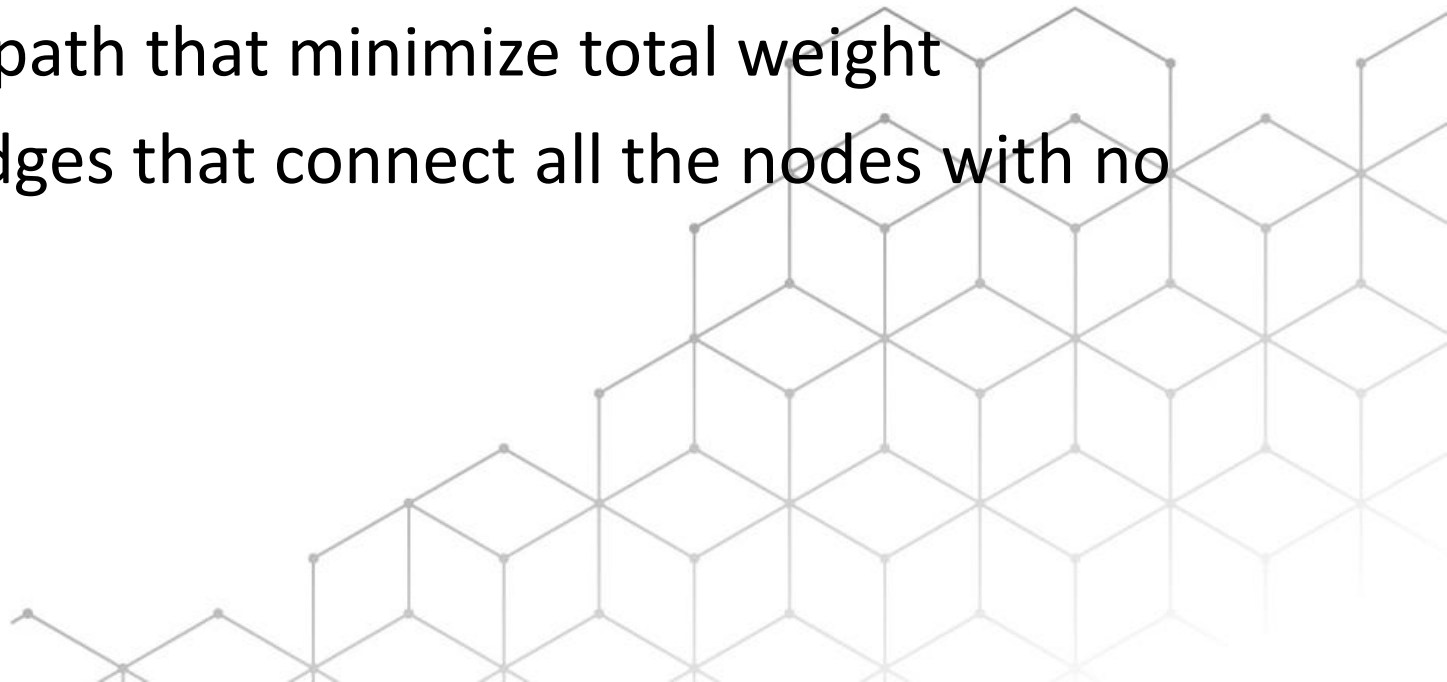
Path, Trail, Walk and Others



	Repeat Node	Repeat Edge	Closed
Walk	Yes	Yes	Closed or Open
Trail	Yes	No	No
Path	No	No	No
Cycle	No (Only start/end)	No	Yes
Circuit	Yes	No	Yes

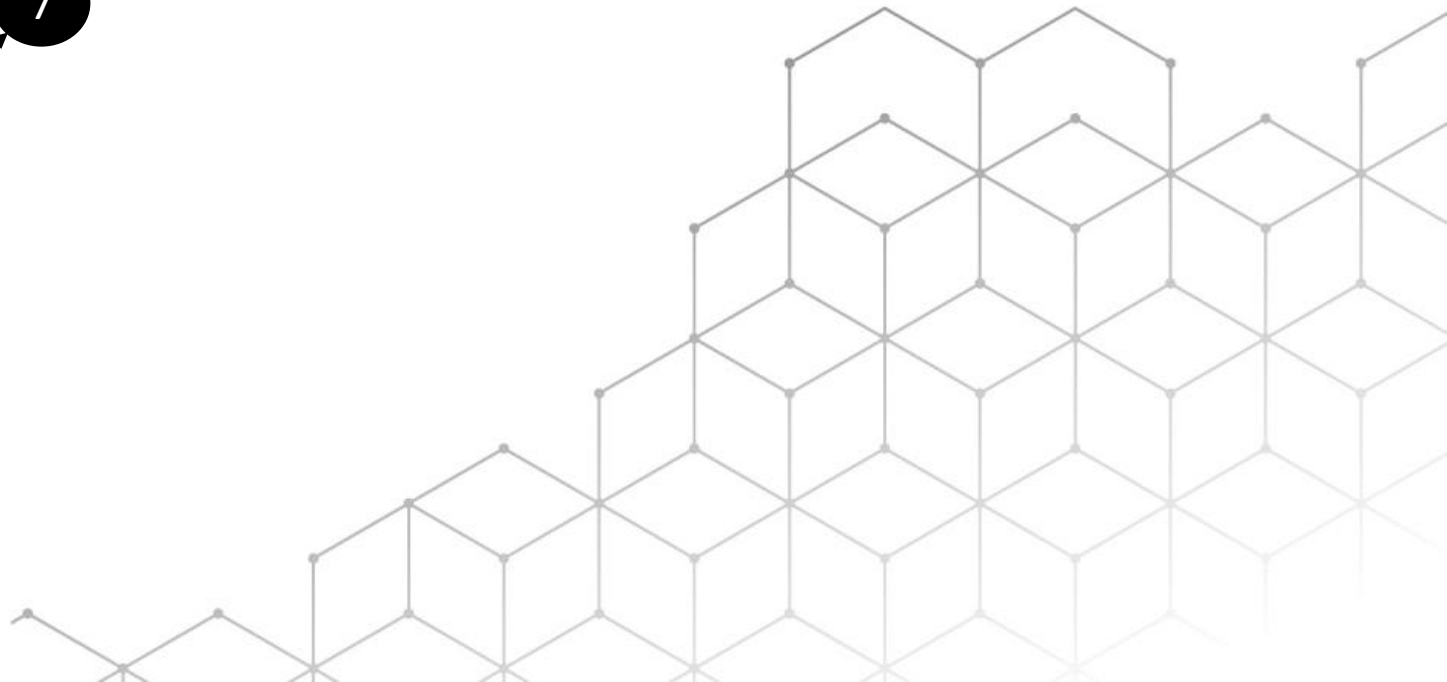
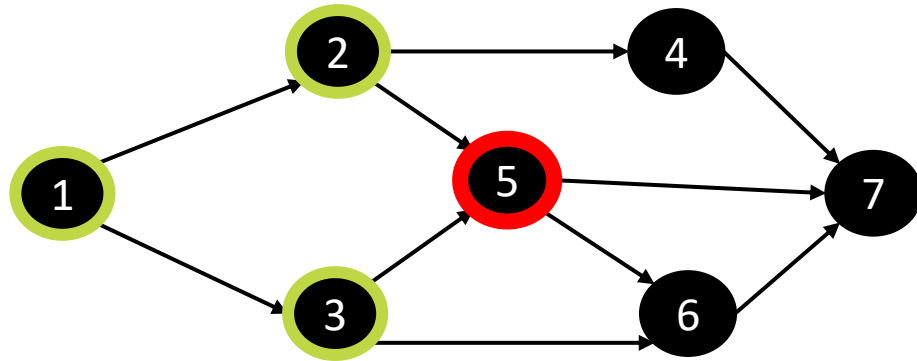
Path Analytics

- Reachability: Can we reach node D from node C?
- Shortest path: minimum edges between two nodes
 - Breadth-First Search
 - Dijkstra's algorithm
- Best path (weighted graph): path that minimize total weight
- Minimum Spanning Trees: edges that connect all the nodes with no cycles and minimum weight

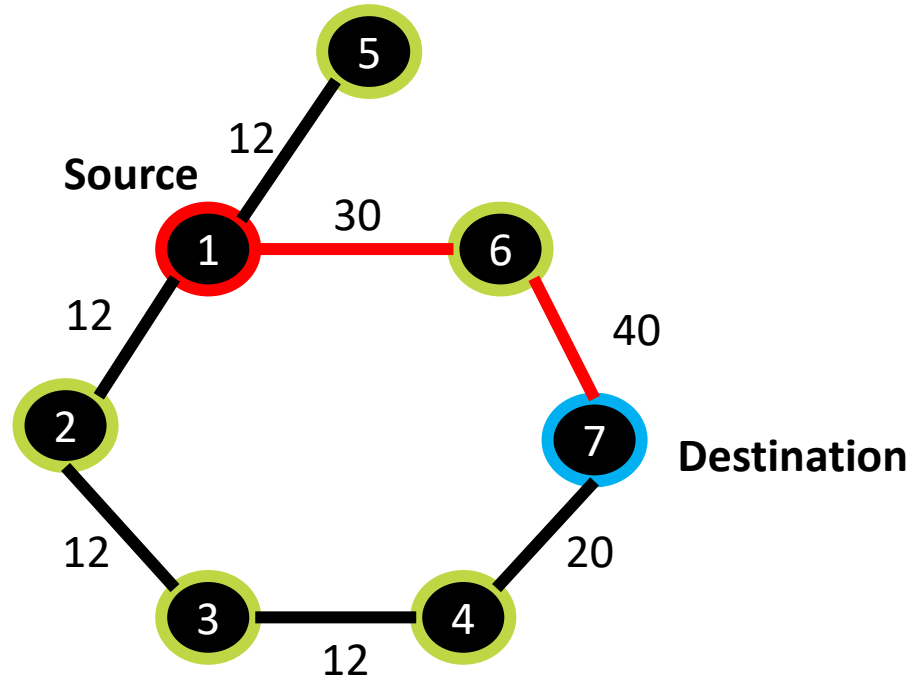


Reachability

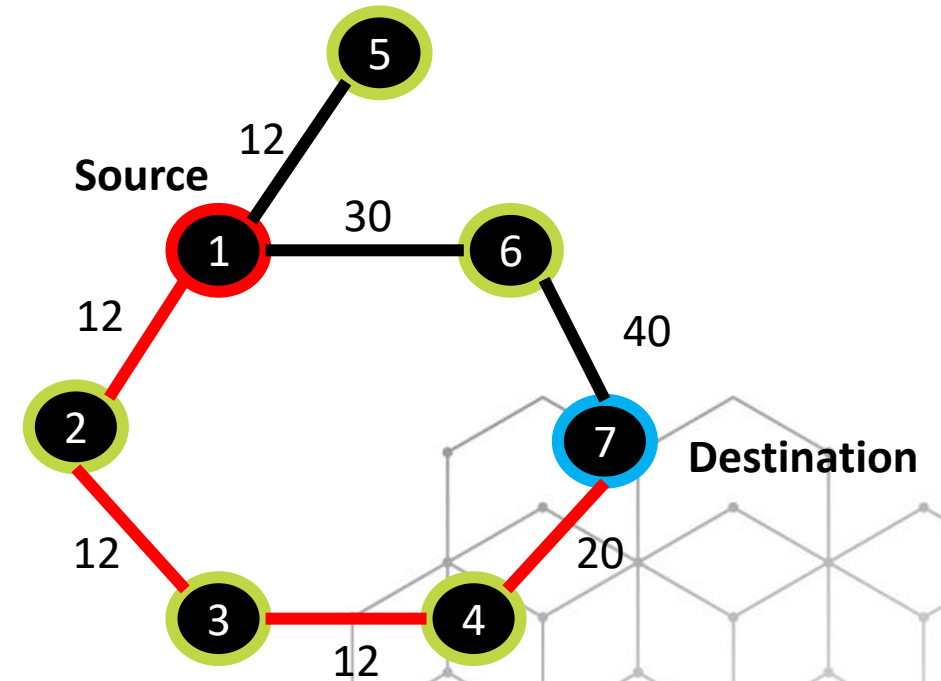
- Node 5 is reachable from 1, 2, 3



Shortest Path and Best Path

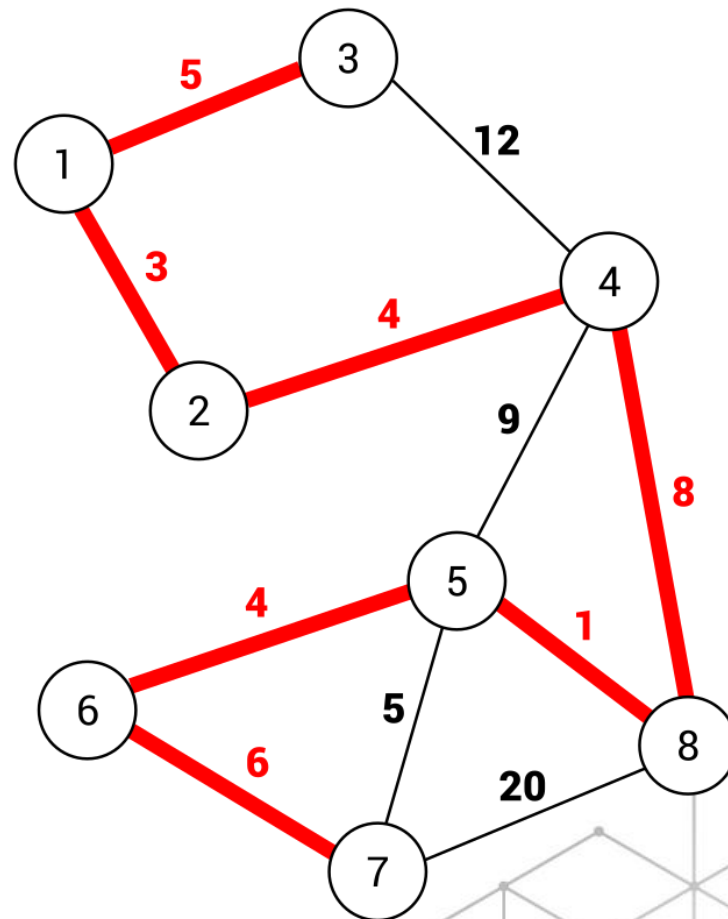
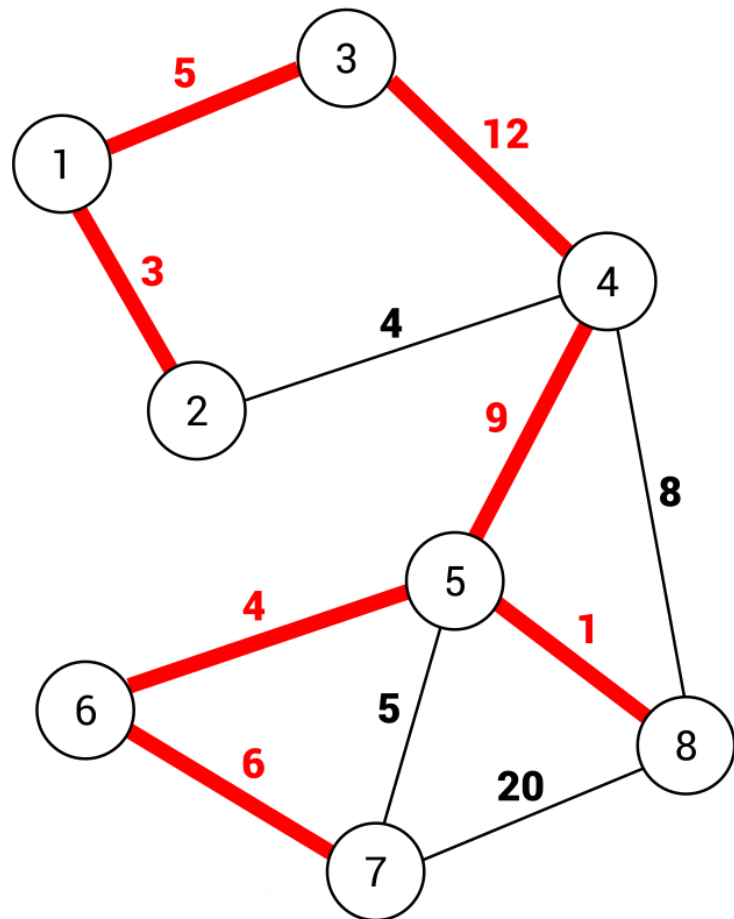


Unweighted Shortest Path



Best Shortest Path

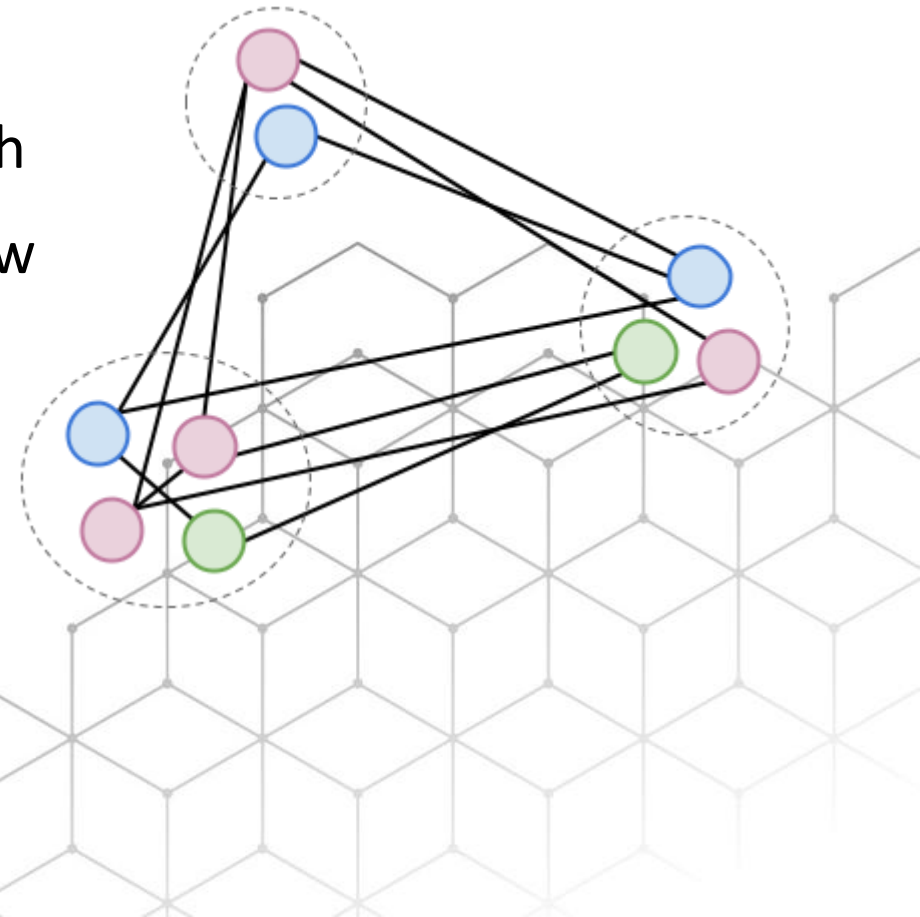
Minimum Spanning Tree



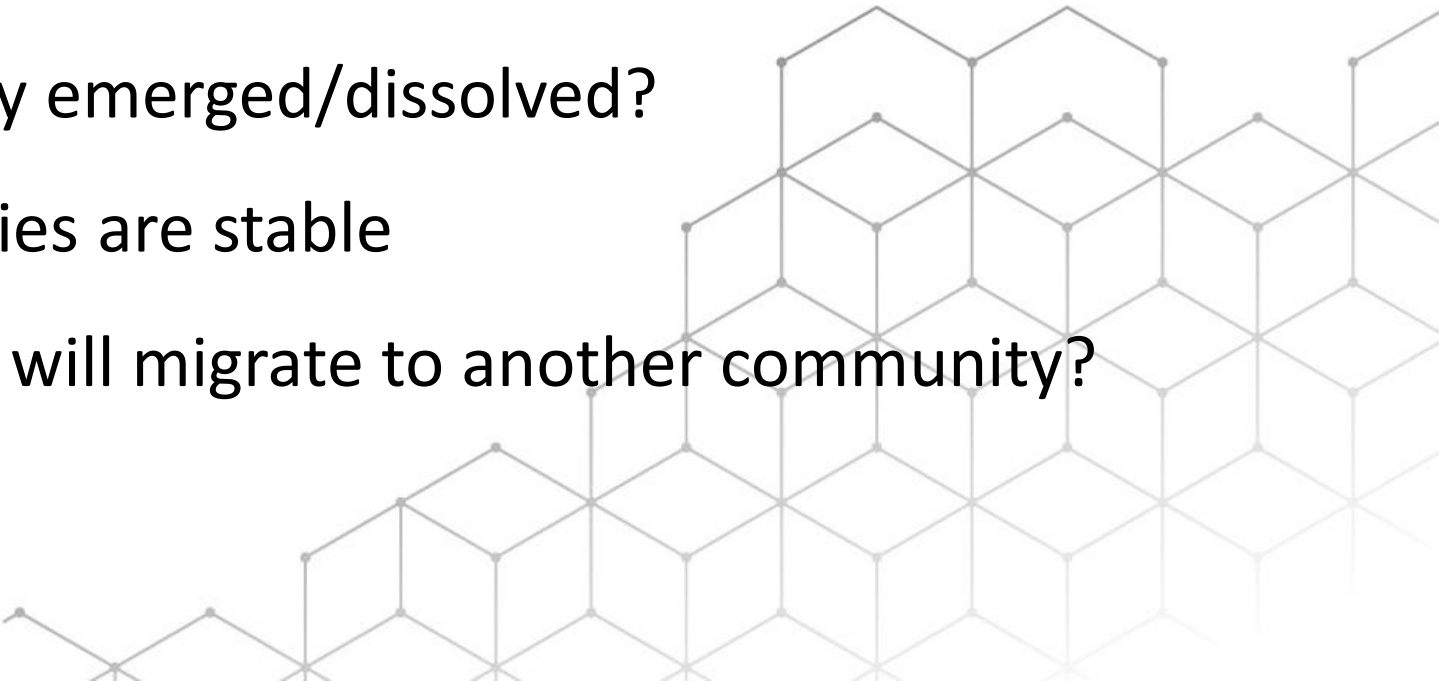
Community Analytics

A dense subgraph (cluster) within a graph has nodes more connected within than outside

- Cohesion – Connectivity “within” the cluster is high
- Separation – Connectivity “between” clusters is low



Community Analytics Questions

- Discover a community
 - Describe interaction with a community
 - Describe interaction between communities
 - Temporal – How a community emerged/dissolved?
 - Temporal – Which communities are stable
 - Temporal – Predict of a node will migrate to another community?
- 

Centrality Analytics

Connectivity Importance

Simply the degree of node X (in and out degrees)

- Queen bees in a community

Closeness Importance

Average length of all shortest paths compared to the averages of the other vertices

Betweenness Importance

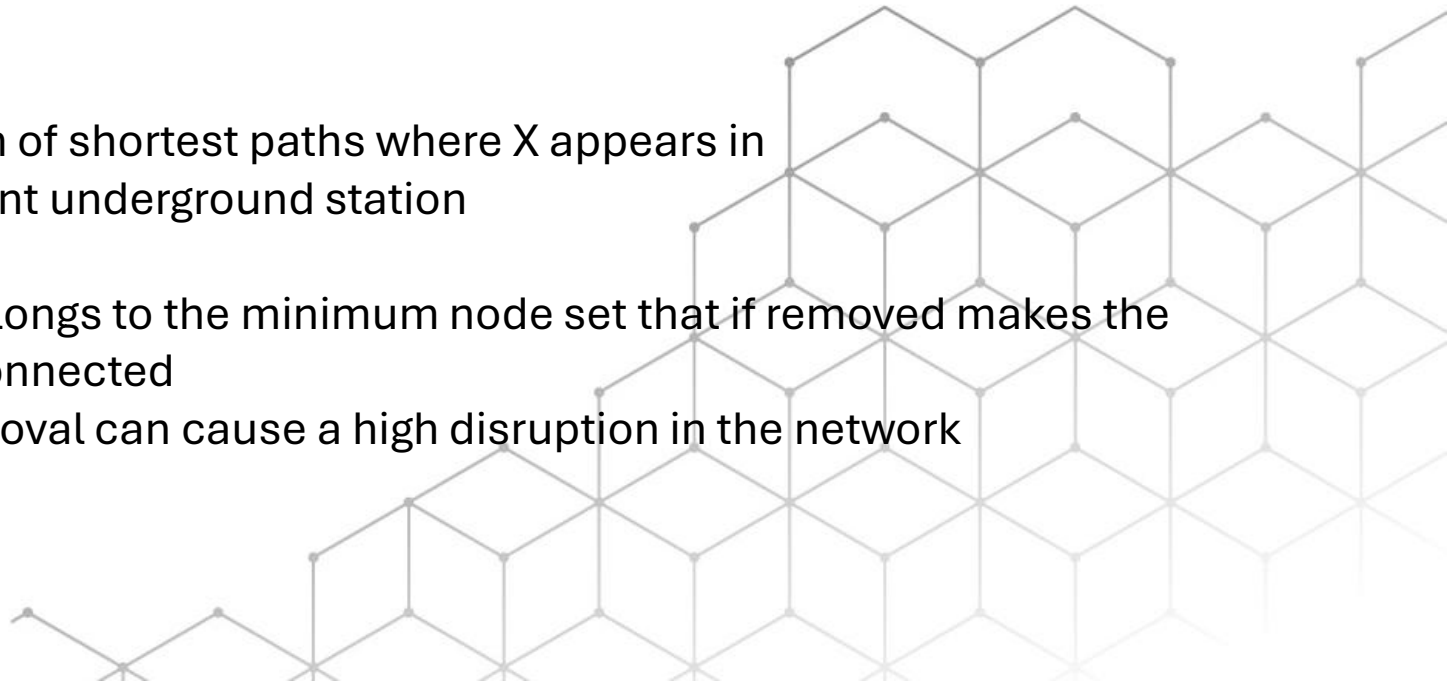
The fraction of shortest paths where X appears in

- Important underground station

Vulnerability

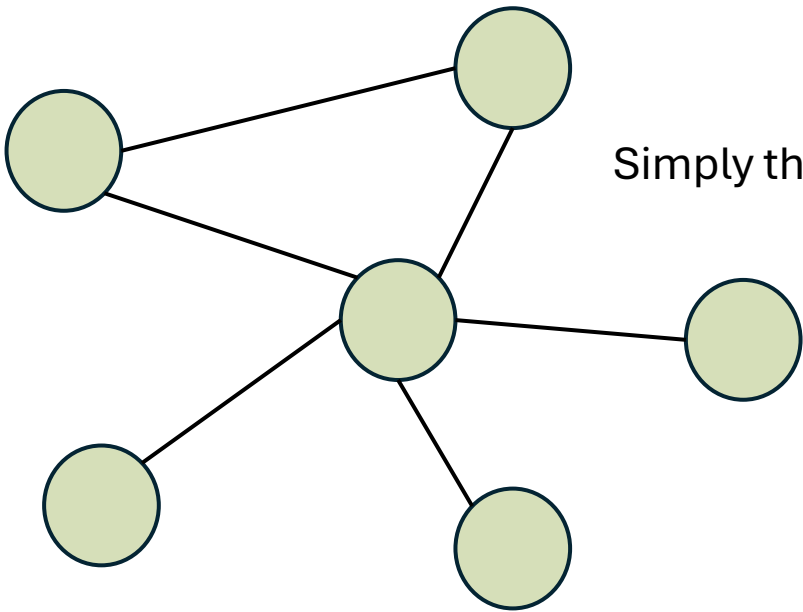
Vertex X belongs to the minimum node set that if removed makes the graph disconnected

- This removal can cause a high disruption in the network

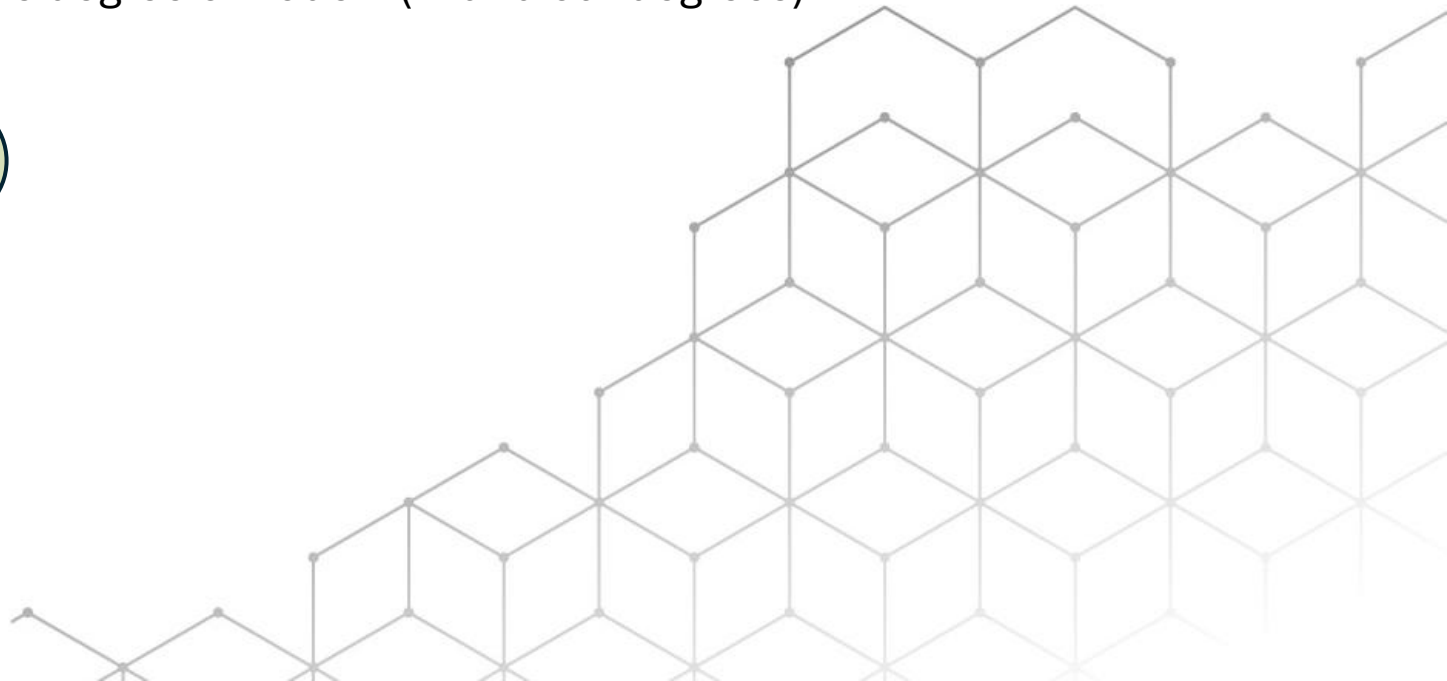


Finding the Queen Bee

- Connectivity Importance: find the most connected nodes

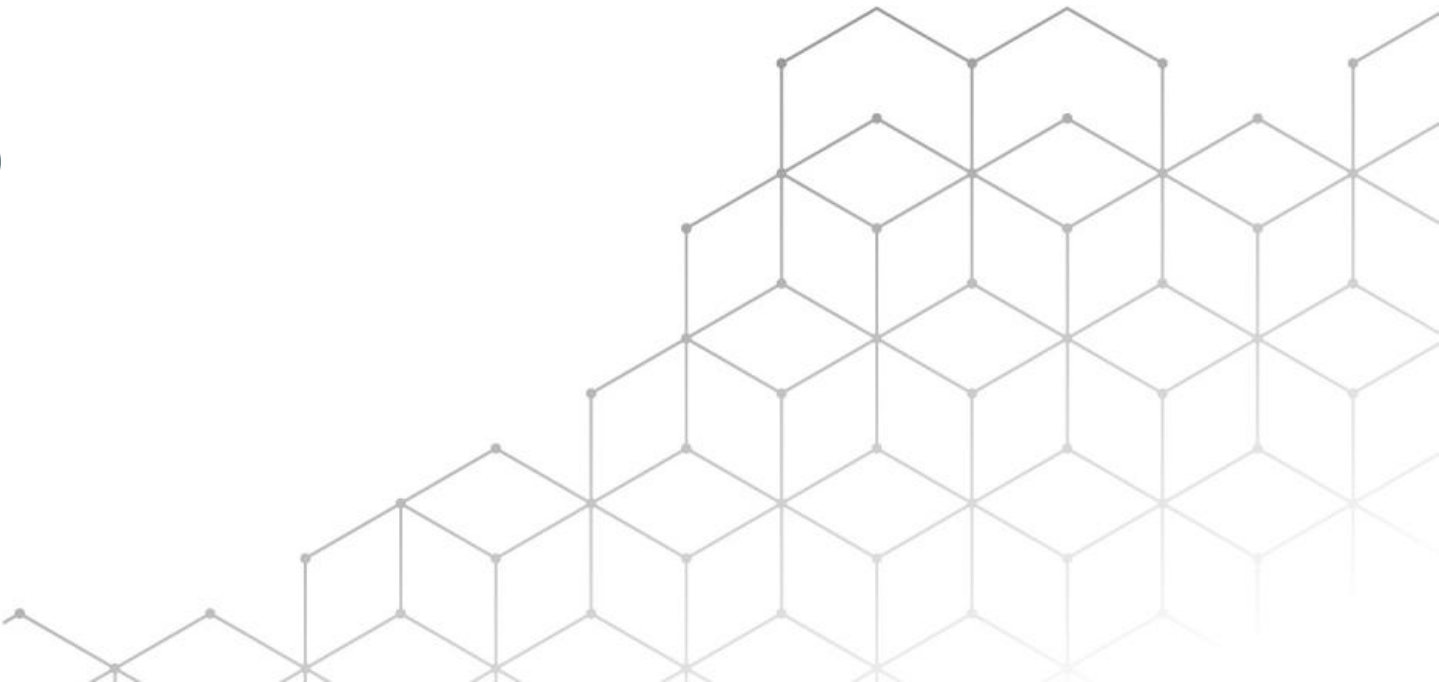
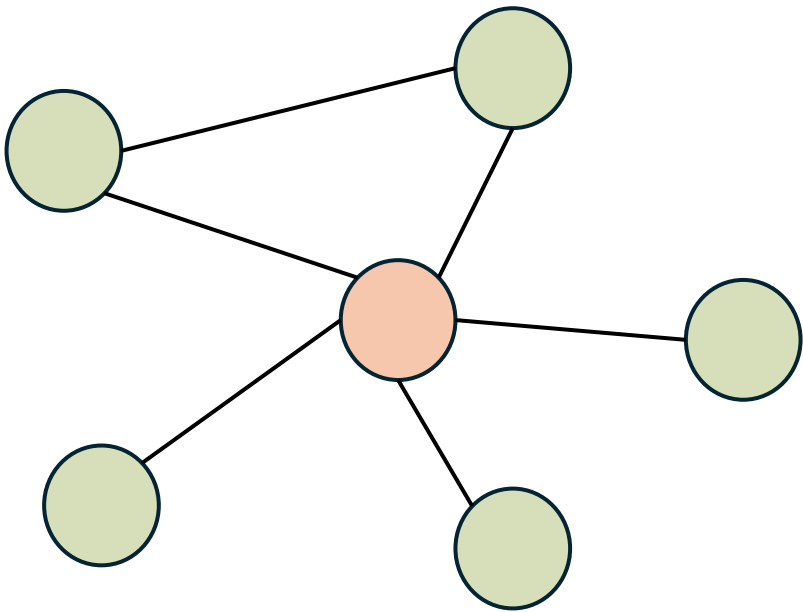


Simply the degree of node X (in and out degrees)

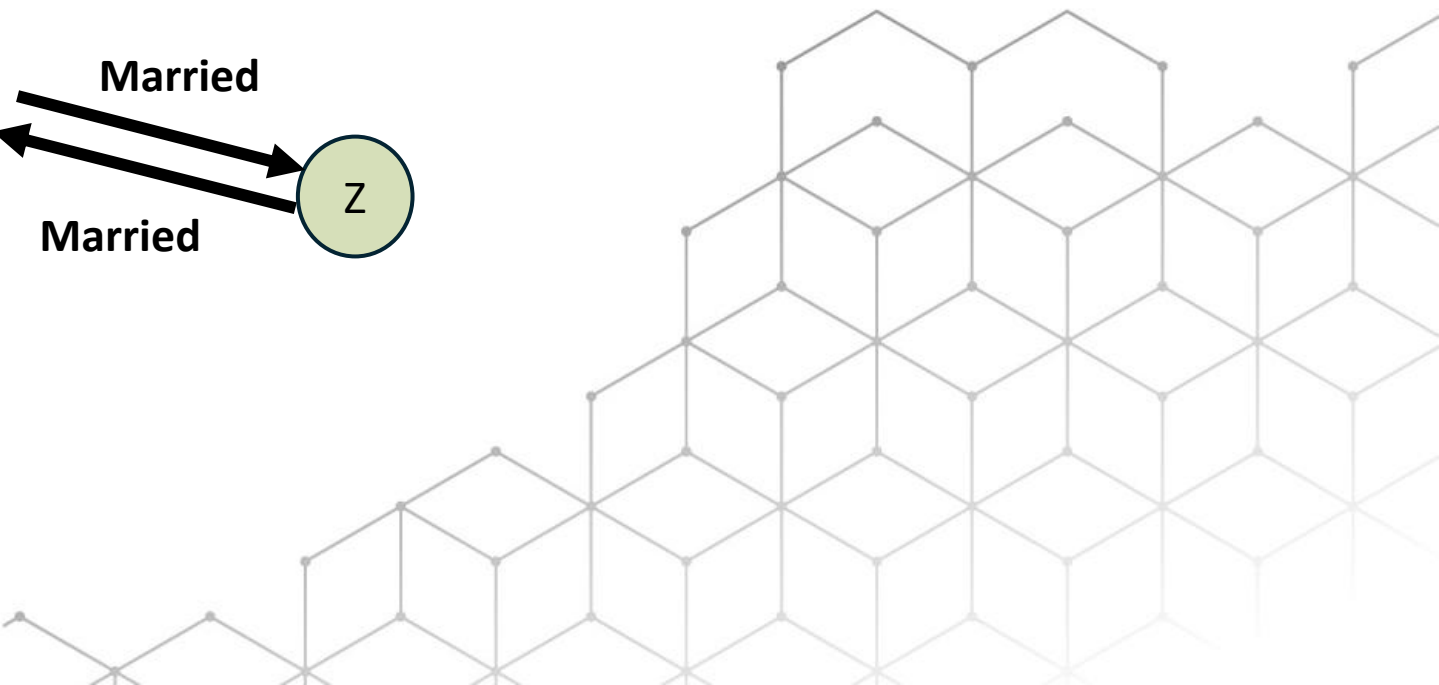
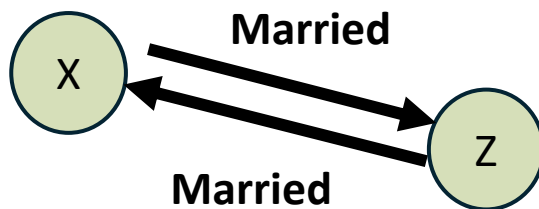
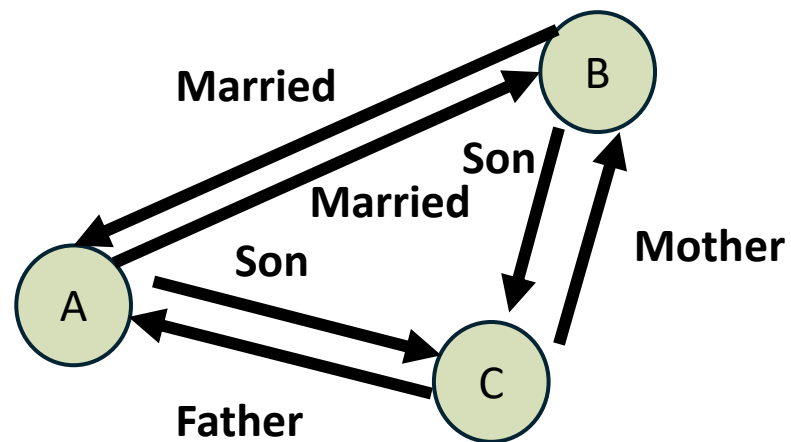
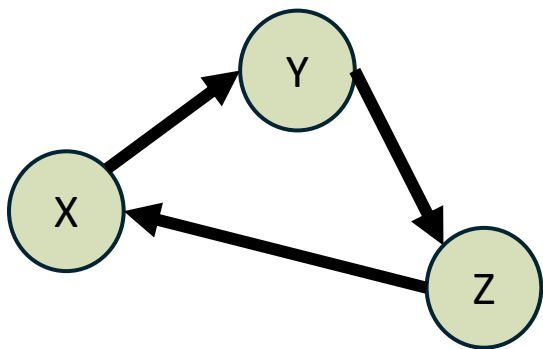
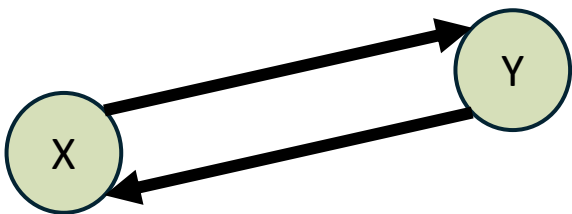


Vulnerability

- Vertex X belongs to the minimum node set that if removed makes the graph disconnected

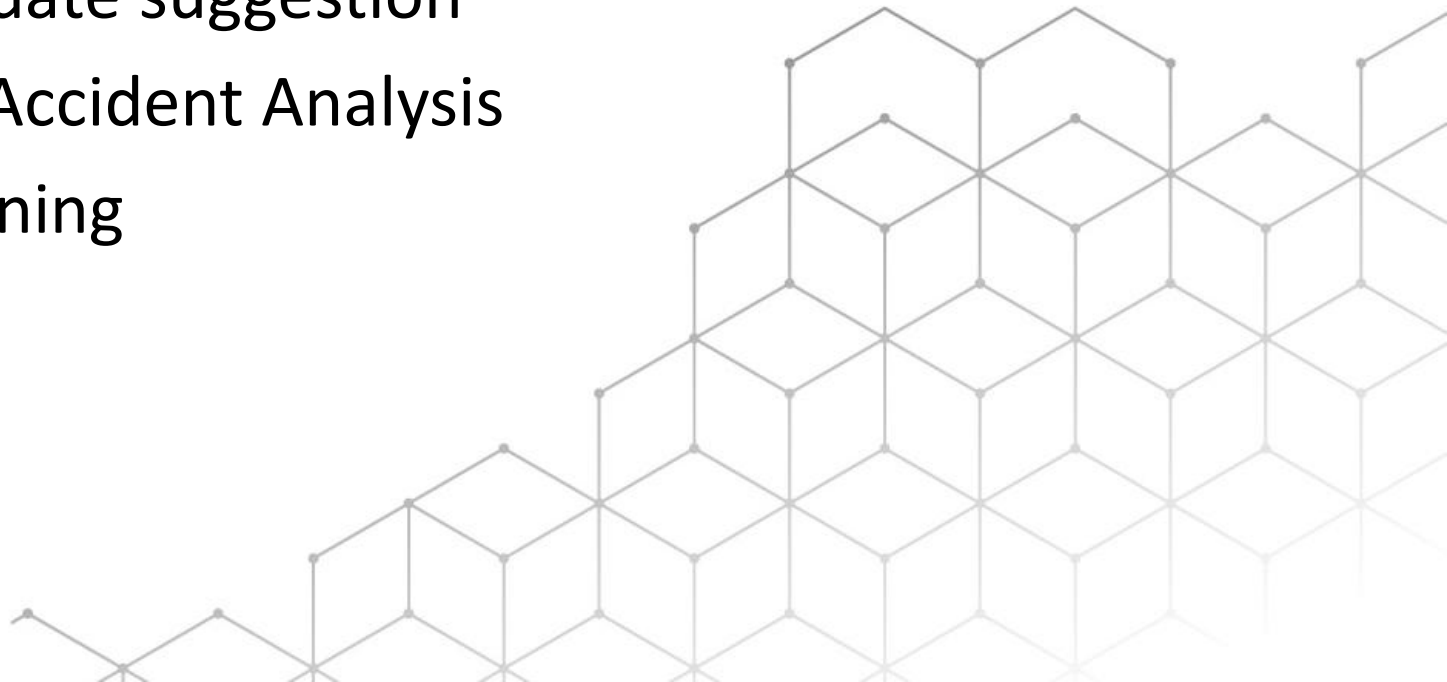


Pattern Matching

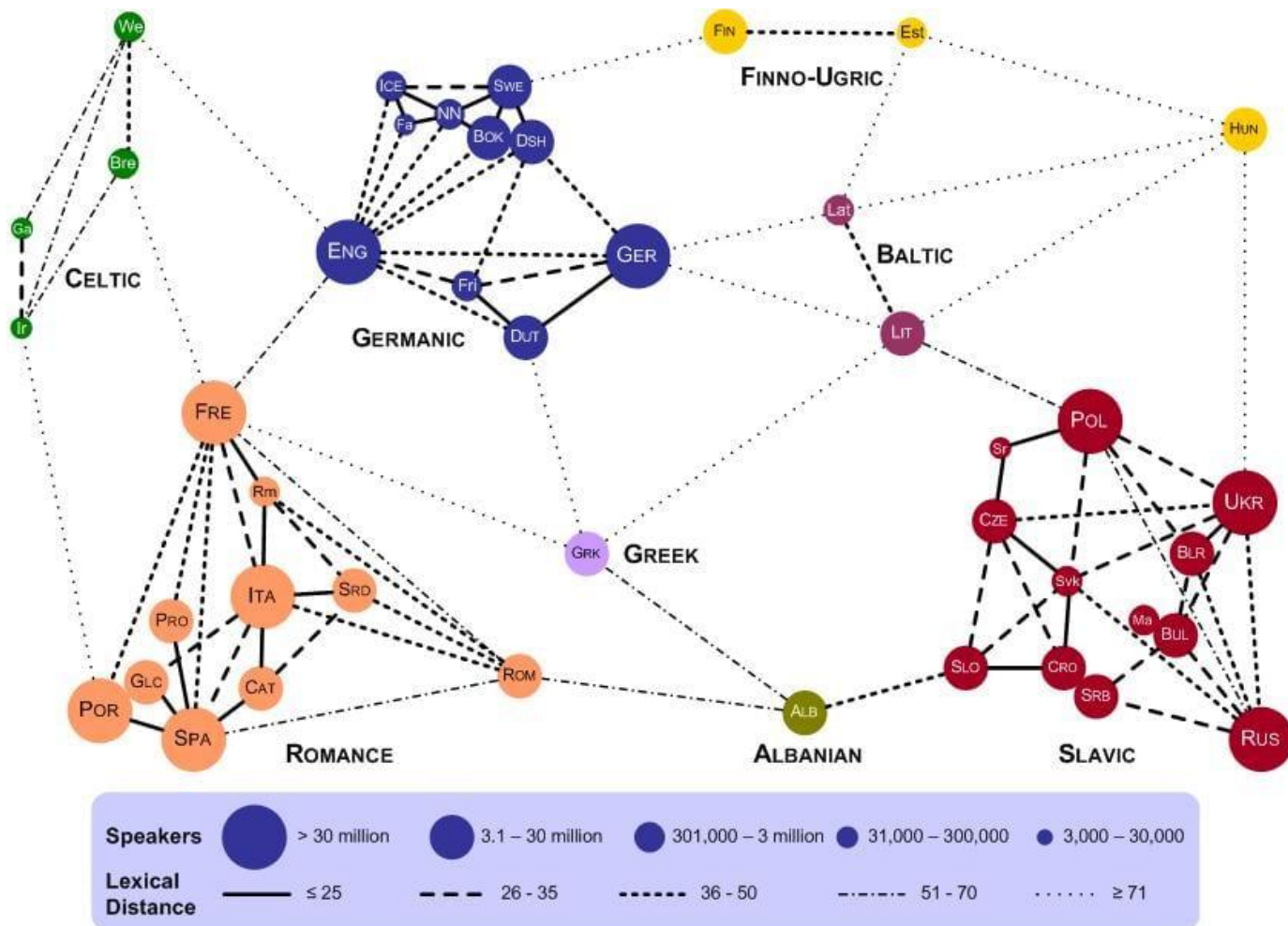


Pattern Matching Applications

- Banking – Fraud Detection
- Security – Threat Detection
- Bioinformatics & Biochemistry – Association Analysis
- Social Networks – Job/Candidate suggestion
- GPS & Smart Cities – Traffic/Accident Analysis
- Telecom – Targeted Campaigning



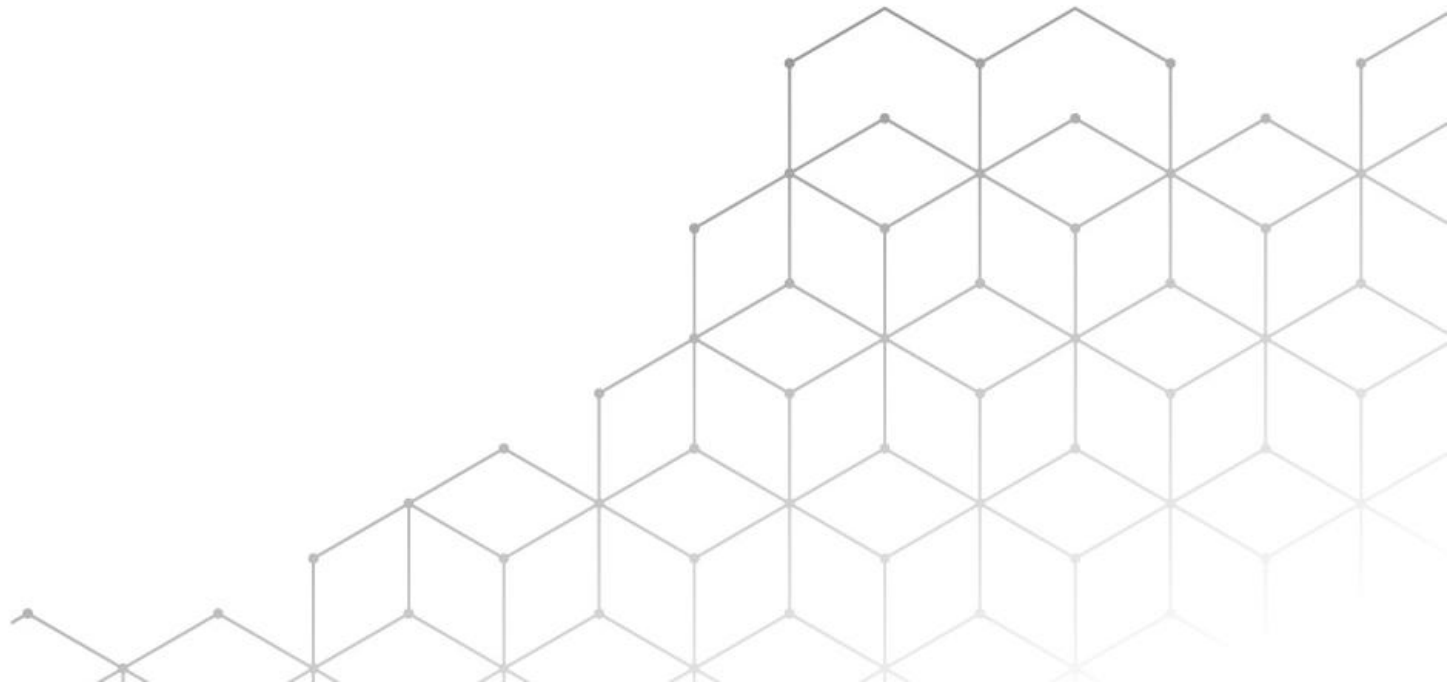
Example of Language Network



Agenda for Today

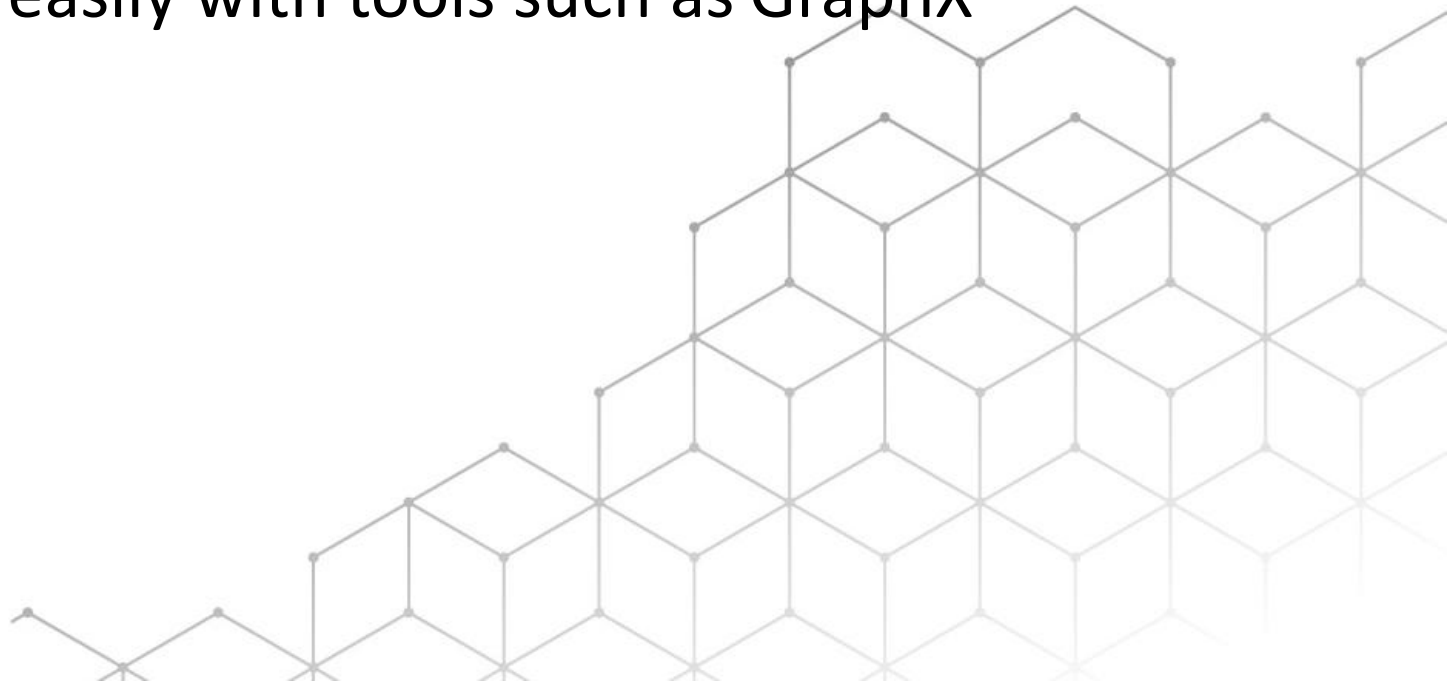


- Overview on Graphs
- Graph Analytics
- GraphX
- Big Data Visualization

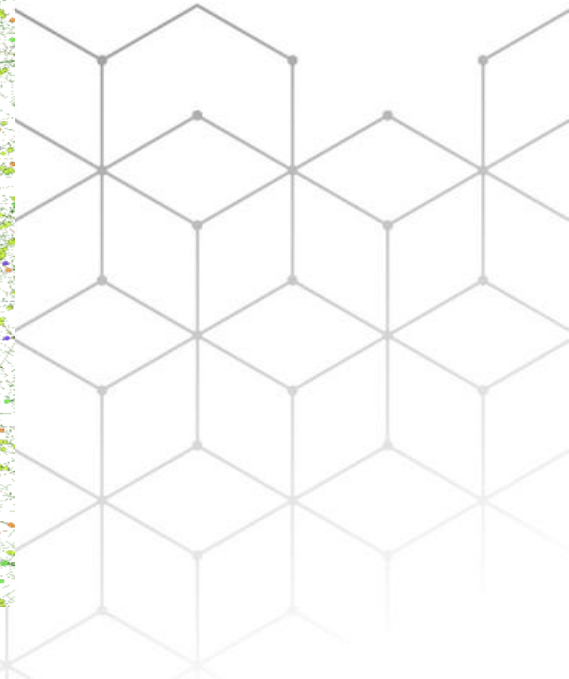


What is a Large Graph?

- A large graph is one that cannot be easily visualized.
- You have to use large tools to calculate the important statistics
- Breaking a large graph down to a small graph is not as simple
- This can be done reasonably easily with tools such as GraphX

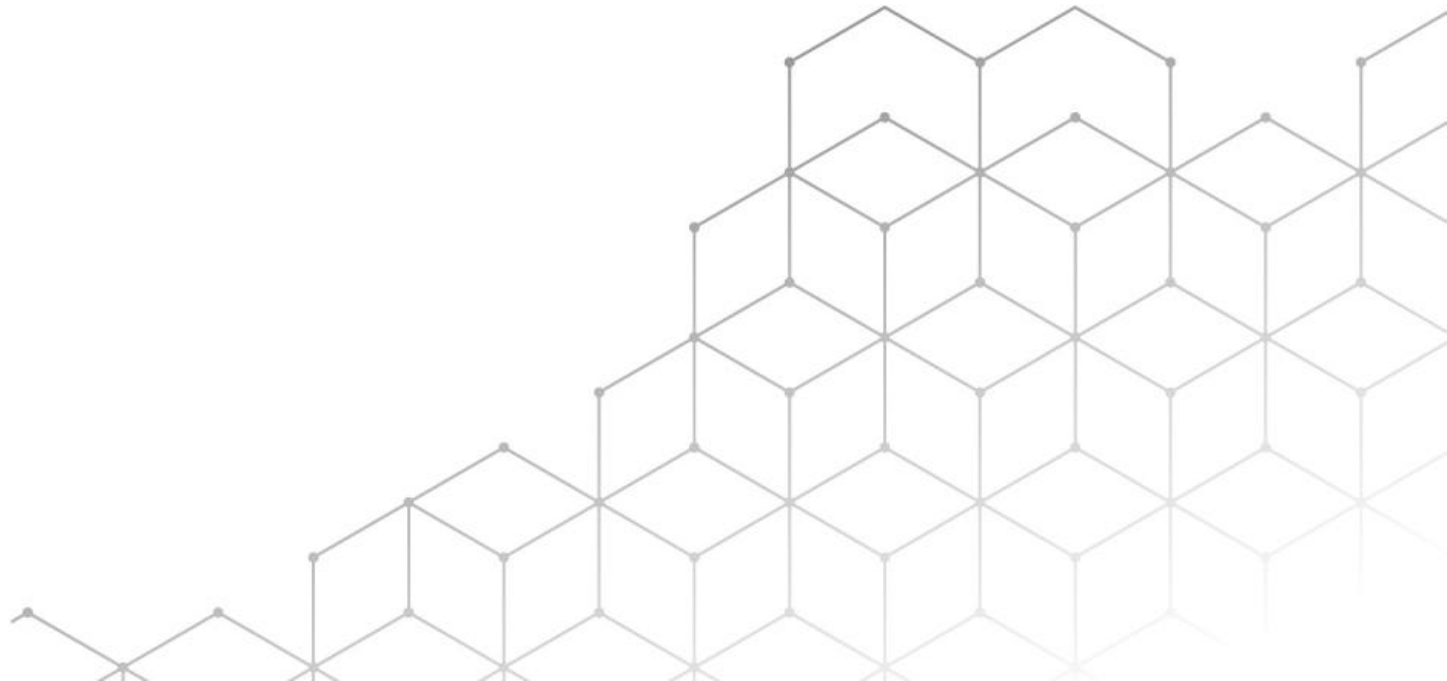


Example of Large Graph



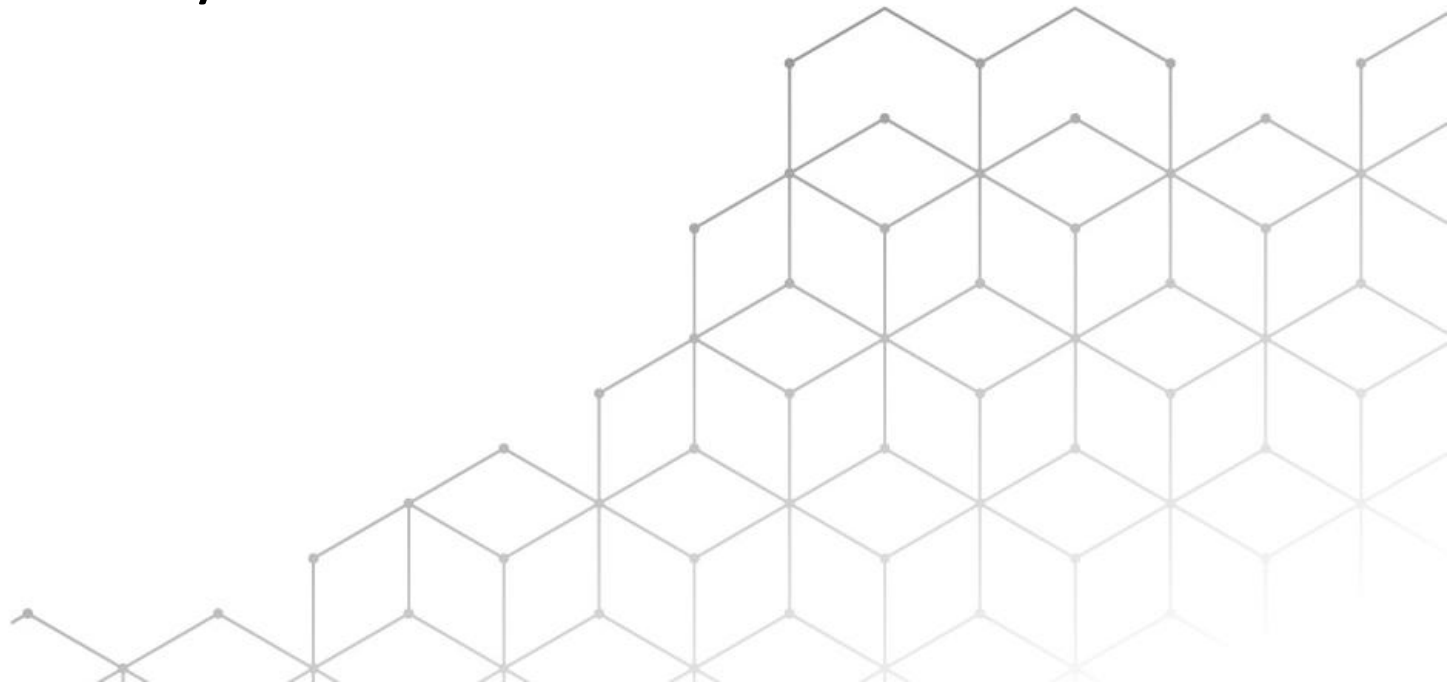
Definition of Graph Database

NoSQL family of data stores that is optimized to store, model, and process data in a graphical form, as well as answering graph-related queries efficiently.

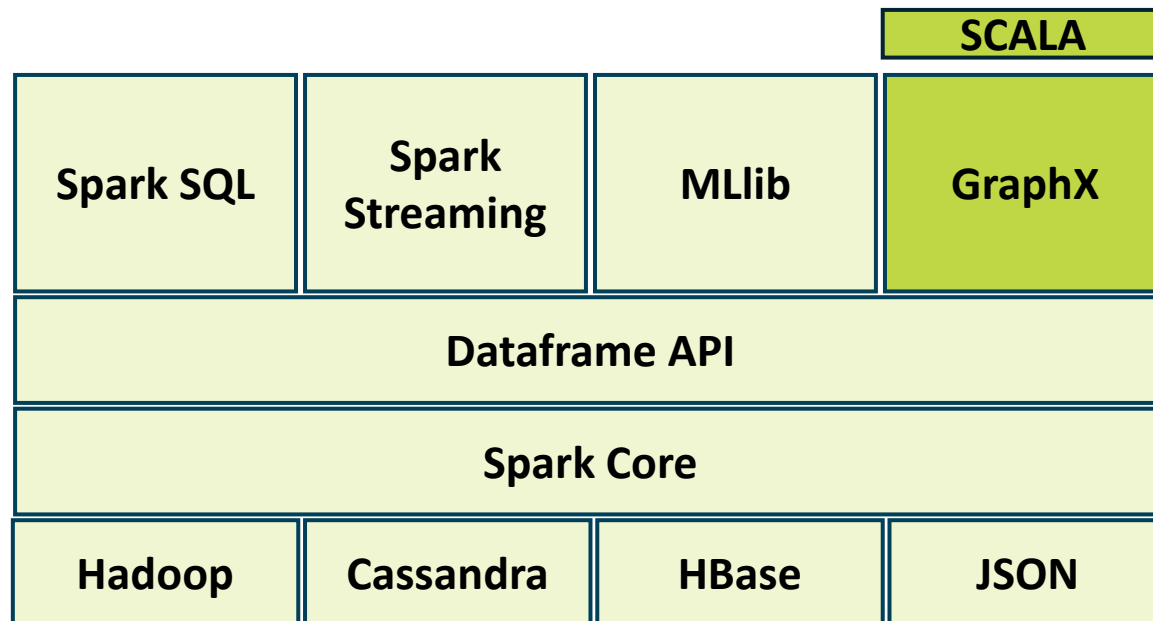


GraphX

- **GraphX** is Apache Spark's API for graph databases and graph parallel computation.
- Spark benefits like scaling, clustering, storage.
- As of right now, GraphX is Scala only.



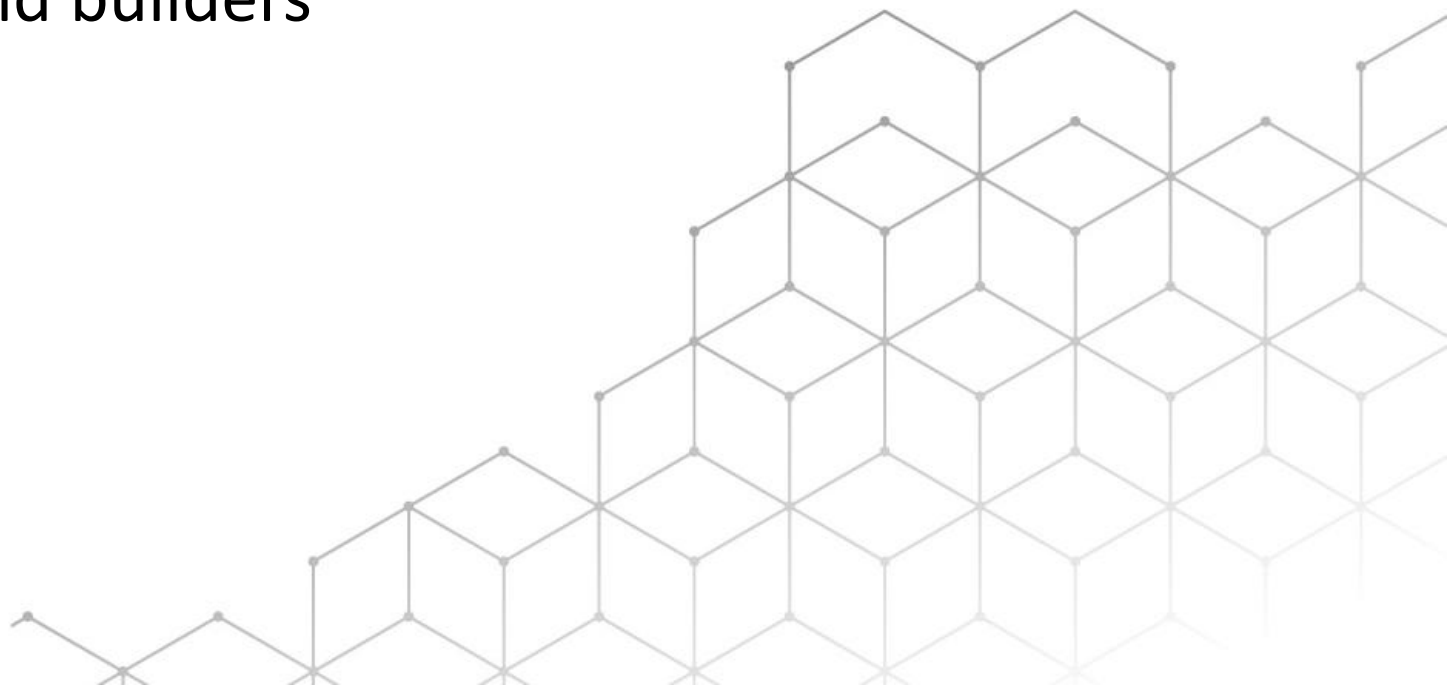
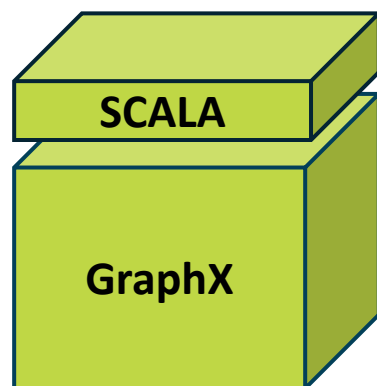
Spark Components



GraphX Description

Spark component for graphs and graph-parallel computations

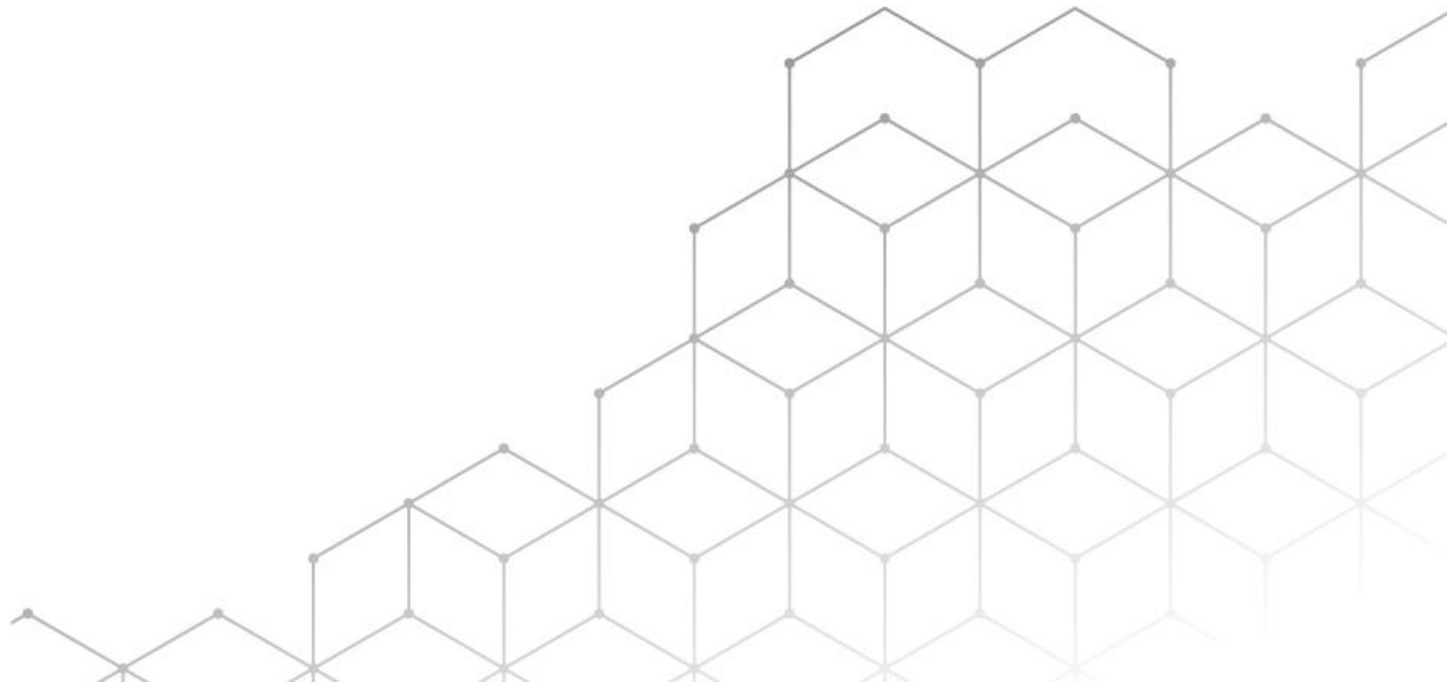
- Combines data parallel and graph parallel processing in single API
- Store graphs as collections (RDD)
- Provides graph algorithms and builders



Agenda for Today

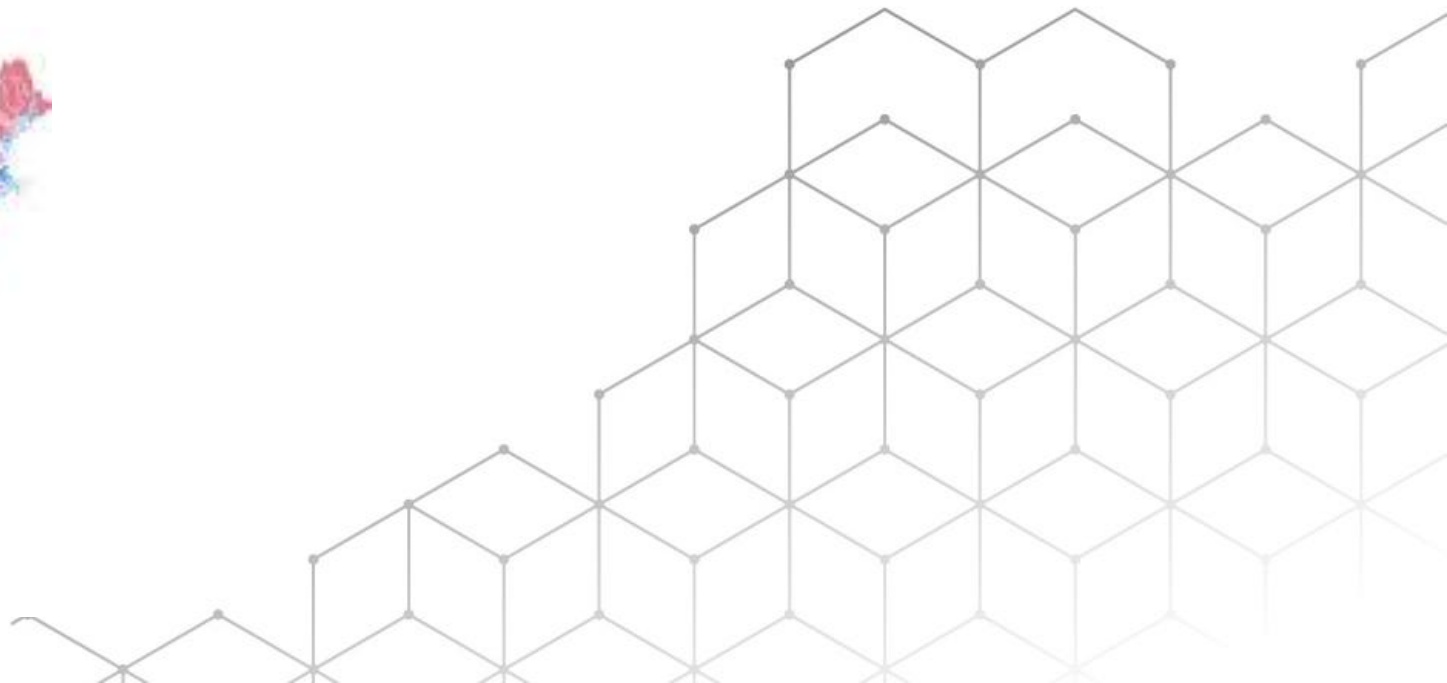


- Overview on Graphs
- Graph Analytics
- GraphX
- Big Data Visualization
 - What is Data Visualization?
 - Benefits of Data Visualization
 - Six Big Data Visualization Tools
 - Google Chart
 - Tableau
 - D3
 - HighChart
 - Types of Data to Visualize
 - Visualization Options



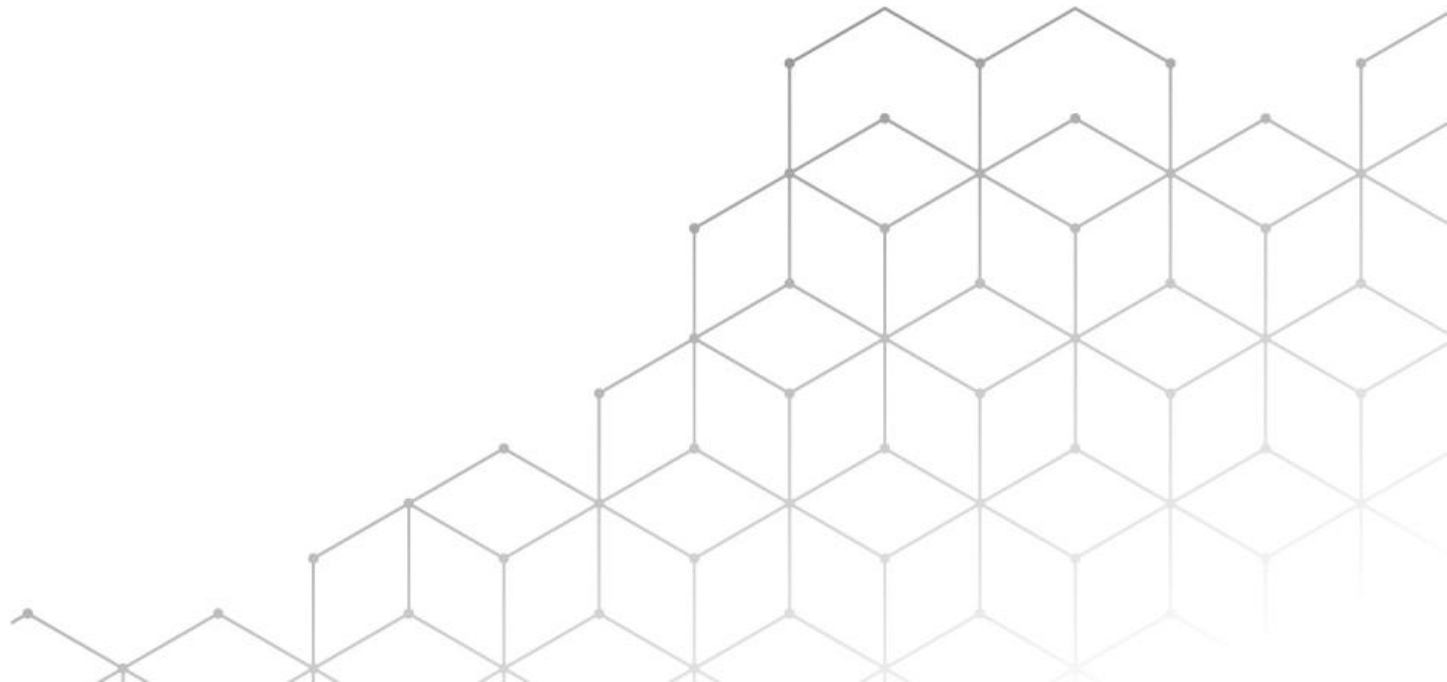
What is Data Visualization?

- Data visualization describes any effort to help people understand the significance of data by placing it in a visual context.
- Patterns, trends and correlations that might go undetected in text-based data can be exposed and recognized easier with visuals



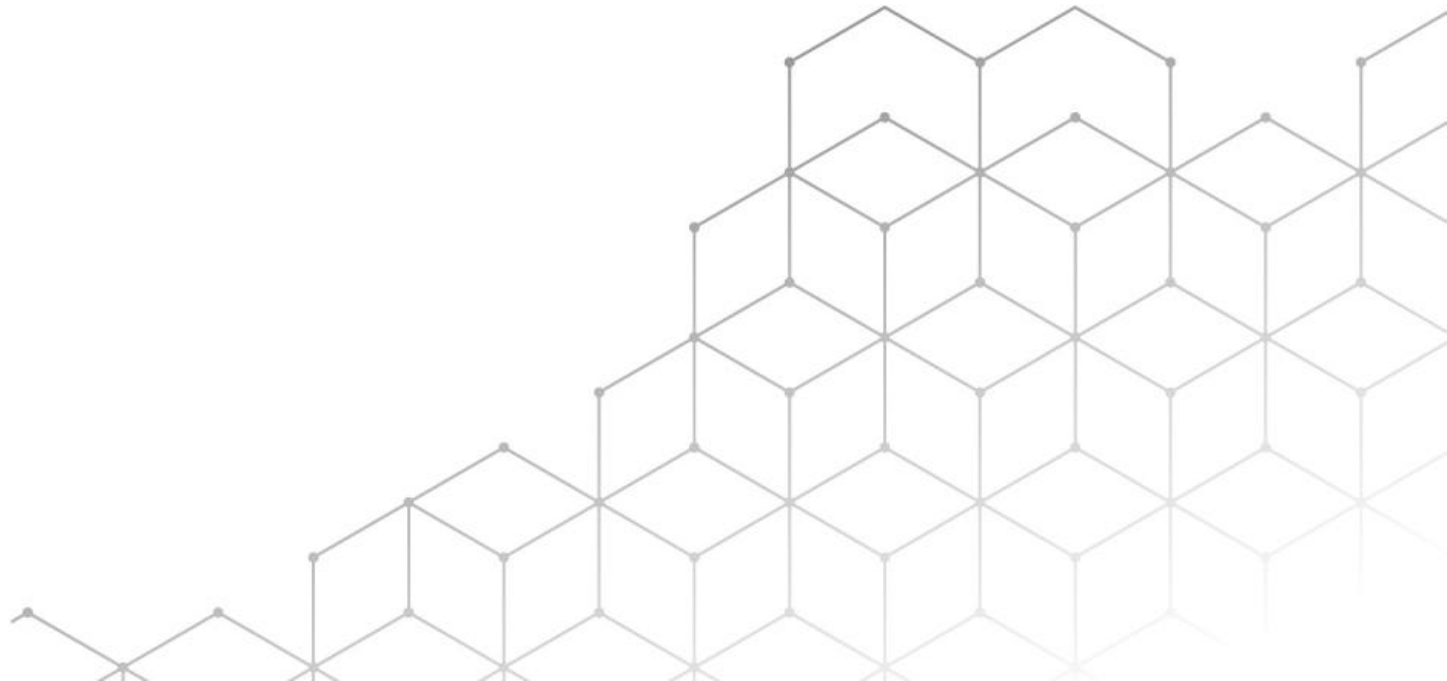
Benefits of Data Visualization

- Faster Action. The human brain tends to process visual information far more easily than written information.
- Discover Emerging Trends
- Interact With Data
- Create New Discussion



Six Big Data Visualization Tools

- Google Chart
- Tableau
- D3
- Fusion chart
- High Charts
- Canvas



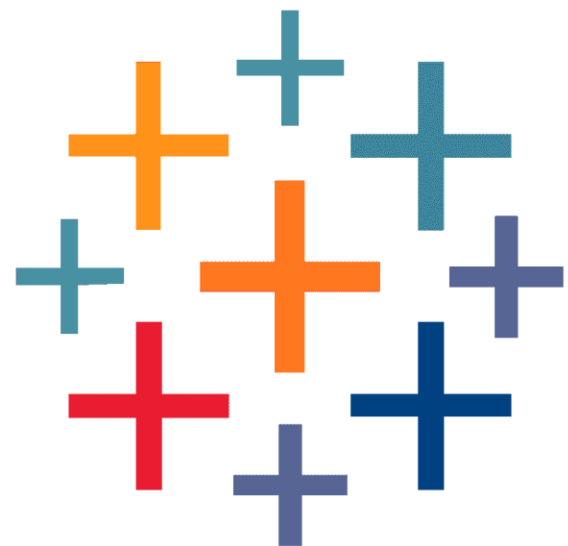
Google Chart

- Renders through HTML and SVG
- Cross browser compatible
- Cross platform compatible
- Chart data can be exported to PNG format
- Can integrate or “endorse” data from other Google products (e.g., Google Maps)
- Strong Google support



Tableau

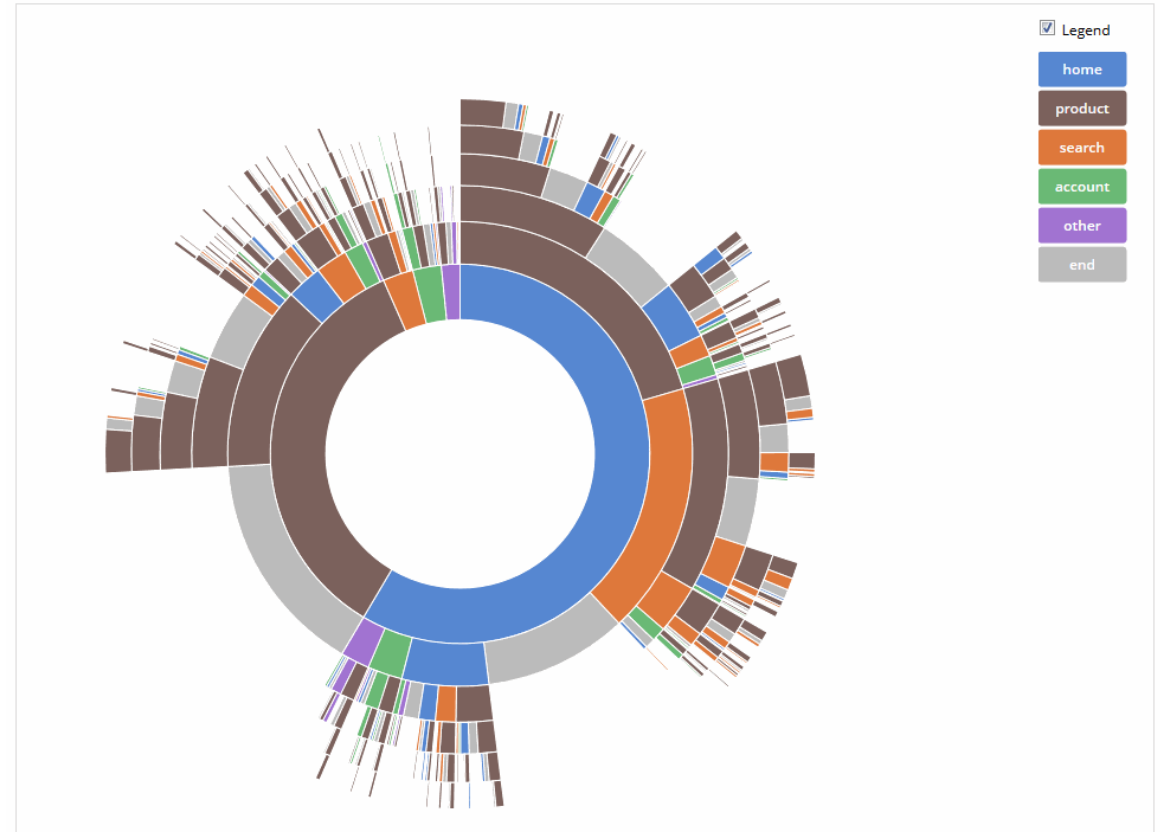
- Handy and lightning-fast tool
- User-friendly, “drag and drop” compatible dashboard
- Can analyze datasets ranging from spreadsheets to Hadoop
- Tableau Desktop is free for students and instructors



tableau

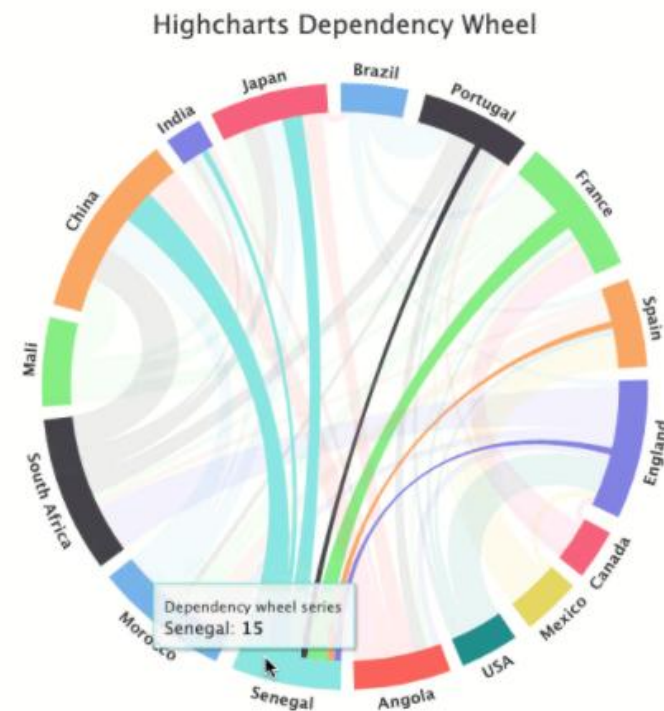
D3 (Data Driven Documents)

- It's a JavaScript library for creating interactive, web-based data visualizations.
- It's extremely fast and capable of handling large datasets in real time.
- Both 2D and 3D rendering are supported.

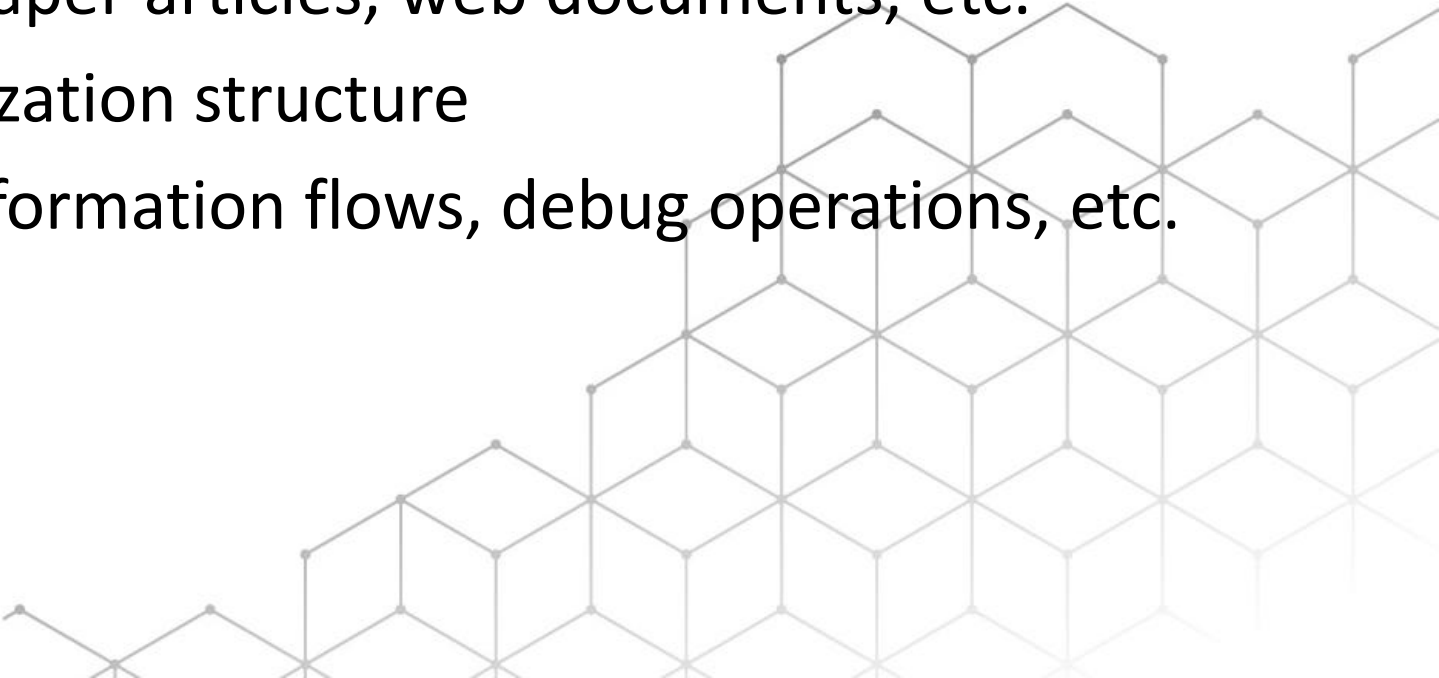


HighChart

- Supports HTML5, SVG, and VML for cross-browser compatibility (including IE6+)
- Offers a wide range of chart types (line, bar, column, pie, etc.)
- Efficient in processing real-time JSON data
- Source code is available for download
- Pricing starts from \$399

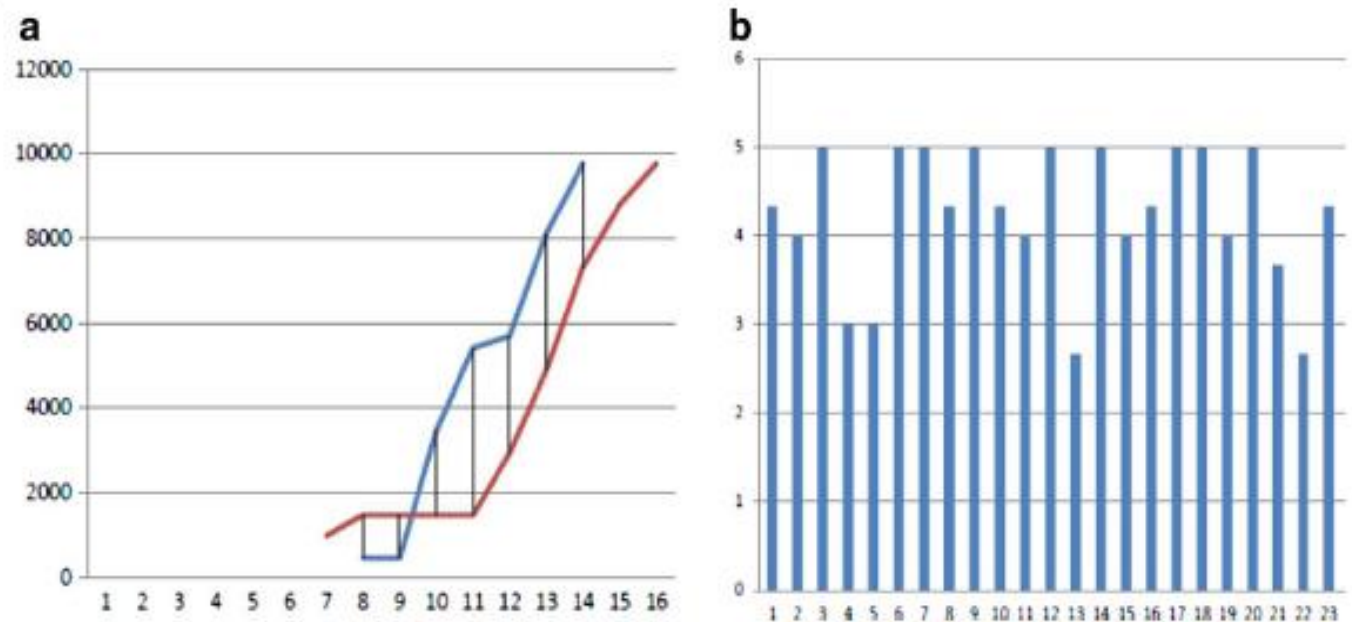


Types of Data To Visualize

- Univariate data: One dimensional arrays, time series, etc.
 - Two-dimensional data: Geographical coordinates, etc.
 - Multidimensional data: Financial indicators, sensors
 - Texts and hypertexts: Newspaper articles, web documents, etc.
 - Hierarchical and links: organization structure
 - Algorithms and programs: Information flows, debug operations, etc.
- 

Standard Figure/Chart

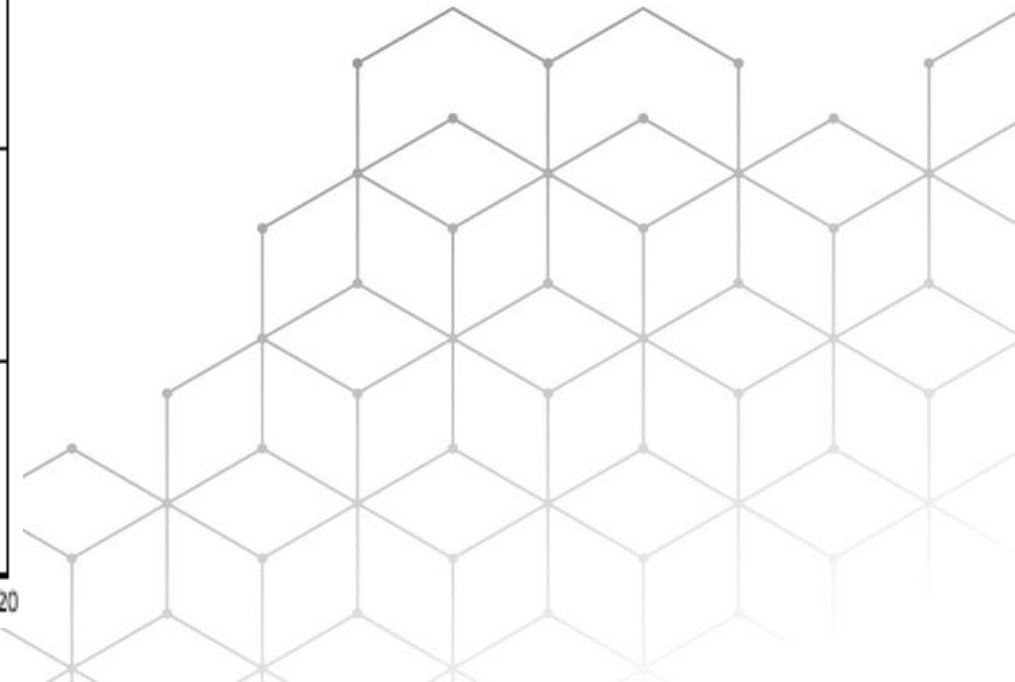
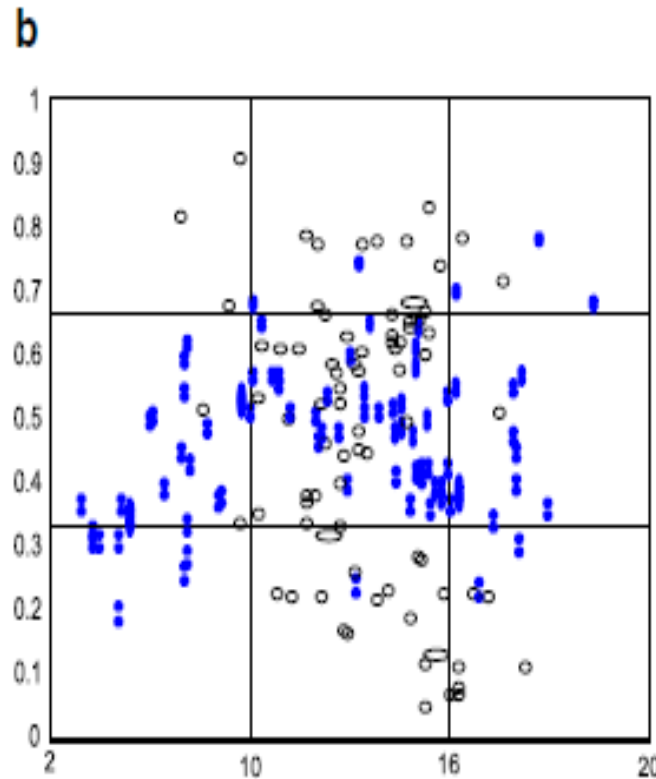
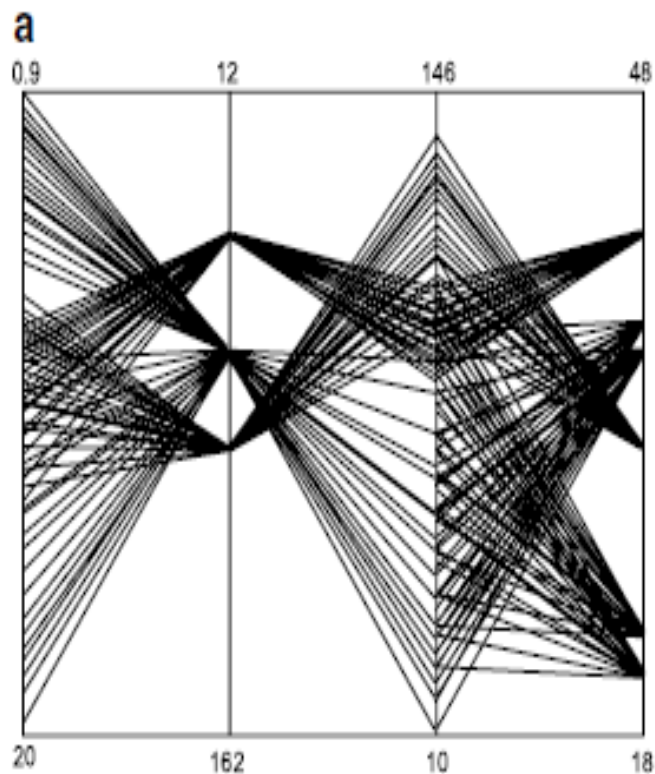
- May be implemented as bars, line graphs, various charts, etc. (see Fig. 5). The main drawback is the complexity for complicated data structures.



Geometric Transformations

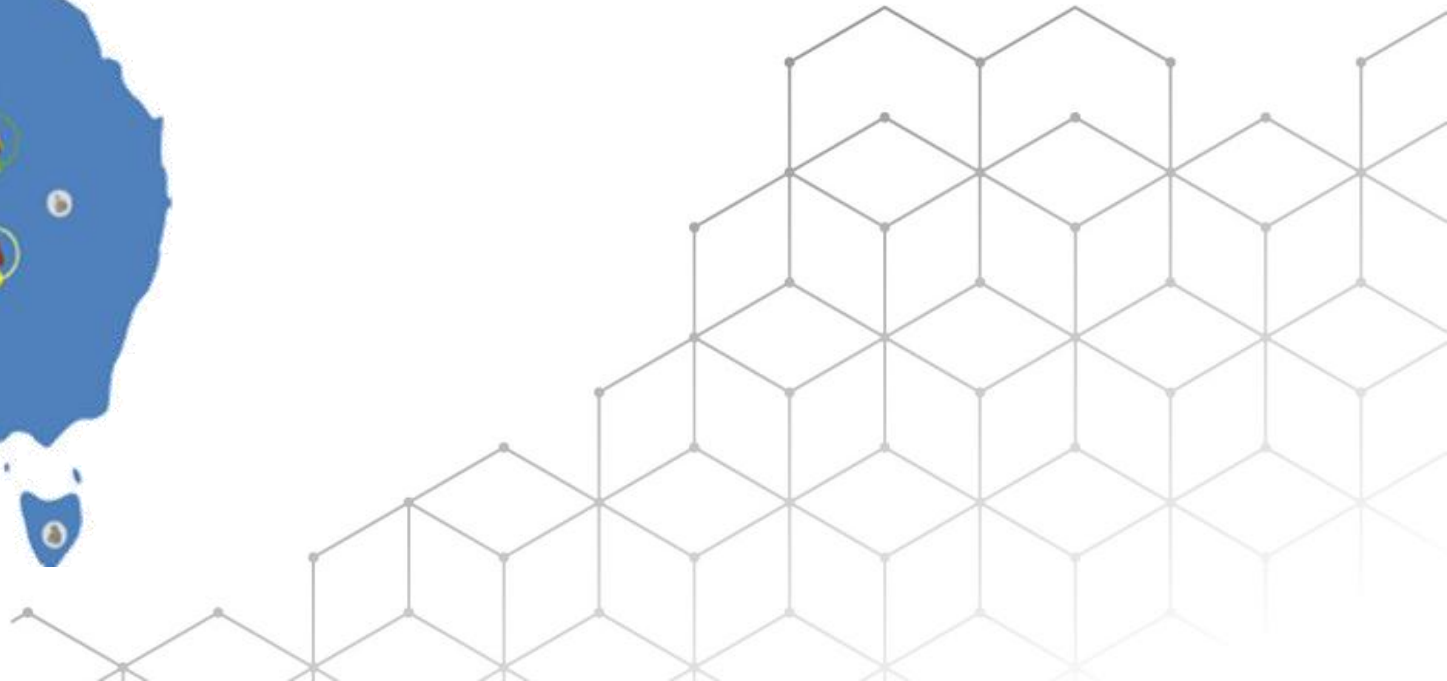
This technique represents information as scatter diagram.

This type is geared towards a multi-dimensional data



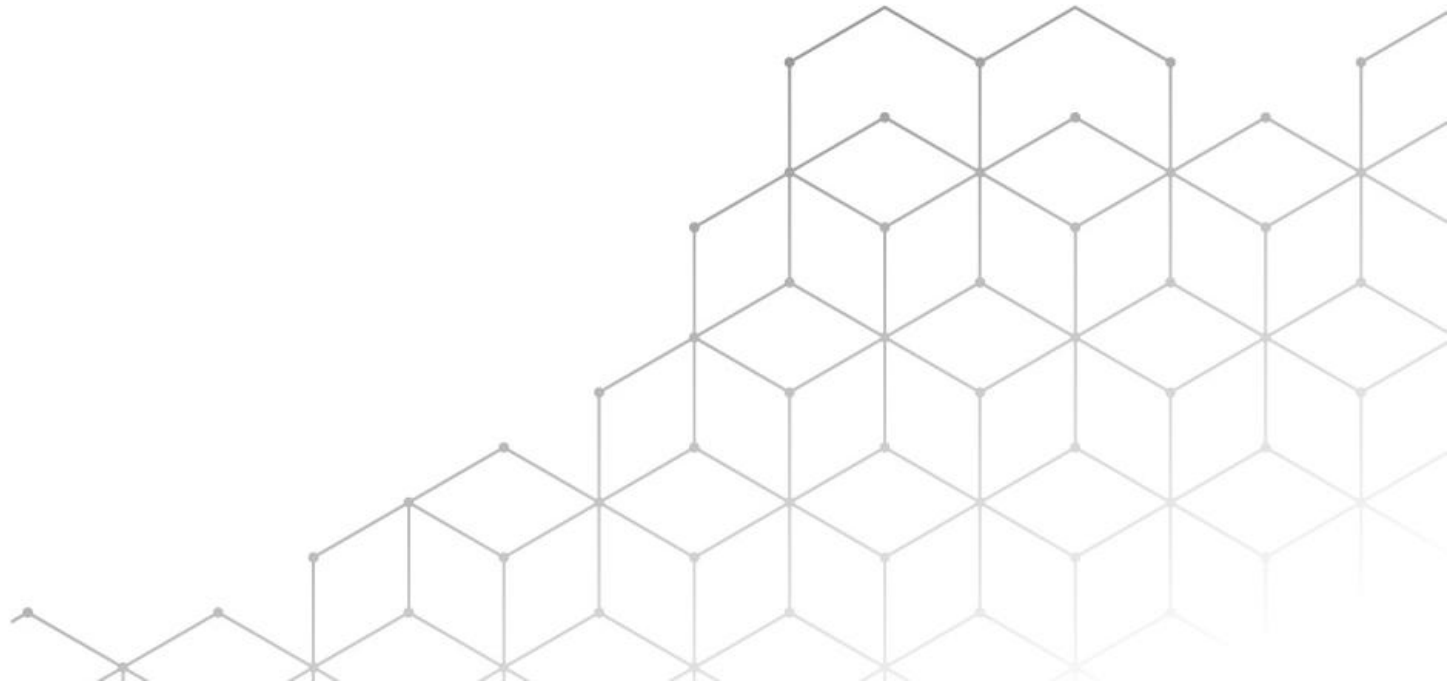
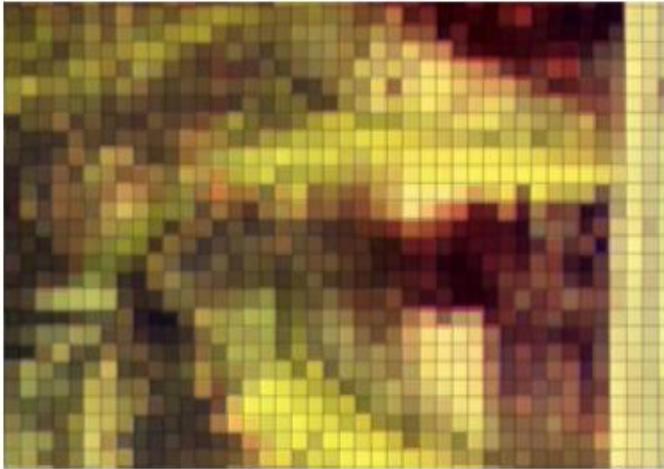
Display Icons

- Displays the values of elements of multidimensional data in properties of images.

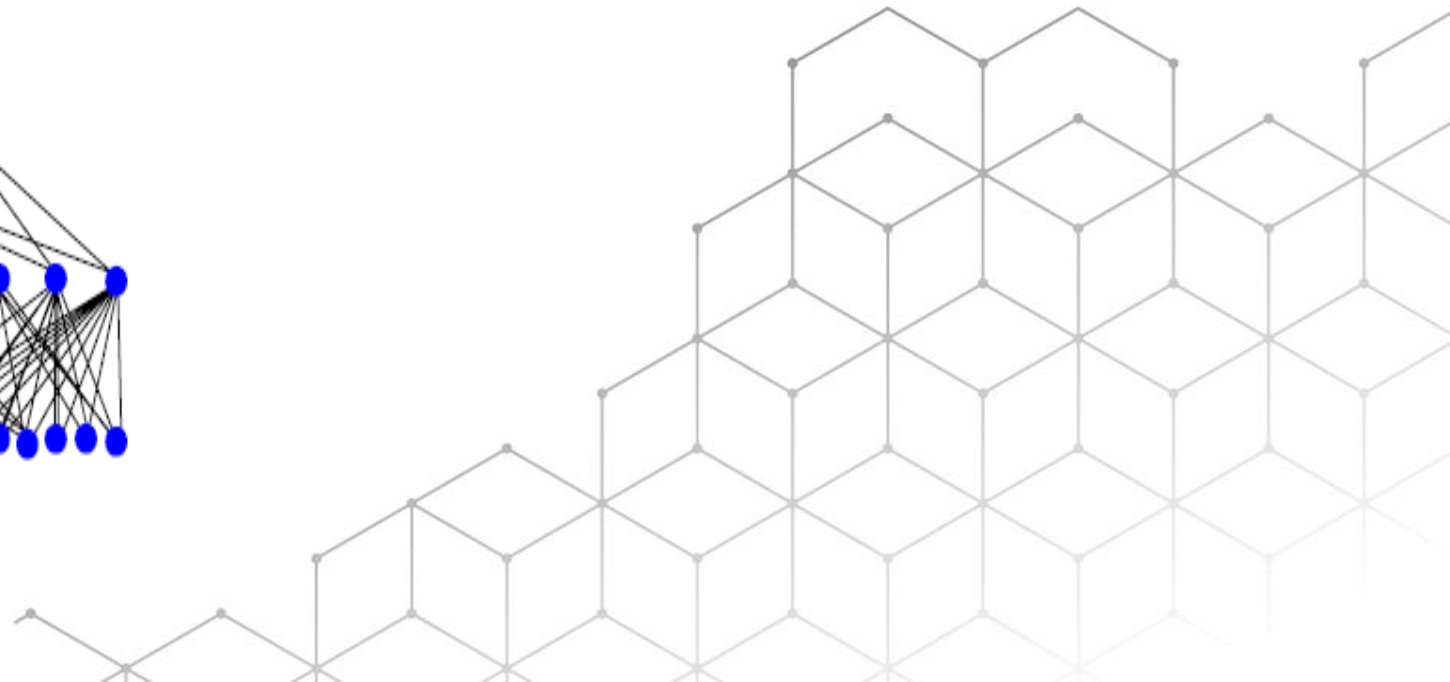
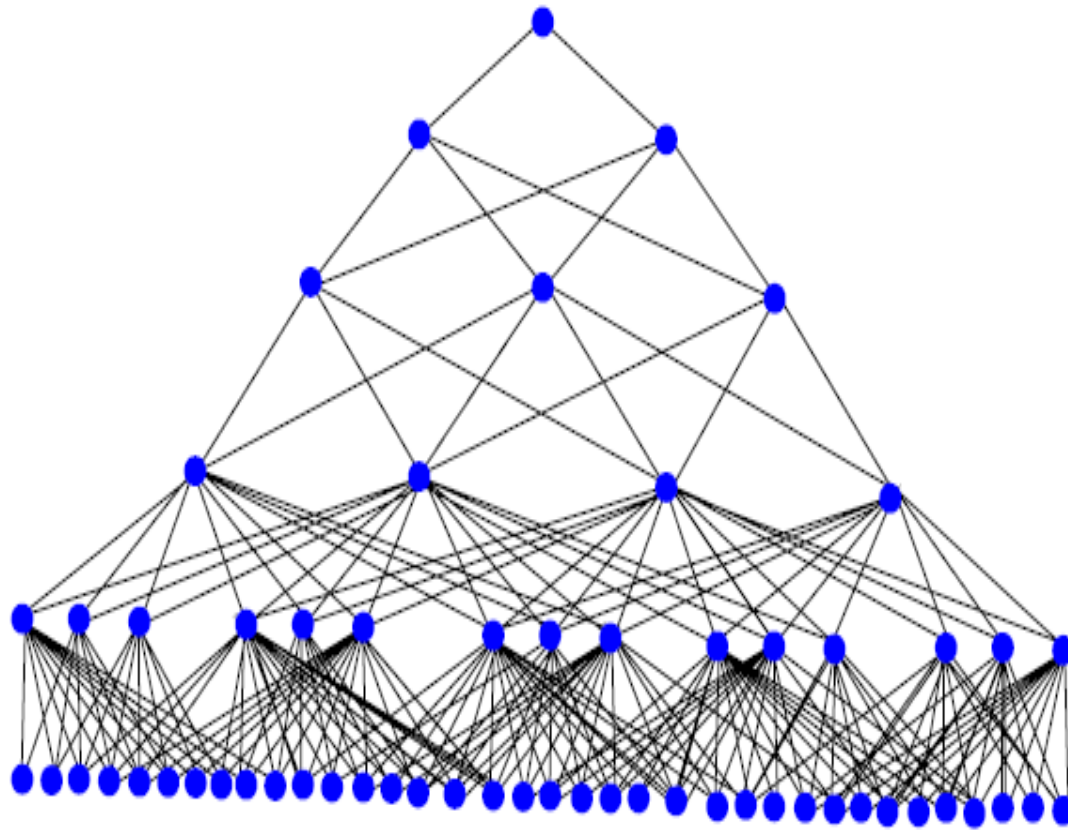


Display based on Pixels

- Display the values in each dimension into the colored pixel and to merge some of them according to specific measurements.
- Visualization of large amounts of data can be reachable with this methodology;



Hierarchical Images

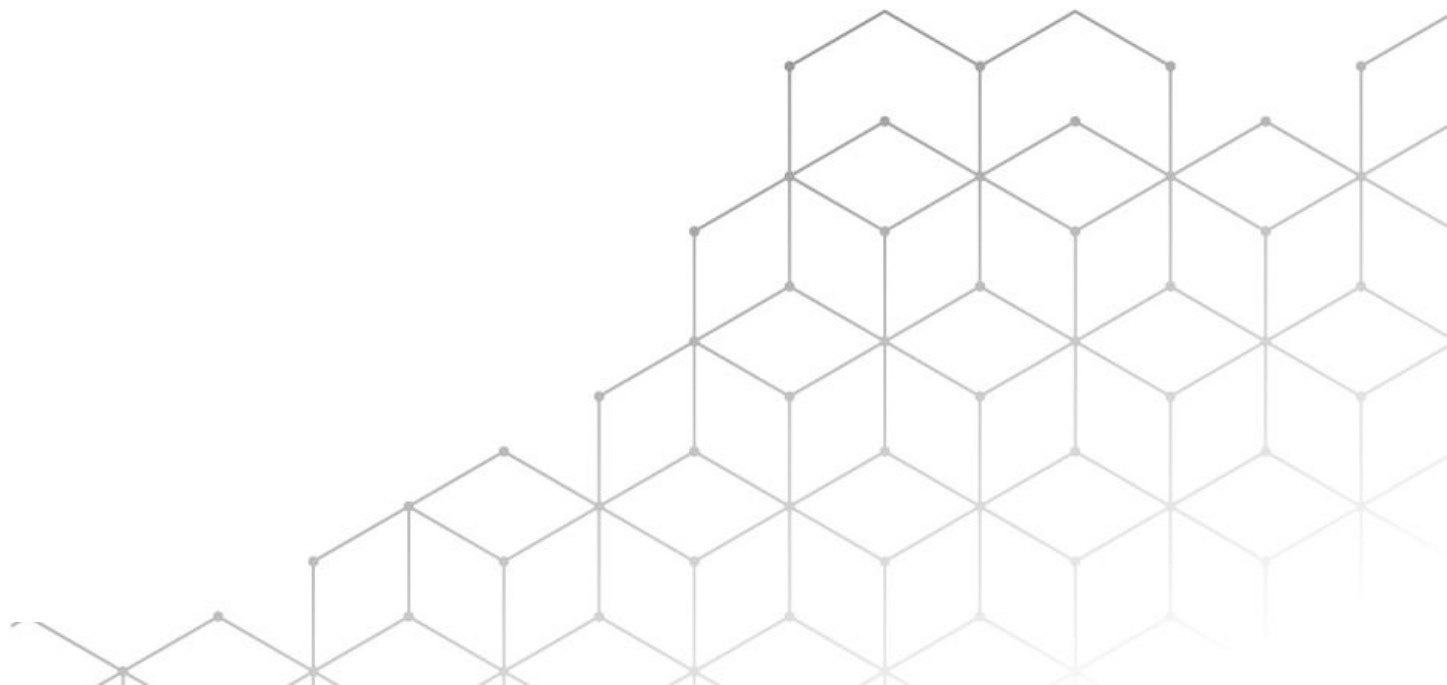
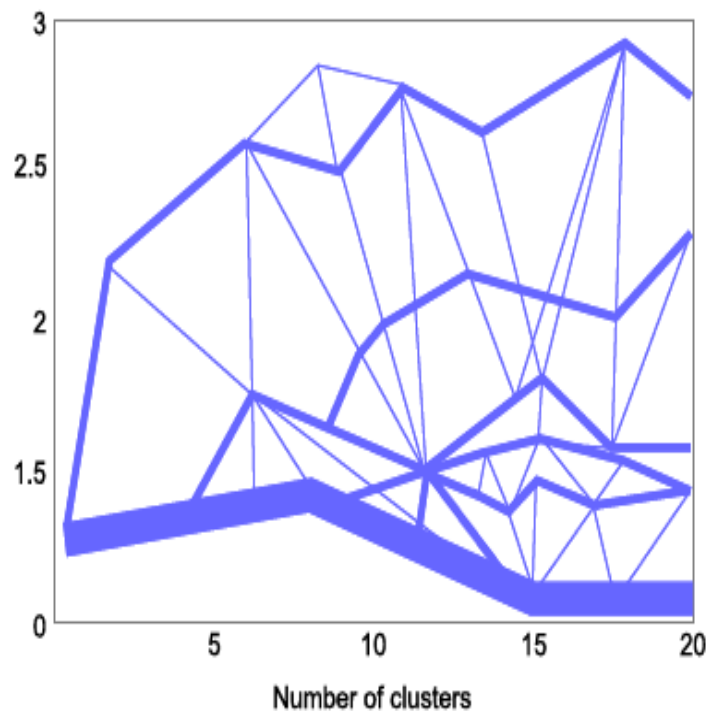


Tag Cloud

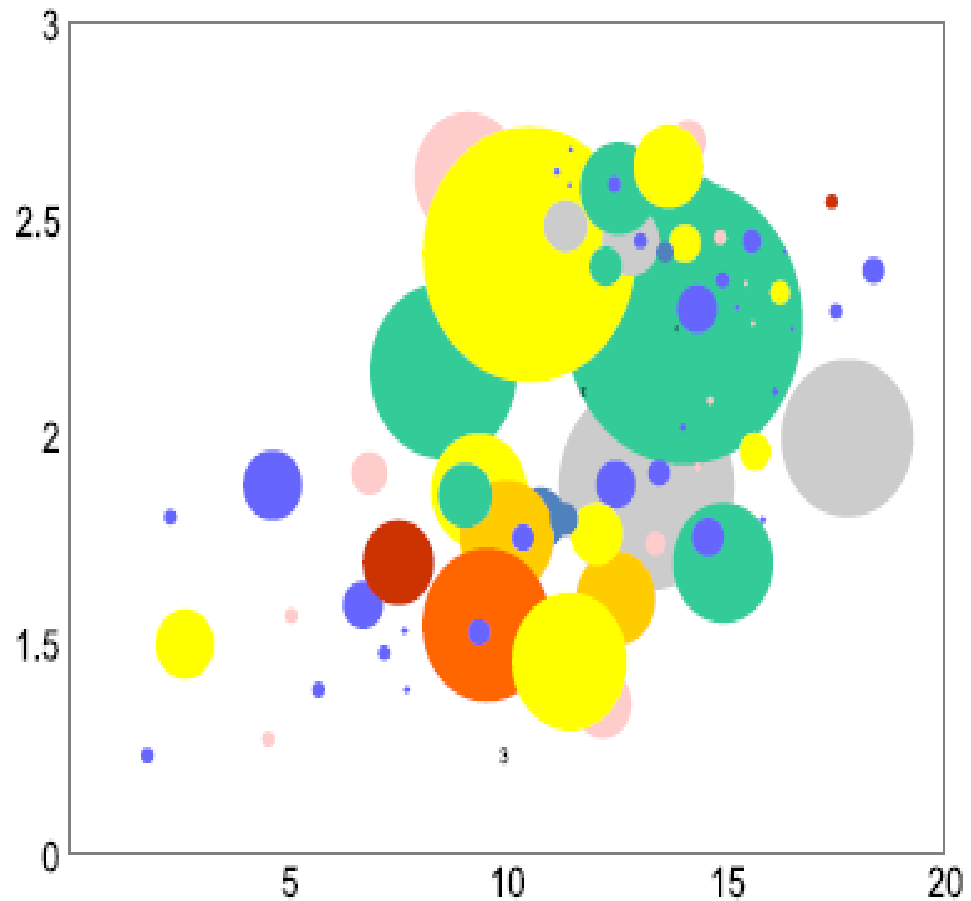


Clustergram

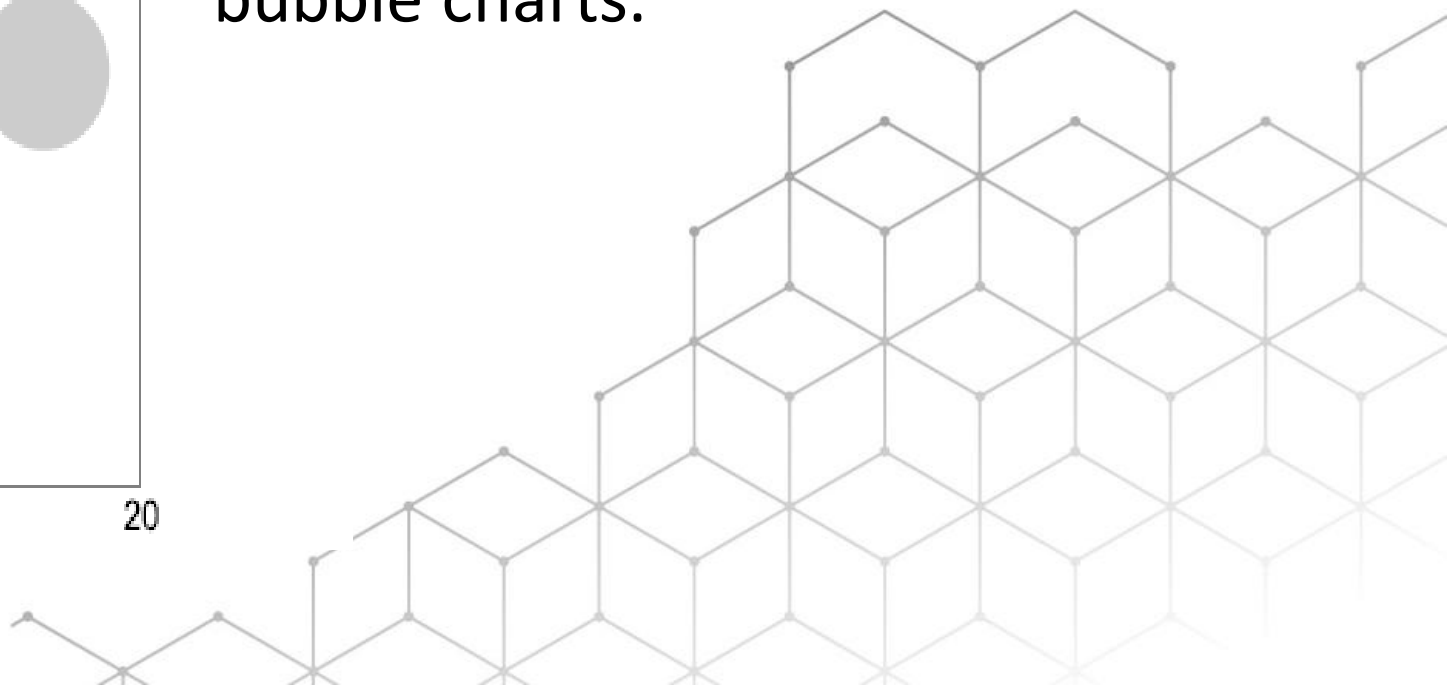
- Clustergram is an imaging technique used in cluster analysis by means of representing the relation of individual elements of the data as they change their number.



Motion Charts



Motion charts allow effective exploration of large and multivariate data and interact with it utilizing dynamic 2D bubble charts.

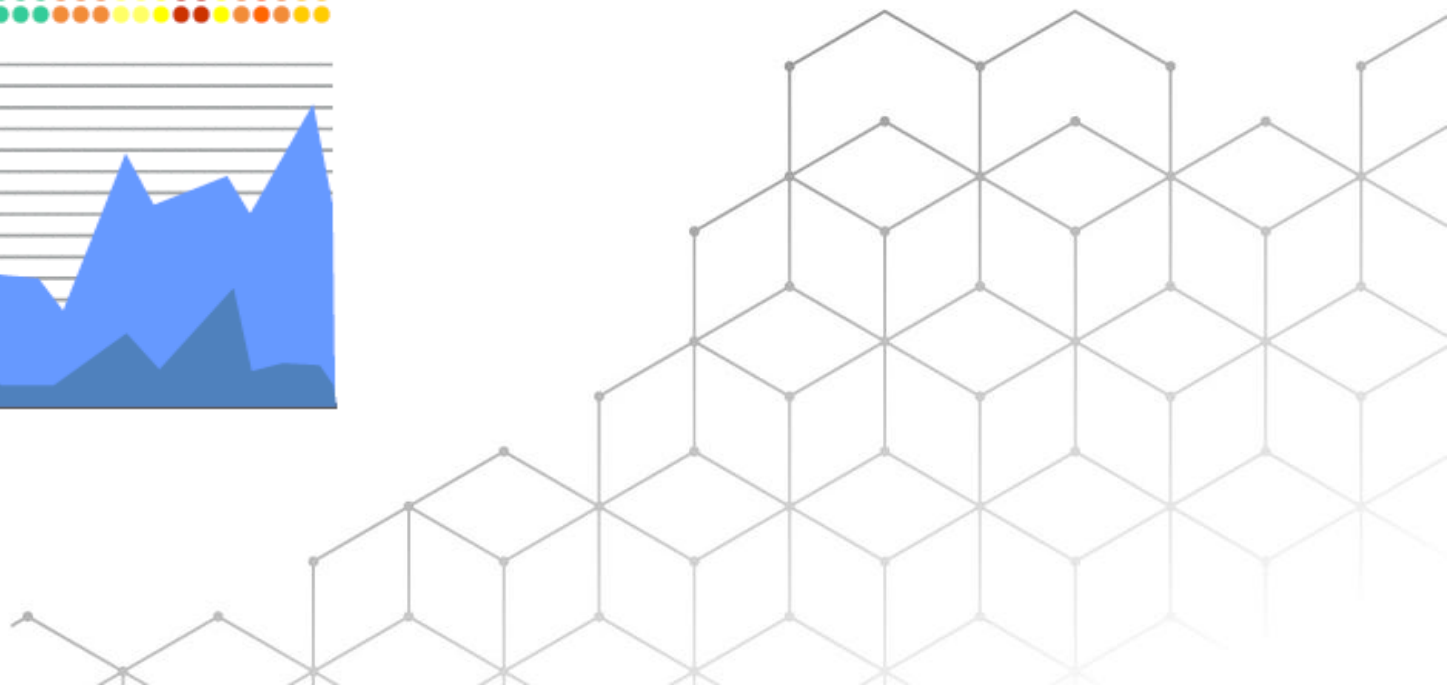
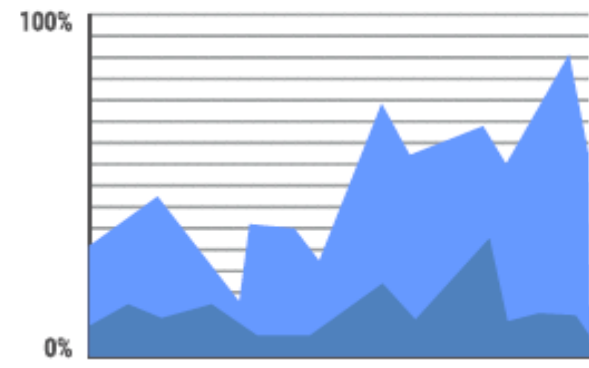
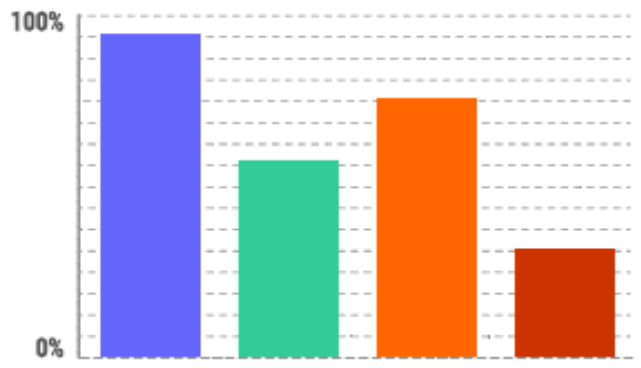
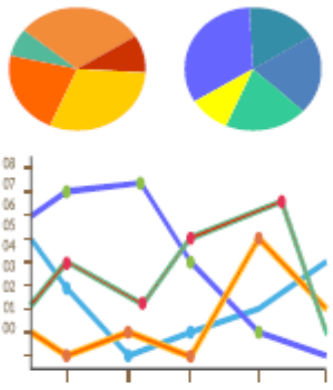


NOVA

IMS

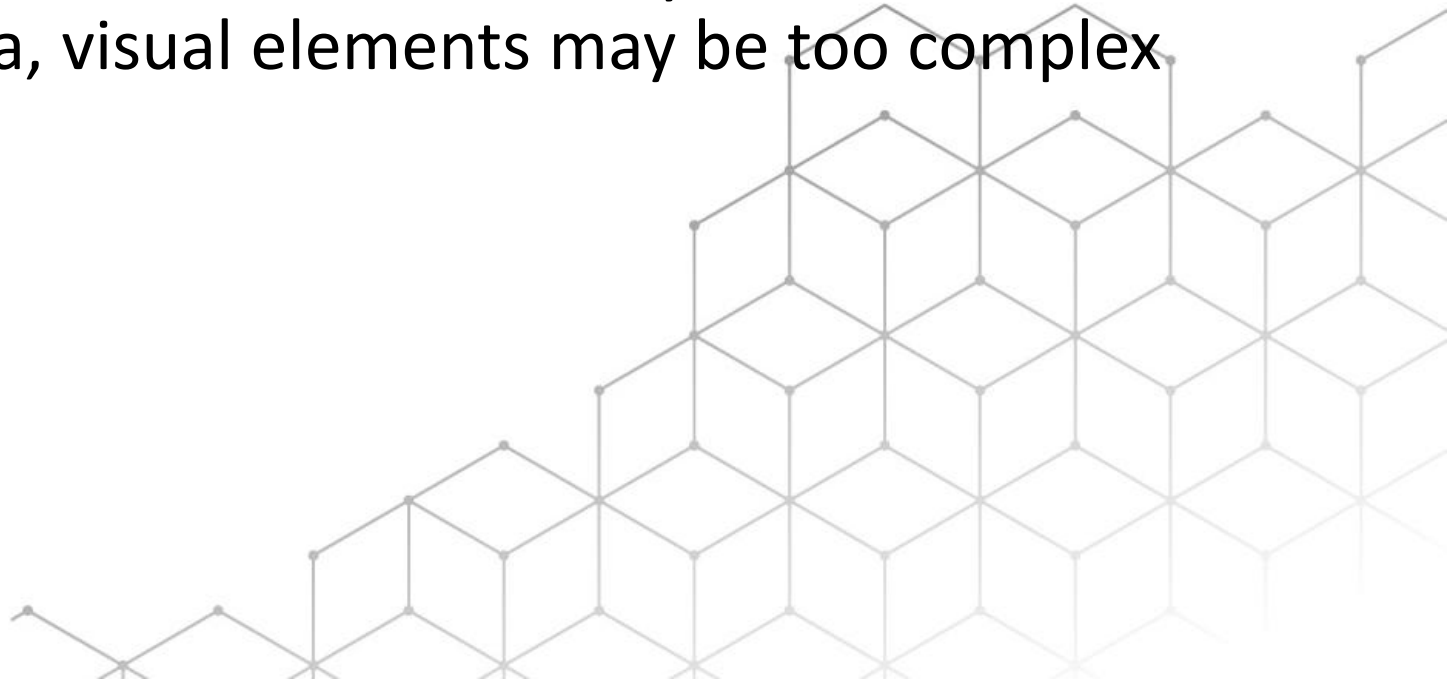
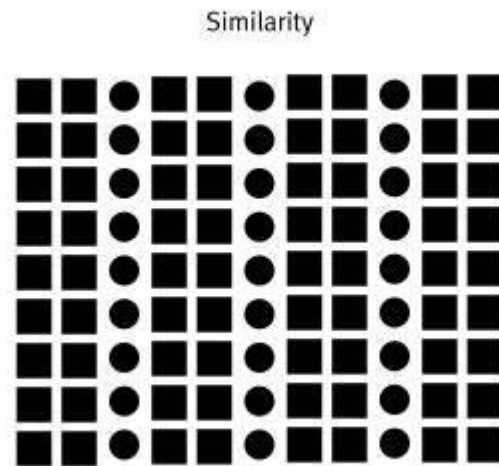
Information
Management
School

Dashboards



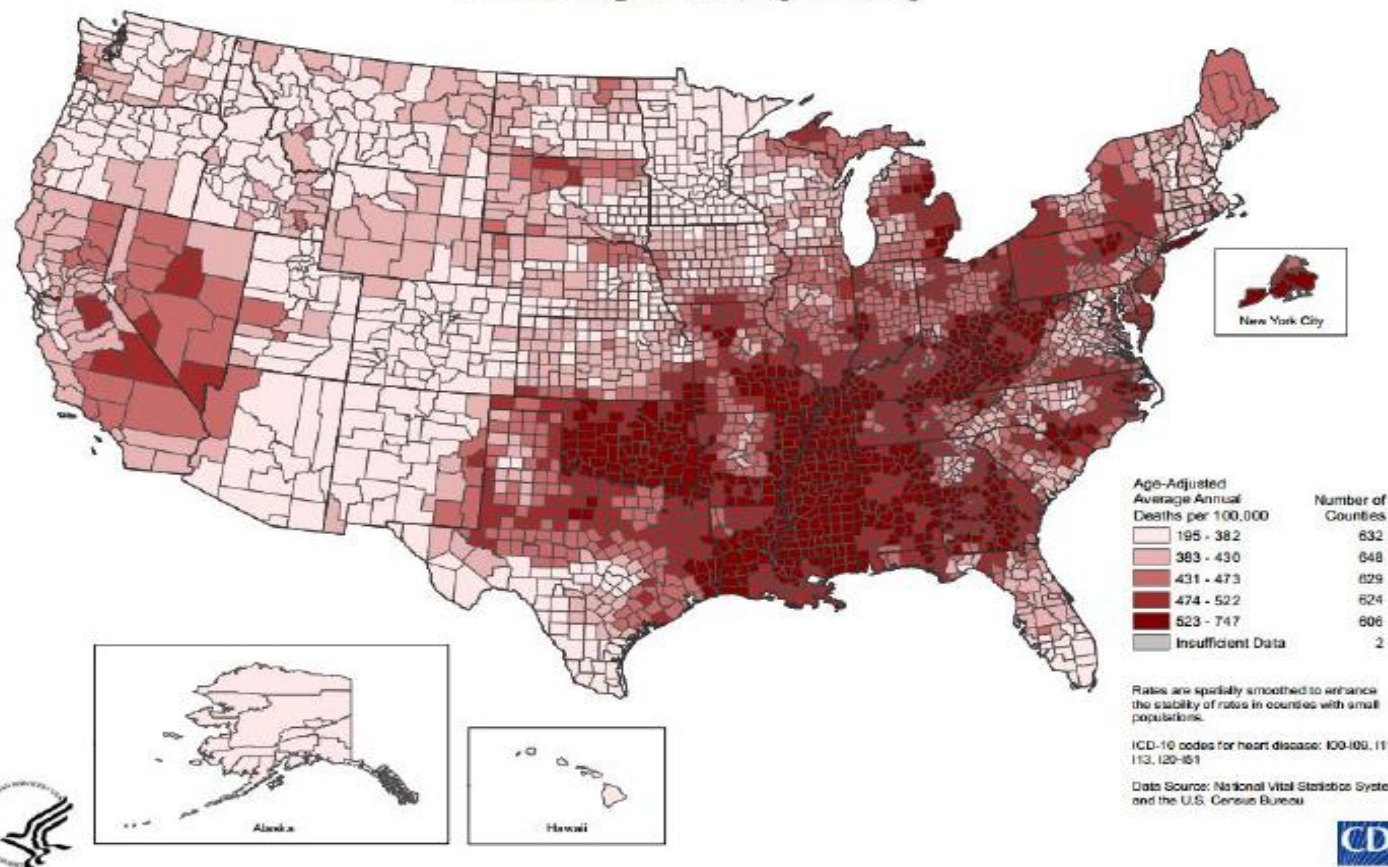
Recommendation

- Nowadays, there are many publicly available tools to create meaningful and attractive visualizations.
- Such as manipulation of size, color and connections between visuals
- Too many colors, shapes, and interconnections may cause difficulties in the comprehension of data, visual elements may be too complex



Thematic Maps

Heart Disease Death Rates, 2000-2006
Adults Ages 35+, by County



Questions and Thank you

